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Agricultural Statistics for Developing Countries

by

K. E. Hunt

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The author acknowledges how deeply he is in the debt of all those whose offices he has invaded, and whose time he has occupied with his questions on this subject. Responsibility for the interpretation which he has placed on the experience in various countries, and the use he has made of it, rests, of course, with him. He hopes that this publication will be of some service, and may thus, in some small measure, say 'Thank you'.

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I. INTRODUCTION

The purpose of agricultural statistics is to contribute to human betterment. The purpose of this bulletin is to complement and support local skills and initiative in improving the agricultural statistical services in developing countries. In particular it tries to provide a link between specialist statisticians on the one hand and agricultural and other field workers on the other, providing the former with a guide to the field problems and organisation which they must take into account in their plans, and the latter with an idea of the setting within which their own work lies.

In the years since the end of the last world war, a great deal of effort has gone into attacking the various statistical problems concerned with agriculture in developing countries. Though in recent years the reports of many of these pieces of work have contained reasonably clear accounts of the methods used, know-how seems to spread rather slowly. The justification for the present publication is this slowness of spread. It tries to bring together these experiences and is, in effect, a third—but completely rewritten—edition of a bulletin of which the first two appeared in the 1950s.[1]

The approach has been to draw on the lessons from a large number of studies, together with more general experience from both developed and developing countries, to provide proposals which seem of fairly general application. These are set within an overall approach which has the aim of getting the best guidance from the available sources of men and money in the agricultural statistics service. Plans which would result in good data for a few parts of the agricultural field and great gaps in the rest do not seem likely to provide the best overall guidance.

No two developing countries are quite alike in the situation which they present to their agricultural statisticians, but there are features common to many. Many have only the results of rather small and infrequent sample censuses to provide a description of the main structure of their agriculture. Annual data are usually available only for export crops. Their agricultural systems are often complex and non-standardised or the land areas involved are huge—or both.

Local people are either not able or not inclined to complete questionnaires, so that all information must be collected by enumerators. Suitable people for training as enumerators are often scarce and so, too, are funds to pay them. Distances are often large and travelling conditions difficult, which in turn makes the supervision of enquiries onerous.

Agricultural statistics in developing countries raise problems of statistical technique and of data processing, but these are not the most difficult problems to deal with. There is relevant experience under both headings from developed countries which can be applied reasonably directly. "Human" problems raise many more difficulties since experience elsewhere can give only very general guidance. Indeed, most of the problems discussed in this bulletin are human problems of one kind or another. Many stem from the difficulty of communication, between one man and another up and down a chain of command or between departments which must maintain liaison on a statistical subject. Others come from difficulties in building up effective and economical organisations and getting people to work within them. To give some examples of these non-statistical considerations, voting of funds by a government for some special, large-sized, statistical project may well depend on the success with which the directors of the statistical services have made clear to their political chiefs the contribution which the data could make to national development. Routine annual statistical enquiries may raise no unusual budgetary problems but they may stand or fall by

1. "Colonial Agricultural Statistics". Colonial Research Studies Nos 11 & 22.

the management skill of the organisers in maintaining morale and discipline amongst the field staff while keeping them moving from sample unit to sample unit, making the records on the timetable required. Again, in farm budget enquiries, the key problem may lie in making repetitive and detailed records without exasperating the informant past the limit of patience. Or a brilliantly designed piece of "statistical engineering", such as a large-scale, multi-stage, multi-phase, stratified random sample enquiry, may fail because it asks more of its operators than they can be moved by loyalty to the service—or patriotism, money, or tough supervision—to give.

It has been said that the establishment of an acceptable statistical system requires ingenuity, hard work, intelligent employment of limited resources, keen awareness of the data needs of other professional disciplines, astuteness in the political world, and probably a reasonable amount of good fortune—and this is a list worth bearing in mind.

Turning now to the application of agricultural statistics. In many developing countries governments are the chief users—sometimes with such singlemindedness that they forget the interest which others are developing. Typically, governments use statistics for various forms of national planning—general development planning, import saving, project appraisal, infra-structure studies, regional planning, agricultural extension programme building, etc. Outside government service there is a great variety of demands. Those of the statutory organisations and large private companies are often quite similar to those of governments. For instance, a seller of fertiliser operating on a country-wide basis is interested in comparisons between different regions in the levels of farm income, in demand prospects for farm products, and probably also in agricultural extension work aimed at improving farming practice. Other users may have rather specific needs. For example, dealers and transport agents who want to form a view of how much of various commodities they will have to handle may need forecasts of prospective crop production some weeks before harvest. Finally there are the economic historians and research workers who are seeking statistics either for purely academic purposes or for research studies which will later play their part in planning and policy formulation For instance, statistical time series are valuable raw materials used by agricultural economists to measure effects of factors which influence the amounts of agricultural products grown or the demand for them.

A relationship of rather special interest to a bulletin like this is that between these agricultural economists and the agricultural statistician. The relationship seems now to be much closer than it was, say, 10 or 15 years ago. Moreover, the distinction between such agricultural statistics as crop areas, livestock numbers, and farm product prices, on the one hand, and the records from "farm management" enquiries on the other, are not so sharply separated now, and rightly so. They are valuable supplements to each other. Generally speaking, agricultural statisticians and agricultural economists have a joint role in defining problems; however, it typically falls to the economist, often speaking for the policy-maker, to define the information required, and to the statistician to identify and obtain the statistics which will provide it, but obviously feed-back between the two is essential.

During the ten years since the previous versions were published, the problems, the availability of senior staff, and the sampling techniques, appear not to have changed greatly. Among other features, there seems to have been a striking increase in the general appreciation of the usefulness of agricultural statistics. More field staff seem to be under the direct control of statisticians with lessened need to borrow staff from the field divisions of the agricultural departments, and there is much more activity with farm input-output and family expenditure studies. Finally, a number of departments have fairly ready access to computers.

The sections which follow fall broadly into three parts. First there is a review of the main statistical needs, the general approach to statistical work which is used here, staffing and an outline of the practical application of sampling methods. It is assumed that specialist statistical advice will be available to those who are likely to use this manual and consequently basic statistical techniques are not described in any detail. Some readers may feel that these more generalised chapters could have been omitted without loss. However, whereas in the 1950's those working with agricultural statistics in developing countries

seemed to want a 'recipe book' which told them to do this, or do that, the mood seems now to have changed and consequently there seems to be a useful place for these chapters. The second part is made up of a group of sections dealing with specific field techniques—e.g. crop area estimation, collection of price statistics, and so on. Finally, there is a group of appendices concerned with technical details of measurement methods, recording arrangements, questionnaire design, and so on.

II. THE DATA REQUIRED

From the documents consulted in the course of preparing this manual it is clear that much more discussion is needed about the agricultural statistics which would be effectively serviceable in developing countries. Practice in temperate countries is still all too often taken as the pattern, and much too little attention seems to be given to the effect of sampling and recording errors on the usefulness of the figures for the purpose which they are supposed to serve. However, it is sufficient here to give an indication of the types of statistics which must be provided for and hence to indicate the collection techniques which must be explained.

Statistics are required for planning the future and assessing the past—and the latter contributes to the former. In development planning, statistics are needed not only when the plan is being developed and various budgets are being evaluated but after it is put into operation in order to keep track of progress. The planners who might use them include governments, statutory organisations, operators of private businesses ancillary to agriculture, and farm firms. If the latter are small they may not undertake any very explicit planning themselves as individuals but a service may be provided for them either by their co-operative society or by a government agricultural extension service.

Ideally one would seek for these and other purposes a comprehensive description in quantitative terms of the agricultural sector set within a framework of national accounts. This would require information on the structure of the industry—the size and organisation of production units, details of land use, numbers of livestock, production of farm products, the quantities sold and the prices realised, inputs [1] in quantity and value hence leading to figures for the income of the sector, and the incomes of the businesses within it. Movements of produce towards users, the distribution of the end use to export and local consumption, would need also to be covered. In order to provide for forecasting it would be necessary to have an appraisal of the availability of resources (for example, labour) and of response relationships—for instance, what change in the output of a farm would follow the departure of one of its workers? Some of these relationships might be estimated from an analysis of time series, others by direct observation. Input-output relationships for fertilisers might very well come best from observations on actual farms.

Where such a matrix of data is available it reinforces a user's confidence in the reliance he can place on a specific item. In a full statistical coverage it would be possible for statistics of subject A to reinforce those of subject B. Where the coverage is thin, however, and those on A are lacking, one is reduced to getting what guidance one can on that subject from inferences based on the statistics of subject B. This is typically the state of a great many developing countries. Usually there are figures on the main structure of the agricultural sector derived from sample censuses at ten year intervals. Data for the years between may be available for a much smaller range of items, e.g. estimates of the areas of one or two important cash crops derived from enquiries using much smaller samples than the decennial census. Between sampling and recording errors it is unlikely that a year-to-year trend can be determined with any certainty. Firm data may be available for certain marketing statistics—for example the quantity of a crop sold for export. These are often valuable, but, more often than not, they represent only a part of the crop, e.g. the citrus exports from a peasant area which also supplies local stallholders. Furthermore, they are not usually publicly available until after the period concerned so they are of little use to the general business public.

1. 'Inputs' is used here as a general term of labour, fertilisers, feedingstuffs, sprays, machinery and other items other than land which are used by farmers in the process of production.

In so far as much of the developing country's agriculture is directed to food production, and in so far as shortage of food is often a cause of anxiety, forecasts of food supply would often be valuable. There are circumstances where it might be possible to obtain forecasts, (for instance, where food supply practically depends on a single crop) but more often the statistical problems are much greater. Subsistence supply is a very difficult thing to estimate and still more difficult to forecast. The distribution of output between various outlets may change from year to year and, generally speaking, the problems are such as to make thoroughly effective forecasting of food supply too costly for practical achievement. In developed countries a great deal of useful guidance on the working of the agricultural sector is obtained by analysis of a time series, but for this purpose reliable estimates, at least annually, are needed of prices and certain consumer income variables.

Without the possibility of obtaining estimates of variables and trends, it is very difficult to be at all rigorous in making forecasts, and it becomes necessary to seek direct pointers to change rather than to depend on comparisons over time. Thus, statistics of new areas planted to perennial crops, of new farmers taking up a crop, of new cultural practices, and similar rather crude indicators, may yet be valuable. If good statistics of inputs and derived relationships bearing on producer response are not available, it is important to include farm management type records in the statistical programmes. The insight they can give into what really happens on a farm, the availability of resources (especially labour), the level of operational achievement, the practical limitations on production, etc., may be invaluable as a basis for making forecasts. This is quite apart from their value for individual farm planning and building of extension programmes.

To sum up, the chapters which follow provide means of dealing with five broad groups of needs:

(a) Basic statistics indicating the structural features of the agricultural sector. These might include statistics of land use, numbers and sizes of farms; production of main crops and forecasts of production of important cash crops; numbers of livestock where these are in commercial production; numbers or areas under tree crops with the ages and the productive state of the trees; farm population, distinguishing hired workers from family workers and those living on the farm from those supported by agriculture; and land tenure. For the broader features of this group, it would suffice if the figures were available at intervals of ten years, but crop acreages and production and livestock numbers should be available at annual intervals.

(b) Output and input data in physical terms, sales of main products through various channels and flows of main inputs into agriculture. It is desirable that these statistics should permit intersector and interregional movements to be assessed. They include prices of inputs and outputs at various stages from the farm gate to retail or export point; indicators of the structure of the marketing system,[1] internal as well as for export products, including some indication of the services offered.

(c) Records relating to individual farm businesses and farm and non farm households. Most of the urban consumer family budgets enquiries will be outside the regular agricultural statistical service. Farm business records should include receipts, expenditures, sales and purchases in physical terms; labour in use with an indication of seasonal usage; capital formation; and degree of self sufficiency. Farm household expenditure statistics should contribute to this and focus the enquirer's attention on the sources of income and the activities of family workers.

(d) Other general indicators of development. These might include data on the presence of new techniques, on farmers' attitudes to new knowledge and to policies and programmes, but it may be extremely difficult to obtain by orthodox agricultural statistics in developing countries some of the indicators that are possible in the developed countries.

(e) The statistical organisation, its size and aims. It is highly important that attention be paid to building up an increasingly effective statistical organisation in addition to collecting a

1. e.g. of the number and size and nature of firms and other agencies operating at various stages of the distribution system.

body of statistics. For example, training should not be wasted; the design of enquiries and the sequence of work should be such as to retain and build up the skills of the staff. The methodology should be improved—it is still possible to come across instances of the same mistakes being made time and again in the same general area. Careful records of experience—successful or otherwise—in the files would help. One particular direction in which organisation may need development is in getting certain data—particularly crop forecasts—*quickly*. This may require looking at the organisation with an open mind and streamlining in various ways, perhaps by putting the work in the hands of selected staff members. Finally, there is ample scope for developing inter-departmental liaison between the statisticians, the economists, the extension staff and the technical staff; the interests of each touches those of the others at various points which are highly relevant to development planning.

III. THE STATISTICAL ENVIRONMENT

The pattern of statistical activities undertaken at any stage in a country's development depends largely on the statistical environment at that time. The main core of this manual is concerned with the application of fairly straightforward sampling methods to obtain basically rather simple items–for example, statistics of the areas of the principal crops planted and their production, this representing a compromise between the needs and possibilities of the present situation. Before going on to the practical details a brief review of the main features of the statistical environment will be useful.

There are few short cuts to getting agricultural statistics in any country. One day aerial survey and observation from satellites may provide feasible solutions to a number of problems, but not yet. Typically, there are two approaches open to the agricultural statistician for obtaining his basic data. A postal enquiry can be sent to the farmer (or other informant) or an enumerator can visit him. The first approach is economical but it will not work if the informant cannot read or write, or if he does not customarily think in terms which conform to statistical practice, or if he is un-cooperative. Further, the more complicated the enquiry, the smaller the proportion of forms likely to be returned. There is also the probability that those who reply under these conditions are different in some important respect from those who do not. Consequently any statistics derived from this biased return must be, to some degree, suspect.

If enumerators are used, they must be paid and provided with transport and subsistence allowances for the time spent in the field. Only rarely can an enquiry be so designed that the enumerator can identify the farm or field in which he is interested without making contact with the farmer who operates it. Even where he can identify his site without such assistance only part of the data needed may be obtainable without the cooperation of the farmer. For example, it is possible to obtain information about the areas of a designated sample unit which are under various crops simply by walking round it and making visual observations or taking measurements. However, if any information is required about the size of the farm of which this is a part, or about its livestock or the equipment used or other items which may not be on open view, recourse must be had to a farmer contact.

Key features of many developing countries are their very large size, the diversity of ecological systems within them, the illiteracy of many of the farmers and the small, often fragmented holdings possibly farmed on a system of shifting cultivation. But much of the experience in large scale agricultural statistical work has been gained in countries which have temperate, fairly 'standardised', agricultural practices and largely European based social systems, or in India where at least the agriculture tends to be non-shifting and where mapping in many areas is good. A number of developing countries are physically small, for example some islands, but seem to make up for this by the unusual complexity of their agricultural or social system and the extreme urgency in solution of farm income problems.

Resources in funds and staff for statistical work are short by any standards, and particularly so with regard to the areas to be handled.

The agencies concerned with collecting agricultural statistics may be the Central Statistics Department, the Agricultural Department with a special statistics and economics section, possibly semi-governmental development agencies which have to engage in statistical work as a preliminary to their development work, various kinds of non-governmental agencies such as universities within the country, and visiting research teams. Within the government organisation the responsibility for statistical work is usually clearly accorded to one or other of the departments but occasionally responsibility is divided–for example, large scale census

enquiries may be carried out by the Statistics Department, but annual enquiries by the Agricultural Department.

Though there is always nominal liaison between the departments concerned it is not unusual to find very little effective contact. Sometimes this arises from friction between personalities or incompatability of departmental organisations, more often from sheer pressure of work which precludes anyone deviating from his formal commitment. In consequence, departments may not be providing the support they might give to one another by realistic sifting of the questions to be included in enquiries, by the cooperative use of their resources from time to time, by the early circulation of data in various forms of consolidation, etc. They may, in fact, be working over very similar areas and asking much the same questions of much the same people.

The original informant in statistical enquiries in the agricultural sector is usually concerned either with producing agricultural products or with trading in them or in requisites. Typically, agricultural production units can be classified into three groups:
i) expertly managed enterprises with formal records and often paid managerial staff. They may be privately owned, for example many sugar plantations, or they may be more or less state enterprises.
ii) "modern" agriculture which may be peasant or family farmer operated, but is distinctly market orientated.
iii) traditional agriculture which may be subsistence or it may have a market for a proportion of its products, either locally, or for export.

The group i) may raise difficulties over cooperation but sampling and recording problems are usually confined to ii) and iii).

In the context of marketing statistics the big traders and statutory marketing organisations correspond to i) of the previous paragraph and the smaller middleman and local traders to ii) and iii). The traders and marketing organisations can usually be easily located and, provided they are prepared to cooperate, there should be no difficulty in getting information from them. The other groups raise many queries of contact and communication, just as do the "modern" and traditional farms of the previous groups.

Many developing countries have statistical ordinances which require firms and individuals to provide information for certain statistical purposes. Often they guarantee they will be treated in confidence. Without such legal backing information may not be obtainable, and, even with it, there are limits to the amount or type of material which informants will supply. Generally speaking, statutory ordinances do not often enforce the provision of financial and income data for purely statistical purposes. This information may be required by governments for taxation purposes but there are usually strict provisions restraining the transfer of information so obtained to any other department. Private agencies attempting to collect statistics must depend on good will and good public relations to smooth their ways. One might expect that informants would be more ready to supply information if they were interested in the results. In fact, in few countries is this much comfort to the statistician and scarcely ever in developing countries. In the latter, the government and certain statutory organisations are probably the main users of statistical material, but they do not themselves contribute data to the statistics service, not being original producers. Associations of farmers may be interested in statistics, and individuals and firms may corporately agree that statistical material needs improving but individually they may not help towards achieving this end. Firms may show resistance or apathy to statistical enquiries, either from statistical illiteracy or from the exact opposite. Both apply in developing countries. Most individual farmers and the smaller marketing firms in developing countries are statistically illiterate; the informant may see no value in figures as such and could not envisage making any use of the results of enquiries. In the case of resistance arising from statistical sophistication, a firm may already be obtaining all the information it needs from private sources and see no point in cooperating with a general enquiry which would have the result of making its competitors as well informed as itself. Again, a comparatively large firm, viewed either nationally or regionally, may hesitate to provide statistical material through fear that, in whatever form the summarised data for the country or region are published, details of their firm which would be useful to competitors may be identifiable.

Most agencies, governmental or otherwise, who set about making statistical studies of developing countries now seem to have access to expert statistical advice that can deal with the design of enquiries. Usually experts are available in the relevant departments as consultants, but if not, the basic technical material is available from reference books. This does not necessarily hold for the field organisation and this accounts for the particular emphasis of this manual.

In the last few years computers have been installed in the statistics departments of many of the larger developing countries and access to mechanised handling of data is available for big enquiries. It might usually be wise to be cautious over access for smaller statistical studies in agriculture and it might be unwise to depend on availability of computer servicing *to a timetable* unless it is known that the agricultural statistics concerned had a high priority for computer time. It may be expected at the planning stage of an enquiry that computer facilities will be available and (partly perhaps for prestige reasons) the organisers are tempted to depend on these facilities. When the work comes to be undertaken, however, the timetable results in exasperating delays.

There is always some promise of new staff and new facilities becoming available to ease the central problem of too few resources for the job at hand. In most developing countries this potential comfort is a mirage. Even if more resources do become available demands are continually being put on the statistics department for data for planning purposes and the ratio of inputs to expected output scarcely increases.

Many developing countries have very few statistics for the period before about 1950, except for quantities of produce exported or imported. They have the results of sample censuses for 1950 and 1960, but usually the sampling fraction was small and the error attaching to the figures was fairly high, say 10% or more. In some instances it was not practicable to complete the enquiry in a single year and different regions were surveyed at different times over a cycle of three or four years. There may have been follow-up studies to provide material on the main items of interest for years between the decennial censuses but often these are patchy and confined to selected sectors of the agricultural economy. In fact, there may be little beyond the acreages on the estate, or large private holdings, and the quantities marketed through statutory channels.

The scope of the response to these conditions lies in one or other of the following directions. Enquiries might confine themselves to only a few of the possible items about which statistics could be collected, or they might deal with only part of the total agricultural producing population. This might be done either by sampling more or less evenly across the whole country or by planned omission of certain regions from the statistical coverage, for example by some form of cyclical arrangement whereby the enquiry is carried out in region A in one year, in region B in the next year, and so on, returning after a period to the first region. Economy might also be achieved by including in the one enquiry (which unavoidably must demand heavy expenditure on manpower and transport), a rather large number of questions so that it serves a multi-purpose objective.

All these approaches have been used in developing countries and a brief mention of the pros and cons may be useful at this point. Undoubtedly some reduction in the cost of a particular enquiry can be made by including fewer items in it, but this device may save much less than a proportionate length of time. The same farms may have to be visited and the same land surveyed whether one crop or ten is included in the enquiry. It may be essential to record the total of a group of crops as well as the figure for an individual one in order to get a cross check on the field operation. Usually, significant economies can be made only by omitting records for items which are heavy users of recording labour—subsistence crops grown in small patches is an obvious example.

The recording of a selection of farms by recognised sampling procedure is a most powerful tool contributing both economy and precision in agricultural statistical work. Efforts are sometimes made to cut costs by omission of parts of a country, not under a formal sampling procedure, but because operations would be too difficult there or because there is little interest in the locality. The general feeling of statisticians reviewing the African

situation recently was that such piecemeal omission should be avoided wherever possible, even a rough coverage being better than none.

There are many difficulties in the kind of enquiry which deals with one region each year until the whole country is covered, unless some kind of continuing local organisation is present. For instance, the sum total of agricultural activity as a whole is never known for any specified date so there is no "bench-mark" for comparative purposes. Again, such studies tend to require field teams to be moved from place to place over large distances and this invites wastage of staff who do not like to be away from their home districts for a matter of years. It may cause difficulty, too, with the language and contact between enumerator and informant.

The general impression of multi-purpose enquiries is that though they may appear to economise in overhead costs as represented by travel and subsistence they are not efficient. It is difficult to control the length and complexity of the questionnaire and it tends to become overburdened with questions. This has its obvious discouragement to informants who may not be prepared to be patient to the end of a long enquiry. Further, different aspects of the field work tend to require a different range of expertise. The kind of man who is good at getting details from a farmer's wife may not be good at doing compass and chain surveys of crop areas. The multi-purpose survey requires each man in a small field team to be expert in all kinds of field approaches, or a big team has to operate, which raises another set of problems.

In practice, agricultural statistics methods that will work in developing countries seem to fall into two broad categories. The one consists of collecting the main structural and commodity data from family farms, whether of a modern or traditional type, and for this purpose sampling techniques are very suitable and are universally applied; the other group of activities consists of three sections, for none of which has the sampling approach as much to offer as it has in the basic family-farm pattern. One of these sections consists of statistics on acreage and similar material for the large-scale private and socialised production enterprises, where postal enquiry is possible provided that the informants will cooperate. There is little need for sampling and there is little trouble in field recording. The second section consists of aspects of marketing and prices and the like, where a good deal of information may be obtained as by-products of records made for other purposes as, in effect, statistics collected in the previous section are by-products of the farm records of large farming enterprises. Sampling procedures have a part to play in marketing statistics, but less than in the production side. The third section includes large scale land use enquiries. Here, too, in principle sampling may be applied but often there is as much interest in the location of the various kinds of land use as in their total in statistical terms, and, consequently, the focus of attention may be as much on mapping as on statistics. The two are capable of being brought together in computer derived maps but not generally from the kind of statistics which can be obtained in the majority of developing countries. These sections in which sampling plays rather a small part will be dealt with towards the end of this manual. The bulk of the chapters which immediately follow deal with the collection of statistical material from the family farm.

It is worth noting that the sampling technique is used not solely because the magnitude of the problem forces selection but because obtaining a complete census is less attractive than it appears at first sight. Recording errors are unavoidable in all census work. Even if it were possible to ensure a complete record the individual figures would rarely be accurate and the total would be inaccurate unless the errors were compensating, those of over estimation being equalled by those of under estimation. Since, within limits, recording errors can be reduced by the careful operation and supervision that is possible in sampling enquiries they would contain fewer errors than those of a complete record attempted without proportionately greater resources. There are not many instances in statistical practice in developing countries where a complete record is obtained. The two exceptions are population censuses and certain enquiries which are made expressly to provide a record of the total universe from which the sample is subsequently to be drawn. These are of the

nature of 'pre-enumeration' [1]enquiries (discussed below) and usually consist of records of rather simple items, for example they might include a list of households in a village without any details of their land or even the number of people in each.

Notwithstanding the extent to which sampling is used and the success which has attended it in agriculture in developing countries there are one or two features which might be mentioned at this point as they bear on certain choices to be made in the designing of enquiries.[2]

Three points might usefully be made here about the results of sampling studies, even when well-planned and executed. First, sample enquiries are usually designed to achieve a given level of precision for broad regional totals, for example, maximum errors might be specified for the national and provincial totals. If only these items are, in fact, required no problem arises. However, increasingly interest is centring on smaller regions and individual localities, and the results of sampling enquiries may be inefficient for this purpose for two reasons. By the nature of the design no observations at all may have been taken in the locality of interest; the selection may have passed over that area and chosen an adjacent one. Even if the area was included appreciably larger errors attach to totals for the smaller geographical units than to the provincial and national totals for which standards of precision were built into the enquiries. The confidence which can be attached to the figures for individual localities may be unsatisfactory for the purposes required, but an enquiry capable of giving the required precision for all local detail may be prohibitively costly. Attention should be given to the amount of 'grain', in the sense in which the term is used in describing a photographic picture, that is acceptable in statistical work. Changes from time to time in census area boundaries, the use of different areas for special surveys from those used by perhaps a census, and the omission of certain small geographical areas from recording under the normal sampling procedure, may mean that comparable agricultural statistics can be assembled only for rather broad geographical areas. These may in fact result in too coarse a grain to give a clear detailed picture. A much better result might have been obtainable had more care been taken in choosing unit boundaries and definitions in the first place.

The two further aspects of the significance of error attaching to samples enquiry results should also be given some weight. Notwithstanding the economy and efficiency of sampling designs it is possible that the minimum error obtainable with the resources available may still be too large to serve the purpose desired, particularly since the error as calculated in most statistical enquiries does not comprise the total error to which observations may be subject. To take a rather gross case, if the weighing equipment which is used for estimating the production of a cash crop in a country consistently gives a reading 50% above the true one the sampling error may be a modest one but a substantial discrepancy can be expected between the estimated crop and the quantity which will be sold. If the error in the equipment is different at different years in the survey, or if the staff, possibly trained by different supervisors, apply their instruments in somewhat different ways in two successive enquiries even the comparison between two years would be totally unreliable.

In conclusion the *continuing nature* of statistical work and the need to take the long view should be stressed. Each enquiry naturally attracts its own special attention. However, it would probably be wise to think of agricultural statistical work in developing countries as rather like the growth of a firm, where it is important to service the investment and to obtain orders and capital to permit continued growth and development. In a manufacturing firm the bulk of the capital may be represented by equipment. In a statistical organisation investment is represented by staff who have been trained, by contacts which have been made, and by sampling designs and field operating procedures which have been tested. A firm would wish to maintain in good working order all the productive equipment and

1. The term is used here for enquiries carried out largely to provide data on which to plan the main enquiry.

2. Emphasis is always placed on probability sampling because this allows a degree of confidence to be assigned to the results of the enquiry. Generally speaking, most studies in developing countries are now, in contrast to the early post-war period, planned on reasonably rigorous statistical lines.

gradually to increase and to modernise it. The same holds for statistical organisations. It is essential that the investment in effort and cash represented by staff who have been trained for one enquiry should not be thrown away but should be retained and built upon for future work. So should the records of tested working experience. Investors are prepared to invest in a firm when they are convinced that there is a demand which the products of the firm could meet. In a similar way, funds for agricultural statistics will not normally come automatically in the required volume; quite explicit efforts may be needed to convince those who control funds that the statistics service is worthy of continuance and expansion of funds.

IV. APPLICATION OF SAMPLING TO AGRICULTURE IN DEVELOPING COUNTRIES

Introduction

Most of the sampling enquiries in the agricultural sector in developing countries have conformed to one basic pattern. Typically the whole national agricultural industry is divided into sections–possibly regions, possibly 'type of farming' areas. This is called 'stratification'. Within each of these a sample of units, of a size which can be readily handled in the particular circumstances, is selected at random. These 'first stage' sampling units might be administrative districts or villages. Typically, it is convenient for certain stages of the work to use first stage units which are too big to be used for all the detailed recording work. For this detailed work second stage units are chosen at random from within the first stage units; perhaps individual farmers from within villages. Sometimes a sample of fields is chosen from those operated by the sample farmers, thus representing a third stage. Such a design is called a stratified, multistage, random sample design. The choice of the proportion of the whole which it is desirable to include in the sample at each stage depends on various considerations, including the variation between units at different stages and the effect on the costs of carrying out the enquiry of different sampling fractions at different stages. Generally speaking, for a given fraction of the agricultural sector actually recorded, a design which is based on recording a few farmers in each of many first stage units is likely to be more costly than one which uses many farmers in each of a few first stage units, because the former requires more time to be spent in travelling per farmer recorded.

Once the records have been made, the process of deriving regional and national totals from them–a process variously termed 'raising' or 'blowing up'–consists essentially of saying that if the sampling fraction was 1/n and the total for the sample is t then the total for the whole is nt. In practice, the computation may not be done in this way, and ancillary data may be brought in, but the statistician is dependent on the data used in the design for the derivation of the totals which are usually the end products desired from enquiries of this kind.

The actual work of recording may occupy very different periods of time in different types of enquiry. It may require only one visit of an hour or two to each unit finally chosen, or practically continuous contact for months, or a series of visits. For convenience each of these visits, in which the enumerator usually moves from sample unit to sample unit across his area, is called a "passage". 'Pre-enumeration' visits may be needed to collect data for planning the enquiry and 'post-enumeration' visits may be made to check details in the main record.

In outline, this is the typical pattern of sampling studies in developing countries. The concept of the stratified random sample has been well tried in practice; the theoretical aspects and the computation of results from such enquiries are dealt with in the statistical text books. The discussion below centres on operating the method in the field under conditions in developing countries.

Frames

Whether a complete or a sample enquiry is to be undertaken it is necessary to know the scope of that sector of agriculture which is to be included. A comprehensive list of one kind or another is essential–it might be a list of farmers for an acreage estimating enquiry or of

ginneries for a cotton buying enquiry—but a list is essential. In sampling work such a list is customarily known as a 'frame'. It is usually necessary to take over for use in agricultural enquiries a list which has been prepared for some other purpose, consequently it tends to be either out-of-date, or incomplete, or both.

Most frames fall into one of two groups:

a) those made up of human or business units or

b) those consisting of areas of land (or their equivalent on maps).

Both tend to have advantages and disadvantages.

Amongst the commonly available human frames are lists of villages, or of farmers selling produce to centralised marketing agencies—complete lists of farmers are rarely obtainable directly in developing countries. One advantage of frames in this group is that they lead directly to agricultural activity, at least to the extent that those on the list are farmers. Further, they tend to lead there by a route which is familiar to the informants. For example, where the list of villages is derived from the ordinary administrative activities of the local and central government, it will seem quite natural to a farmer for an enquirer to come to him through the local administrative headquarters. Again, practically all kinds of information which can reasonably be sought about farming can be obtained through the farmer if he is prepared to be cooperative. This means that a frame which is keyed to the farmer is suitable for enquiries on all kinds of subjects. In contrast, area frames are elegantly suitable for certain kinds of estimation where the cooperation of the farmer is not required, but if the statistical service is later to take in other kinds of information which can only be obtained from him an entirely new approach, with its staff training, would be necessary.

A typical frame would consist of the map sheets for the country concerned. The means of using them to obtain a sample is described in some detail later in the chapter, but essentially consists of choosing, by an appropriate method, sample segments from sample map sheets. When the appropriate areas are visited on the ground, the necessary information can be collected. Clearly such a frame remains perpetually up-to-date. Its disadvantages include the fact that there is no simple and reliable means of telling without visiting all the areas whether or not they include agricultural activities. In a number of developing countries there are large areas of undeveloped land without farming and it may not be at all easy to devise a scheme which will retain the advantages of this frame but avoid staff visiting many areas without agricultural interest. Again, though the technique can be readily adapted to directing attention to farmers, it loses some of its economy in the process. Moreover, viewed from the standpoint of the farmer, particularly an illiterate farmer, the approach is a strange one. The enumerator turns up with a map and measuring equipment, in the process of locating on the ground the selected map segment and thence proceeds to a contact with the producers. This is in total contrast with the lines of responsibility and authority that are well-known in local administration and may well raise fears of compulsory purchase of land.

The process of location on the ground is not by any means straightforward in cases where the maps are on a fairly small scale. It is probably impracticable to operate these schemes with anything less than about a 1:25,000 map. There is the obvious difficulty of locating the field equivalent of the chosen map segment and, to judge by Japanese experience, there is some appreciable risk that field staff will, where a margin of uncertainty prevails, move segments to areas where recording needs less effort.

In principle, it is, of course, possible to use one type of frame for one part of an enquiry, and the other type for another, and some rather complex techniques of overlapping frames have been devised. Though it is quite practical to use two frames it may involve difficulties in accurately drawing a boundary line between the areas being handled by each.

Of the frames actually used in developing countries in recent years the records of previous population censuses have been amongst the commonest lists used for agricultural enquiries. It is unusual for an agricultural and population census to be combined and, since the population census has a more general application, it is usually undertaken first. The records, at least down to village level, are usually good. The household heads may have changed with passing years but this can usually be dealt with by other means. Perhaps the commonest disadvantage is that the smallest unit recognised as an entity for population census purposes

which may differ considerably in size from unit to unit. It is, therefore, usually essential to provide for the grouping of these units within a reasonably standard "primary sampling unit". The provisions must also be made for a substantial expenditure of time in locating the primary units on the ground. A unit may still exist in complete conformity with the census record of a few years back, but there may be disagreement on the part of the inhabitants about its boundaries. There is, however, a special risk here in that, whereas originally the whole area might have been covered by a complete population census, a use of the list as a frame may mean that only a single village or district in a locality may be included in the agricultural enquiries. If the inhabitants of only that district are consulted about boundaries they may indicate positions which are different from those corresponding to the records of the census. It means that the authorities on all sides of a chosen sample unit may have to be consulted before its boundaries are cleared. Otherwise an error might be introduced if it were assumed for purposes of 'raising' that the relation between the unit as recorded and the country as a whole was equal to the ratio between the census record and the country total.

Rather cruder problems have arisen from time to time; for instance the original field records for a census have, on occasion, been sent for tabulation outside the country concerned and have not been returned; blocks of field records have been lost in moving; and necessary identification data may have been lost though the record books were retained.

Among other fairly comprehensive and serviceable frames which have been used from time to time are various kinds of tax lists and more general administrative lists. Typically, local administration provides for some kinds of hierarchy of units leading perhaps from the whole country to provinces and sub-provinces, then to districts and sub-districts, possibly to parishes. The smallest unit may consist of communities of a few hundred people. Generally, such lists are reasonably complete and reliable down to the lowest or perhaps the next but one level. Here, too, it may not be possible to locate the boundaries of the lowest administrative unit without local enquiry. Tax lists were formerly much used in default of other frames and they have the advantage of providing a line of contact which is generally familiar to most of the people concerned. The disadvantages include the odium attaching to taxation, the fact that widows and old men and other minorities are probably excluded from the list although they may be still farming, that there are multiple taxpayer families, and that the final units are often of very unequal size.

It is best, when starting out on agricultural statistical work in developing countries, to begin with the clear realisation that there is no hope of finding good frames ready made. They have to be built up by painstaking work. Such lists as are available from various sources should be taken, and an organised effort made to check and develop them either completely so that there is a record of the whole of the relevant universe, or partially by methods that are outlined below. This approach, namely that a frame is something to be built and kept in continual repair, representing a valuable capital asset, is essential for good statistical work. As a pointer to what might be possible in particular situations, some less usual bases for sampling might be noted which have been used in developing countries in recent years.

In Libya a study of tomato production was based on the lists obtained from processing plants of the holdings having contracts with them. Incidentally, the sample size was based on the quantity contracted by the various holdings. Sheikships were used in an enquiry into durra production in the Gedaraf area in the Sudan. In Malawi those selling cotton at buying points were interviewed, a card completed giving the size of holding, labour used, and other crops grown, and these cards provided the frame for a sample enquiry. A tea survey in Uganda used as a frame the list of those to whom tea seedlings had been issued in a development programme. The lists of areas for which agricultural instructors are responsible have been used as a frame for a crop estimating study in Ceylon and a number of other countries. In principle this ought to have the advantage that each such unit should represent about the same volume of agricultural activity.

Care must always be taken that the checking of frames is not carried out in too narrow a manner. Thus it may be necessary to make a systematic and broad-ranging check of exceptions to the content of any frame, and in doing so it might be well to explore the matter with a wide range of experts in agriculture in the neighbourhood, not least those with

an interest in land tenure. For instance, state enterprises will not normally be listed in any of the frames referred to above, except the area frames. In Iran for example, some large private farms which were not in villages had to be listed separately and brought in to the coverage of the enquiry. Absentee landlords who farm by hired labour would be missed if sampling is, in effect, from a resident farmers list in whatever way it may be built up. If they have tenants in occupation and these are treated for the purposes of the enquiry as farm operators then the land would be correctly represented. In general it is probably worthwhile in preparing frames to make out a list of all large farms and their locations even if many can be later deleted as already covered in the standard frame being used.

The approaches outlined in the previous paragraphs should result in a serviceable frame being available. It may be that, initially, only the skeleton is actually prepared but the steps for filling in the details should be clear for completion during pre-enumeration work.

The process of designing an enquiry requires the organiser to weigh the facilities he has available to carry out the enquiry, including the amount of staff time available, the supervision and other skills, the amount of transport and the facilities for tabulation. The planning process first requires a general review of the kinds of procedure which are open (for instance random or systematic sampling, simple or stratified sampling), the hierarchy of sampling units that the frame involves, whether equal or unequal sampling fractions are to be used, and what method is to be used for raising the sample material to national level. It is against this general background that choice of procedures will be made and steps taken to develop strata and select samples.

Stratification

In statistical terms the aim in choosing a basis for stratification is to get as big a variation as possible between the mean values for the various strata of the item being recorded. This is helpful because this inter-strata variation can be 'analysed out' of the total variation shown by the individual records when assessing the 'error' attaching to the estimate for the country as a whole. Thus in a sample scheme to estimate maize area per farm, and hence total maize area, suppose that the country is divided into climatic zones and a random sample of farms drawn from within each. If maize area per farm is very different in some zones from others the variation in the mean values of maize area per farm between strata will be large. When the source of variation is removed from the total variation it will represent a valuable reduction in error.[1]

Obviously, if several items are being recorded one basis of stratification may not be equally effective in error control for each. For example, the area per farm of a particular crop may differ considerably between climatic zones, but equipment per farm may show little inter-zone variation but a considerable variation between strata based on size of farm.

Happily no bias is introduced if stratification procedure proves not to be successful in this aim, only a smaller control of variance will be achieved than was hoped. The commonest bases of stratification in the experience of developing countries appears to be (a) administrative units, (b) ecological/agricultural zones, (c) village or farm sizes. These are not intended to be in order of preference since often they are used in combination. The advantage of stratification is usually quite clear, e.g. a Lesotho enquiry found that stratification by administrative units gave a gain in precision of 12% and stratification by ecological zones a gain of 25%.

In principle, one would expect that the best gain from stratification would come if the strata could be some kind of ecological or agricultural zone. These are usually well known, in

1. For those unfamiliar with the statistical approach some insight into the usefulness of stratification may be obtained in entirely non-technical terms along the following lines. If there are parts of the whole which differ a great deal between themselves in respect of the item being recorded then, if stratification is not used and the sample is chosen directly from the whole total of farms, it might happen merely by chance that the chosen farms bunched in some groups while in others few farms were chosen. This could result in an estimate of the national mean a long.way from the true one. If, however, the whole is separated into groups and sampling is from within groups, the result is unlikely to be so far wrong.

general terms, in most countries, and maps are available to indicate, at least generally, where the several zones are located. The difficulty is that there is usually no indication in the records from population censuses, etc., to be used as frames to show to which zone each should be assigned. To make this allocation normally requires a very considerable amount of pre-enumeration enquiry. As has been indicated already in the discussion of frames, the actual location on the ground of the smaller units, whether they are administrative parishes or sub-districts, or even larger units or segments of a census record, often presents difficulties. It is usually quite easy to regard the higher units in a hierarchy of census divisions or administrative divisions as strata and to sample within them. However, each of these higher units customarily contains more than one ecological zone and consequently part of the potential gain from stratification is lost. Moreover, it may well mean that the opportunities for analysis of the final records to give information on an inter-zone comparative basis are prejudiced.

It is very much to be desired that agricultural ecological zones be used for stratification purposes in agricultural statistics enquiries. This is dependent on inter-disciplinary liaison between the statisticians and the field experts—a liaison which is, in the end, well worthwhile for both. The availability of aerial photographs can help considerably in this work since the general pattern of zone distribution can be determined in relation to communities and roads in greater detail than may be possible from maps made from ground observations.

Among other bases of stratification which have been used in developing countries are a number which are related either to acreage per holding or holdings per village. Thus the number of adult males at a previous population census has been valuable in stratifying villages within one or other of the administrative units next above in the hierarchy, the number of adult males tends to vary fairly closely with the number of holdings. In other areas stratification has been by the number of working oxen in the village, after villages had been divided into strata depending upon whether the land was owned by large landowners, small landowners or a mixture (Iran). Sometimes the necessary data can be obtained by an exploratory enquiry of feasible size and cost. In one such study, where very little was recorded about the agriculture, the maps available were divided into segments of 6 kilometres side and one tenth of these were chosen at random. These were sent to their localities and the local agricultural experts were asked to give details of the type of cropping to be found within the areas concerned, together with various views on the size of farm, contact with market facilities, etc. etc. These items were used to stratify the chosen "six kilometre segments" for subsequent sampling procedures. This method illustrates two useful features to bear in mind in stratification work, namely, first, that it is possible to obtain supplementary data by special enquiry to use as the basis of stratification and, second—perhaps more important—fairly general qualitative information may be perfectly adequate as the basis of a stratification scheme which will achieve very useful improvements in precision.

Design

Some examples of sampling procedures which have been used in developing countries may illustrate techniques better than a general exposition. A very straightforward example is that of an enquiry in Turkey for which the frame consisted of cards, one for each village, which had been prepared for the 1960 census. Each card showed the total population, the geographical location, and the main economic activities. For technical reasons villages greater than 5,000 in population were dealt with by a separate procedure and the cards were removed. The remaining cards were then sorted into nine strata representing the agricultural regions. Within each of these nine strata the cards were sorted by economic activities. Within these, villages were arranged by province as an administrative convenience to get neighbouring groups as close together as possible in the records. Within each province these were distributed in order of population. Finally it had been estimated that the required sampling units would include about 39 persons, so the number of sampling units to be assigned to each village was obtained by dividing the population in 1960 by 39. The accumulated numbers of sampling units were listed for the whole of each province and the

required proportion selected systematically with a random start.

In a small farm survey in Malawi all villages in an area were assigned to one or other of 58 equal strata according to the size of village and ecological zones. One village was then chosen from each stratum and 21 families from each village for final interviews.

In a study of date production in U.A.R. the frame was built up by pre-enumeration enquiries which took the form of the preparation of sketch maps of trees in the orchard areas of the villages concerned. These individual plantations—which varied considerably in size from owner to owner—were combined to give units of similar size—actually of 250-350 bearing trees. This standardisation of size was essential to improve the efficiency of sampling. One-fifth of these units were then chosen at random for the final survey. In an enquiry in the same area on citrus production much the same procedure was followed except that the villages were divided into those above and those below a certain area of citrus production.

In the Sudan a frame consisting of sheikships was divided into three strata on the following bases. One consisted of those near Gedaref town of, in the main, big holdings with some degree of mechanisation and a high degree of durra production. This, in fact, included nearly 340 sheikships with 18,000 cultivators. From this, 18 sheikships were chosen and 6 cultivators per sheikship. The second stratum was made up of sheikships in the hilly area and the third of those which remained. A similar sampling procedure was followed in each. The working procedure consisted of preparing cumulative lists of cultivators from the sheikships and drawing a sample from this list.

In a Kenya enquiry, districts were stratified by agricultural divisions and within these divisions smaller administrative units were selected at random—perhaps the simplest of stratification procedure. Some parts, however, had records which permitted sampling systematically from within land consolidation registers. An example of an even simpler procedure was provided by a groundnut enquiry in Gezira, Sudan, which consisted of simple random sampling of cultivating tenants from the block records of the operating agencies. This was a case where the volume of production to be handled was heavy and niceties of design had to give way to pressure of time. Moreover, the ground layout was such as to make agricultural production fairly consistent over the whole area and the gain for stratification might therefore be expected to be slight. (One might contrast this kind of situation with its complete records and fairly standardised system of production with the very difficult problems set by non-static agriculture—e.g. the problem of estimating the crops of a semi-nomadic people who move to a good soil at the cultivation seasons, and then live in encampments which are largely hidden in high bush. It is very difficult to devise an estimating technique which will handle this kind of situation).

These outlines in the preceding few paragraphs are concerned with human based frames. A fairly typical type of enquiry using area frames is that which disposes a 2.25 acre sample unit, of square shape, per half square mile grid on maps. The location within each half square mile unit is determined by random coordinates. This area of slightly over 2 acres in a number of enquiries of this kind would embrace perhaps 10 to 15 separate plots of crops. It usually means that maps appreciably larger in scale than 1:10,000 are needed for satisfactory operation of this kind of enquiry, even if facilities are available for enlarging the maps.

The possibility has been explored of using 1:25,000 maps with the assistance of aerial photographs to divide up the ground into segments which are clearly identifiable by reference to roads, streams and other physical and generally immovable features. These are then coded on the map and a sample drawn from them by any suitable procedures. The chosen 1:25,000 sectors are then expanded photographically to twice this size and sent to the field staff concerned for sketching in the plots found upon them and identification of the crops being grown.

Cluster sampling might be included in this general group, too. In this procedure groups of points are located according to a formal sampling procedure and steps taken to identify the crop on the land where each point falls. The number of points falling on the several crops then indicates the proportion of the total area which is occupied by the various crops. There is some experience of using clusters within about 1.5 to 2 hectares for this purpose. The

advantages of points over grid is that only identification at these few selected points is needed, and recording of areas within selected sample quadrats is avoided. The purpose of grouping points in a moderately confined space rather than scattering them more or less at random across the whole of an area, is that the travelling costs are appreciably reduced, even though the precision, for a given number of points examined, may be somewhat lower.

It is, of course, entirely possible, as mentioned under the discussion of frames, that a different sampling procedure might be used for one part of the country from that used for another.

It is, in principle, desirable to provide for an 'interpenetrating' sampling procedure when drawing up the design of a large enquiry. A typical procedure might be as follows. Suppose that primary sample units are to be selected within each of a number of administrative districts. When drawn each unit will be assigned in respect of district A to group A1 or A2; for district B to groups B1, B2; district C, groups C1, C2; and so on. When organising field work, one team will be assigned to samples A1 and B1, another team B2 and C2, and another team C1 and A2. Thus, in each district there are two estimates of the situation from two independent observers. This provides a means of making some check on the accuracy of the individual observers. It also gives the opportunity to halve the size of the enquiry after the design stage has been reached without upsetting the balance of the enquiry. Such curtailment of scope is not desirable but is occasionally forced on the organiser by difficulty over staff or problems in field operations.

Arrangements for drawing samples

The physical arrangements for drawing samples vary with the situation almost infinitely. A simple case has been noted above where the records were available in card form which could be sorted physically into the respective strata and sub-strata. A somewhat similar case may arise where record books from censuses are available. Care must be taken in the planning stage to find out what is the actual physical location of records and arrangements made for decentralising the sampling at lower stages in the hierarchy. It is probably worth while, unless the cost would be prohibitive, to get the process of selecting sample villages and sample farmers completed in a round of visits in advance of those necessary for making the actual observations. This means sending field staff to the appropriate lower stage units where relative centralisation of records ceases. Local information can then be obtained to supply the frame for that particular stage of the enquiry and the instructions for sampling can be carried out and farmers identified. This allows instructions for dealing with the selection of reserve sample units to be carried out, uncomplicated by instructions relating to actual enumeration. Each of these may present sufficient problems for the field staff to occupy all their attention and the attention of the supervisors without tangling the two.

It is usually especially important at this stage to give attention to the non-resident farmers, to small farmers, to joint operators (for example, brothers in matrilineal societies) and to districts on the edges of towns.

V. STAFF

Introduction

In very broad terms the agencies with which we may be concerned in agricultural statistical work include:

a) government central statistics departments,
b) government agricultural departments, especially those sections of them concerned with agricultural economics, and with agricultural extension,
c) private research agencies, principally academic.

The work which has to be dealt with, again in broad terms, includes

a) the planning of enquiries,
b) the field management,
c) field supervision,
d) tabulation and editing work on current records,
e) the tabulation and summarisation of final records.[1]

The kinds of staff concerned in this work include

a) highly trained professional statisticians and statistician-economists, with interests spread across all kinds of statistical work, without any necessary specialisation in agriculture,
b) working statisticians and economists with experience of the general management of statistical enquiries,
c) extension staff of agricultural departments with special skills in agricultural fields and with particular local knowledge,
d) field supervisory staff of a kind trained within the department,
e) various categories of field enumerators and recorders.

Again in very broad terms, there are four ways in which organisations, particularly government departments, can obtain the staff they require. They may

a) establish a permanent body of staff which they will move from place to place and from project to project as required,
b) borrow staff from other departments for a limited period; generally this means that a section with statistical responsibilities borrows the service of extension staff,
c) appoint staff for a particular enquiry and discharge them when it is over,
d) train staff for a particular project, and perhaps in some other basic skills, use them intermittently but pay them a retaining fee to keep them in touch.

The majority of the discussion which follows assumes that, wherever possible, staff will be engaged on a permanent basis and will increase in value as they gather experience to the steady improvement of agricultural statistics.

However there are occasions where other expedients must be resorted to. The use of agricultural extension staff for statistical enquiries seems less common now than a decade ago. It is only to be expected that the extension department, with its own programme expected of it, will be reluctant to divert staff to time-consuming field studies. It generally seems easier to arrange for liaison and joint workings in the early days after setting up of an extension service and possibly it is in these conditions that circumstances are likely to press the statisticians into wanting to use them. Generally, it is probably best to think in terms of their helping with problems of stratification, local identification of agricultural zones and

1. Though the distinction may not be clear-cut in all cases, there is a great deal to be said for avoiding the term "analysis of results" for those steps which are directed to preparing the data for the user. It is preferable to confine the term "Analysis" to cases where the data are being used to show relationships or to test hypotheses.

boundaries, rather than with day to day supervision or recording extending over significant periods.

Where it is not possible to think in terms of permanent or semi-permanent staff for enquiries, useful work can be done by primary school teachers on a retained basis. Where the field work can be fitted into vacations, senior students in University Agricultural Departments have been used for statistical work of fairly high quality, senior students in agricultural schools for field supervisory work, and graduates of high schools for clerical work. These, however, are all expedients. Statistical work, at almost all levels, should be regarded as a career in itself or at least an important stage in progress to a career, and permanency should be the aim. Furthermore, the aim of an organisation setting out to develop sound statistical work should be to get men with good training on to the established staff, bringing into establishment men who have proved themselves in temporary work, and gradually building up a reliable, well-founded body of statistical expertise. It is as important to have this available at the lower level of the service as at the higher.

The amount of staff required for the various stages of study can only be determined by calculation of the work to be done in the light of the field conditions in which it has to be done. Some of this will emerge in the course of the details of techniques in later chapters but two points might be made of a general nature here.

First there is the very great importance of designing enquiries so that they are immune to variations in senior staff. Sometimes it is unavoidable that one staff member should be primarily responsible for organising and executing but, quite often, more risks of disruption if he is not available are incurred than are necessary. One safeguarding action is to make sure that a careful diary of the whole project is maintained with a complete record of instructions and actions undertaken. Another is to arrange, wherever possible, that the Director takes into his active confidence one or two other Senior staff members, if not those directly involved with the working of the enquiry then those which are directly involved in the department's responsibility for this kind of work. If the main Director leaves, it is then still possible to keep the momentum of an organisation going until new arrangements can be made. To deal with a section which has "run-down" because of loss of drive is almost worse than having to build up a section where none existed before. These risks occur at all levels but the farther up the scale of responsibility one goes the greater the risks and the greater the opportunities for safeguarding them to some degree.

The second point that is worth emphasising at this stage is the paramount need to provide for sufficient office staff for editorial scrutiny, current tabulation, and final tabulation. Attention tends to centre on the field work because the risks of hold-ups from local obstruction, transport problems, and so on force themselves upon the planners' attention. However, if there is not the capacity to handle the processing of any data as soon as it comes in, then it is impossible to keep up to date with the checking of errors. This has two grave consequences. One is that when tabulation is finally undertaken at some later stage, a large number of forms or questionnaires will have to be either arbitrarily edited or abandoned because of inaccuracies or uncertainties. Perhaps more important is the fact that if errors are not followed up straight away the valuable stimulus for field staff provided by the knowledge that someone cares about the material they send in is lost.

Scarcely less important than the need to keep abreast of current office work is the need to provide for adequate tabulation facilities for getting the material into publishable form in a reasonable time. If a great delay ensues between the conduct of the enquiry and the availability of the results, not only is useful guidance not available but the goodwill is foregone at least amongst the agencies and firms who might find the data useful.

The calculation of requirements for these various stages will depend in part on what facilities are available for handling tabulations, whether they have to be done on desk machines or whether computer time can be obtained. If there is any doubt at all about the provision of computer facilities it is highly desirable to budget in the original plans for sufficient office staff to carry out the required work. In any event, it is important to provide for sufficient to carry out the various editorial and initial office checking procedures. It might be found worthwhile to try to make ad hoc arrangements to get one or two skilled people to oversee the forms as they come in, even if their services are dispensed with for the

rest of the period of the job. For instance, somebody with field experience of an agricultural nature with some general statistical knowledge might offer advantages which would take a long time to achieve by formal training of staff with less intellectual experience. Possibly an agricultural officer or a good agricultural student of perhaps third or fourth year in the university might serve this purpose. In these plans one has to take into account the probability that the ordinary routine checking staff and those carrying out basic clerical functions will change fairly frequently and retraining may be necessary. In the original budget necessary equipment must be provided for and also actual room in which the work is to be carried out and records stored.

Mobile or home-based field staff

In principle, the staff who are concerned with the field recordings up to field supervisor level might operate either as centrally based teams which can be sent where needed within the whole country, or they may be based in their own home localities. Naturally, locally-based people are more liable, unless there is a substantial body of enquiries going on, to be used intermittently, but in many countries now there is so much work in hand that it is possible to think of staff members in permanent employment on statistical work living in their home areas.

Among the advantages of using people in their home bases are–

a) travelling costs should be reduced
b) there is no language problem and no problem in introducing the staff to the community,
c) they are well placed to know if people are omitted from local frames or fields from local farmers' records
d) if they have some skills they should be in a position to help people in the locality in the course of non-survey activities and thus increase the public goodwill.

On the other hand–

a) such arrangements are inflexible: well-trained staff may need to be used more fully than is feasible in their own locality
b) if the field worker has any feature attached to him which makes him disliked in the community then this will affect his statistical work
c) it is difficult to maintain discipline amongst reporters based in their own area since if a man is dismissed he is there in the path of the successor and, even if he is not actively obstructionist, the new man has difficulties to overcome.

The decision in a particular case must rest on a balance of advantages and disadvantages and the best approach may be to concentrate on guarding against the more acute sources of embarrassment. For example it may be possible to get the advantages of local knowledge without the disadvantages of local obstructionism by appointing someone from a locality at a little distance from the one in which he has to work. This could also have the advantage that he is not in the village at times when he is not required for actual observational work. It is at such times that sources of friction tend to be built up.

Selection

With rising academic standards in all countries, it is likely to be possible to have fairly objective means of assessing whether an applicant is educationally of sufficient standard for the work required However, if there is any doubt about this a test which includes:-

an interview (to get some idea of whether the man is in the least suitable for the post);
dictation (to see whether he can take down a simple passage and record reasonably effectively);
a short essay on some familiar subject (to indicate whether he can marshal his ideas sensibly);
an arthimetic test suited to the kind of information he is likely to have to handle, giving particular weight not only to the correctness of the answer but on the methodical way in which this problem is attacked and worked through.
memory tests (to show that he can make a reasonable attempt at remembering the kinds of things which are to be seen around a farm homestead, for example).

It should be remembered that suitable tests of these kinds are probably readily available from educational departments in the country concerned.

Once employed, staff may have to turn a hand to a range of types of work but it might be worth distinguishing, broadly, two groups, namely the repetitive enquiry in which a team moves from one village to another over quite a considerable area, and the arrangement in which the enumerator stays within a village for some considerable time, perhaps even for a full year. Only rarely can any kind of field statistical work be undertaken without contact with human beings, and reasonable tact and respect for the social courtesies of the communities concerned are essential characteristics. For the travelling enumerator good health and a certain persevering toughness, intellectual honesty and a capacity for hard work are needed if he is to make a good showing, month in and month out, under conditions which are often rather arduous. Resident enumeration staff and, to some extent, the supervisory staff of such enquiries, have a special need for that tact and innate decency which can allow them to stay on and be reasonably welcome in a community when operating a life which is quite different from that of the local people. They need to be as unobtrusive as is consistant with getting the necessary attention and, perhaps above all, they need to avoid taking advantage of their apparent leisure to disrupt the household and farming operations of the people with whom they have to deal. It would also be useful, since repeated enquiries of similar character are tedious for the informant if the enumerator could have the skill to encourage informants to talk about their social activities and their agricultural work and be able to summarise from this flow of natural conversation the items which are required for the purpose of the enquiry.

Occasionally, it is possible to use staff members of an established organisation (e.g. a marketing board or a settlement agency, etc.) for field work in collecting agricultural statistical material. It may be better for this purpose to try to use a member who is known to informants but is not locally resident—for example, somebody who operates on a regional or sub-regional basis and send him round to visit a number of places, even if it occupies an appreciable amount of time, rather than to use a much smaller proportion of the time of each of a lot of staff members lower down in the organisation's hierarchy. Whether this is feasible will probably depend on the interest of the organisation in the results of the work.

Training

Training may be required at a number of levels. A statistician may need training in the special skills of census work, a man with a good agricultural degree or economics degree may require training in the main skills of a statistician, staff may need training for supervisory duties in the field or in the office; local field staff may need basic training; etc. Any or all of these people may need follow up in-service training to take account of modifications in the practice either of an organisation as a whole, or of details of a particular enquiry which are being made because of initial experience. In all cases, in the conditions of developing countries, special care should be taken to search out opportunities for getting the best increase in the skills required for the minimum disruption of organisation and expenditure of funds. Thus a serviceable member of a statistics organisation, at fairly senior level, might get a very useful increase in his own self-esteem and competence to work in a developing organisation if he could have perhaps a one-month specialist course in some aspect of his work at a college or department somewhere within the geographical region where he is to work. This, with a little attention all round, could be achieved with little more than the cost of his maintenance and short-distance travel expenditure. At that particular stage in his working life it might be quite pointless to give him the opportunity of a one- or two-years course, with only partially suitable emphasis, in some country on the other side of the world (though later he might benefit from such a course). Similarly it may be a good deal more effective to bring in field staff to some local centre that they can reach for a day, to work them hard for that day and send them back again to their base, than to go through the escalating costs of arranging residential courses of some length. Of course, these costs may have to be met during the basic training for staff who have not previously had any experience in the work required, but even this may merit close scrutiny.

There seems to be at least some evidence that interviewers and enumerators who get a very short training make no more errors than those who get one of two or three times the length. If this is so, the explanation may be that in the longer course they get enough attention to get somewhat above themselves, or that they get information which is not really relevant to their job and merely confuses them, or that the trainers in the longer sessions spend time on material which is of no direct constructive benefit to the job in hand. There is also a financial point which affects particularly temporary staff. If a period of residential training is provided, even a week, with expenses paid, there is some temptation for staff to take it and drop out soon afterwards so that the costs of maintaining a trained staff with this system may be unnecessarily high. The more elaborate the training planned the more temptation there is to have fairly large numbers of trainees at the sessions and therefore to have them coming in from a considerable distance. This is an accumulating set of embarrassments.

For the purpose of this manual, training of staff above the supervisory level can be omitted and the following notes relate primarily to training of basic field staff and of clerical and supervisory staff. It must be remembered that the concern is with adults and, notwithstanding selection tests, there will be bright boys, average boys, dull boys and bright ones with blind spots, in any group who are brought together for the purpose. Consequently, it should be the aim to teach only what the students want to know for their own purposes and the instructor who is vividly appreciative of the actual field job to be done will get far better results than a man–or a succession of men–who have mainly academic acquaintance with particular facets of the whole training. The old saying that "you should tell them what you are going to tell them, tell them, and tell them what you have told them", should in most cases ensure that no more is included in any sessions than they have a reasonable chance of absorbing.

The length of training needed is difficult to appraise because different studies have managed quite reasonably well with very different periods for training their field staff. In part, the amount needed depends on the amount of supervisory time that can be deployed for day-to-day control of the actual enquiry. One might say that the basic skills of using the selection methods required in a stratified, randomised, sample enquiry, locating the necessary sample units on the map, preparing local frames, making contact with farmers, cross-checking the details of agricultural information provided by informants, measuring fields and weighing crops, etc., might be taught in, perhaps, a week to those who already have a good agricultural college type of knowledge of the industry. Others with much less specialised knowledge might require a month or more. However, enquiry into the tea industry in Uganda illustrates the difficulties of generalising. In this case the enumerators were chosen after an interview and an examination in arithmetic and general knowledge, and were sent to a Farm Institute for two weeks training. The first week included the way to enter information on records, how to measure plots, and use scales, the way to approach informants, and how to deal with their usual objections; the second week was spent in trial interviews in the morning and analysis of the results in the afternoons. In two weeks this basic training was completed and a number of the group were chosen to act as senior enumerators in charge of checking. The staff then immediately went on to a fairly complicated enquiry in the field.

It is essential to emphasise, however, that when the length of training is very sharply cut it is essential to provide for steady and patient supervision in the earlier stages of the enquiry. It is one of the losses in a very brief period of instruction, especially with minimal opportunity to follow the results of the students trying out the practice in the field, that one does not know in which particular direction they have failed to grasp the basic principles. If these gaps are discovered early, either by the field supervisors or by the editors of the forms when they come into the office, no great loss occurs, but if the period of training, the scope of supervision and the promptness of editorial work are clipped, then disaster is almost certain.

For those working in the field training should include at least the fundamentals of the following:-

a) basic arithmetic operations.

b) operations with fractions decimals, proportions, calculating percentages.
c) all units of measurement and their conversion. Practice should be combined with instruction so that they are thoroughly familiar with calculations of heights, distances, areas, volumes, cost per unit, and so on. This can be limited somewhat if their activities are to be on a very narrow front but anyone likely to be involved in general-purpose farm recordings will require all these items of experience in the course of a season.
d) some basic attention to report writing for all those who are going to have to give verbal accounts of what their informants tell them. Here perhaps the most essential features are to stress the need for headings, paragraphs and enumerated notes. It is arguable that the enumerator should not write down *in extenso* information he gathers, but should be required to put it into various places on a standardised form. Probably this will usually be true but there are still administrative reports to write which are the easier to handle if they have some reasonable form about them.
e) general ideas of the agricultural practices likely to be met with, the names of the crops and livestock and their systems, stages of growth, harvesting times, spacing distances, general levels of yield, practices in the processing and disposal of crops. Here again a short selection of these may be sufficient for staff engaged on particular narrow fields.
f) the use of maps for locating position and the use of large-scale maps for sketching the position of plots as a record of crop areas.
g) the use of prismatic compass, chain, wheel and pacing for the purpose of making accurate field maps. The use of weighing equipment, as provided, and the accurate designation of the state of products which are weighed.
g) the basic skills in approach and interviewing. The informant is always liable to look askance at people who come asking him questions, fearing to meet an official though perhaps hoping to meet a human being, and it is desirable that his hopes should be fulfilled more often than his fears. Interviewers should be encouraged to classify their informants into several main groups. There are belligerent ones who may adopt this attitude partly out of total irrelevant views on politics or economic policy, etc., or because they hesitate to admit certain features of their agricultural activities. However, an informant may be difficult simply because the situation of the interview means that there is an audience–a situation which is a discourtesy on the interviewer's part. Those who are cooperative may be the rambling type which have continually to be brought back to the main point, or have to be attended to patiently with the aim of picking up the relevant item as it turns up. Others may be highly forthcoming to the point of speaking the truth with some lack of seriousness and tactful cross-checking of such informants may be essential. The hesitant person may be just ignorant or he may be seeking a precision of statement which he finds hard to achieve.
h) the rules of a particular enquiry for which training is required should be laid down. Thus the enumerator should know to whom he is responsible at all times, whom he should consult if his instructions do not cover a particular situation, and what types of case should be turned over to his senior staff for handling or for special instruction.
i) the detailed arrangements about travelling, subsistence arrangements, claims, report forms, and so on, should be gone over so that there should be no misunderstanding on the employees' part.

In the course of this instruction it would be worthwhile to watch for two kinds of special skills emerging. One, the man who is by temperament suitable for a role of greater responsibility in local supervision, and the other, one who is capable of special skill but not necessarily in the supervisory direction. For instance, a man may be particularly skilful in surveying with compass and chain and it might be worth noting him as a possible specialist in this field who could with advantage do all this work for a number of field groups. This arrangement has been found preferable to keeping all staff on this kind of work in a number of enquiries. It is particularly true if a mixture of skills is required–interviewing, surveying, crop weighing, and so on in each group.

All the instruction given in the basic training course may need reinforcements by the supervisors when the teams are in the field, and some further training may be necessary at

this stage arising from changes in instructions which follow from initial experience with the enquiry. This often gets less attention than it deserves since it sets a number of problems to both supervisors and field staff. When the organiser of a survey sees that some change should be made in instruction, for example a change in definition, or a change in the way that an entry should be made on an established form, or even the introduction of a new form, there is usually some resistance from the field end. This may be just laziness on the part of those concerned but a substantial element arises from the difficulty they find in getting the idea of what is wanted over to the enumerators or to the farmers. This is particularly true when dealing with an enquiry that continues in the same villages with the same informants over an appreciable period.

It is important to take up with supervisors in a rather formal way a planned change of this kind, if it is at all possible to arrange, since this approach may bring to light other means of achieving the same end which are as satisfactory to the organisor and less disrupting to the enumerators. Even if the change has to be enforced, however, the fact that they have been consulted is likely to make for rather better cooperation on the part of the supervisors and the enumerators than would otherwise be the case.

One of the pieces of experience that it is important to obtain from editing completed forms and to store for future reference, is an assessment of which parts of an enquiry and what kinds of question the enumerators find most difficult to handle so that these items can be looked at specially in a future enquiry.

The training for supervisors should have a somewhat different emphasis from the training for enumerators. Clearly they should be fully conversant with all the procedures which the enumerator has to carry out and should be able to do them at least as efficiently as the enumerator. They should be able to identify departures from proper practice and to give the necessary instruction and advice for improvements. However, in addition to all this it is important to give the supervisor something rather more than he actually needs for carrying out his job and this is particularly true if a continuing and developing body of statistical staff is to be constructed. It should be made clear to him that each of the levels in the sharing of responsibility—the statistician, the supervisor, the enumerator—has an essential part to play and these parts should be clearly explained in such terms as are suited to the man being dealt with. His part in the whole should be made clear to him. He is not merely a postbox as might at first sight seem to be the case in some types of organisation; his role is a critical one in achieving high standards of statistical work. It is absolutely essential to stress the need for precision and economy of operation, and to indicate the general commonsense in the whole pattern of the statistical work. Where it is possible to give him the opportunity of a little extra training, with the status which this brings even if it is not absolutely essential for the work he is doing, such an opportunity should not be lost.

Staff management and control

There are two sides to this. On the one, the staff have rights and duties and, on the other, the organisations have rights and duties, and both have valid places. The situation with regard to salary and allowance, terms of appointment, etc., should be made quite clear to the staff on appointment, and it is highly desirable that the status and training of staff should be recognised quite clearly in the salary they get. Responsibility and extra training should be reflected in extra pay. It is also highly important that the contact between the head of a statistics organisation and the staff at various levels and in various geographical locations should be effective and not merely formally operative. Some countries have found it very well worthwhile to arrange meetings at regular intervals of all the regional supervisory staff in the field, and the cost of these meetings was explicitly provided for in the budget. They have been regarded as having a value in contact which no indirect communication can supply. Again, time should be taken by some representative of the central organisation to visit each locality regularly so that a contact may be maintained and queries thrashed out. The file for that area should be taken out with the visitor and letters received and carbons of letters despatched checked one against the other to make sure that no action has been omitted which should have been taken and no important letters have gone astray in other

directions. This has sensible working value and tends to avoid an impression that queries sent up the line get lost if they are inconvenient. Where corrections and revisions of statistical material have to be made at any stage then it is extremely important that the opportunity provided by meetings of this kind should be taken for consultation on this subject. Reconciling or amending statistics by correspondence is slow and unsatisfactory at best.

Given that communication in both directions is effective and that the pay and conditions of staff are defensible–it is too much to be hoped that everyone will be satisfied of course–then the central organisation must ensure that it has disciplinary control throughout the whole regional and local set-up. If a mistake has occurred it is essential that it is not passed over and, further, that it can be followed up to the point where it originated. This does not necessarily mean that every arithmetical error or minor variation in the application of a procedure would be the subject of a reference back down the line, but it does mean that an opportunity should be taken to correct any general misuse or misapplication of instructions. A clear distinction should be drawn in this between genuine errors and 'cooking' results. It is not unreasonable to dismiss instantly on 'cooking' being found, but leniency with genuine errors, at least until in-service training has been used to the full, is desirable.

The arrangement for payment of temporary enumerators and interviewers need some attention. It is a good policy, if it can be arranged, that an interviewer is not paid at the end of the week until his supervisor has checked his records. This has two advantages–one, that he will be encouraged to get his material through to the supervisor on time and the other that there is a strong incentive to be accurate enough to satisfy the supervisor. Faults of a grave nature can be met by stoppage of pay. More generally, if it is necessary for temporary enumerators to be paid off before their work is checked it is important to make sure that there is some lien on them to do repeats. This may be feasible enough in a case, for instance, of primary school teachers who are already in government employment or students who are attached to a college where some discipline can be applied.

The position of enumerators on long-term enquiries which keeps them in the locality for some weeks or months is a matter of particular delicacy. It must usually be necessary for him or his supervisor to convince the farmers with whom he is dealing, and the farmers' wives, that the enquiry is a sensible one and to convince neighbours who might otherwise chaff the farmer for working with the government and silence the barrack-room lawyers in the locality. By the nature of his work he is likely to be in rather frequent touch with the farmers' family and he could easily be at the home when the farmer is away at the farm. This is liable to produce misunderstanding, if nothing more. It is for the enumerator, under the supervisor, to ensure that he does not get too involved with the informants, that he does not get into the habit of talking to the farmer except in the express purpose of the enquiry, or of accepting entertainment at the farm home, or talking to young female members of the family. The social consequences of getting too involved are obvious enough but it is also found that enumerators who become much involved with their informants are more liable to guess what to put down than to press in an objective way for the exact quantitative detail required.

A problem of particular difficulty faces the supervisor when his enumerator reports a refusal by a farmer to cooperate in spite of all available explanations and tactful pressures. In quite a lot of instances it is not difficult for a supervisor, or a more senior member of the statistics organisation, to go along to the farmer concerned, talk to him as man to man, making much of the farmer's importance, allowing him kudos from the attention of a senior visitor. The exact details of such approaches vary and they may very well be successful in getting sufficient cooperation for the purpose of the point immediately in hand. There is a grave temptation to deal with the problems of this kind in this sort of way, it solves a problem and it tends to give the senior staff member concerned a sense of satisfaction in handling other people. However, it puts the enumerator who failed in a very difficult position. It is frustrating to him personally because he was not able to use the same approach and it spoils his status in the locality by being overriden in this way. Moreover, it is liable to produce some friction between the people who are prepared to play with the enumerator

and the man who stuck out and got increased attention. Explicit care should be taken to maintain the position of the enumerator who has got to live with the situation after his senior has departed.

It is too much to hope that the senior staff in the control organisation will be able to find time to travel extensively, but before closing this section the very great benefit of personal contact should be underlined. It has been found in numerous instances in developing countries and it applies equally effectively in developed countries. Visits from a statistician to a neighbourhood more than once notably increased the proportion of a sample questionnaire which was returned. Visits to voluntary crop reporters has increased the care with which they complete their routine questionnaires. If a chance presents itself for contact of this kind it should certainly be taken and if it does not present itself at least every twelve months, or so, an occasion should be explicitly created.

VI. FIELD ORGANISATION

Introduction

When the shape of the enquiry begins to emerge a skeleton timetable should be prepared, running from the initial planning and outstanding work on the frame through selection of field staff, their training, allotment to bases, publicity for the enquiry and the various stages of its actual conduct. Obviously the length and formality of such a programme will vary with the job to be done and some parts–training, for example–may be non-existent if it represents a further stage in an already established programme.

Public relations

Keeping the importance of statistical work before senior members of government, other departments, statutory organisations, and private firms has already been stressed. In general, care should be taken to avoid springing the news of any major development in statistical work on those who might be directly or indirectly involved. There is much to be said for the "drip-drip" method whereby the idea is introduced in an exploratory manner well before the plans are in any way finalised and a note taken of the response that the idea evokes. This may give very useful guidance on certain aspects of planning and may show the direction in which further information should lay its stress. It is important that all levels of a statistical department should foster good relations with other departments and agencies. These other contacts may include agricultural, labour, and commerce and industry departments, and general administration together with statutory organisations for marketing agricultural products, and firms trading in farm products and in requisites. When the plans have crystallised to a reasonable degree information about what is planned should be circulated to those concerned. It may be that the proposals will have to be pushed ahead against the opposition or indifference of some of those concerned. This will generally not be any the more powerful for opponents having known of the plans at a fairly early stage. On the other hand, most people will react violently at having been presented with a *fait accompli.*

It is possible in some environments to present statistical work as having an independent or neutral image, with some gain in goodwill. This is particularly true of studies carried out by non-governmental agencies, such as universities and philanthropic organisations. In many developing countries it is practically impossible to convey this picture of independence and no great weight should be attached to achieving it.

Before field work actually begins key officials should be provided with a project outline. They may not in fact read beyond the opening paragraphs–so it is important that the main pattern and aims should be clearly stated right at the beginning–but it is a courtesy and it prevents their being placed in the embarrassing position of having it first mentioned to them more or less publicly, and having to cover up the fact that they knew nothing about it.

Those directly involved in the operations of an enquiry should receive a letter at a suitable time before the work starts setting out the aim, making clear what value can be looked for from the information so gathered, and stressing the fact that the personal affairs of the individual providing the information will not be divulged to anyone, including other departments of government. Typically such letters should go to civic leaders in the neighbourhoods concerned, chiefs, church leaders, etc. In a number of population census enquiries the local churches have been asked to read prepared statements from their pulpits in the weeks before the enquiry. The statements have been graduated to bring interest to a head immediately before the study has begun. There may be occasions where village

meetings can be called at which such letters can be read. Though the method does not seem to have been used recently there are instances of useful information being actually collected at village meetings. They have the advantage that everything seems to be above board and therefore acceptable to the community and the information can be cross-checked by the comments of neighbours. Obviously one needs certain social conditions and rapport between the statistical staff and the local people before this kind of data gathering meeting is possible, but opportunities for information meetings are much wider.

Considerable care should be given to making contact with those from whom information is sought and in convincing them that there will be no disadvantage to them in becoming involved in the enquiry and indeed they will be contributing to the development of the country. Any honest argument which has local force should be used to introduce the study. National planning may be accepted and may form a sufficient argument; there may be problems about getting crops moved at harvest and it may make sense to make forecasts of the crop in order that better preparation can be made for transport; or where, for instance, a programme of subsidising a crop improvement measure such as crop spraying is in force, it can be argued that crop yield estimation must be carried out in order to form a basis for organising such assistance.

The people to be convinced may extend beyond the actual farm operators. Often enough the farmer's wife is the person who answers many of the questions and she must be convinced of the sense of cooperating, too. In any enquiry that is to last more than a single interview it is important also to make sure that neighbours and others will not undermine the preparedness of the informant to continue with the study. It is worth taking a good deal of trouble to start off these farmer contacts on the right foot. In a large scale, but relatively straightforward, enquiry such as a crop area estimating survey, the enumerators must make all contacts. It is essential that their training should have stressed the importance of their contact with the public and that they should do it with patience and courtesy. Where studies are to continue for longer perhaps, involving several "passages" through the country at different periods in the crop year, or even continuous residence of the enumerator in the locality, much more is at stake. On the other hand this usually means that there is a little more leisure in organising the enquiry and the supervisory staff are more ready to hand. There is a lot to be said for each association between enumerator and informant being started rather formally by a senior member of the survey staff.

Not only should the occasion be taken to make a good general impression but the authority for making the enquiry should be clearly set out. In the more complicated enquiries it will almost certainly be voluntary and this, and the confidential nature of the information, must be stressed. Again, no promise of any kind should be made about the farmer getting any return for this cooperation and this should be underlined. Generally, the most that can be offered is a certificate that the farmer has cooperated.

Some of the problems that come from resident enumerators with a different working day from farmers have been indicated in the training section above. It is extremely important for supervisors to keep their ear to the ground over these matters and to step in before any major friction can build up.

The last few paragraphs have been concerned with the relationship between the statistical field staff and the general public. It is equally important to maintain good relations with the field staff of other government departments. One element may be merely a matter of maintaining good relations which will allow mutual help and cooperation in the future. A somewhat different feature may be the desirability of keeping the nature of cooperation between two departments on a comparable basis in different localities. Where, for example, a statistics department and the extension staff of an agricultural department are jointly working on an enquiry the compromise under which the work is split may be left to local initiative. This could mean quite considerable variation in procedures and a good deal of variation in the soundness of the data from district to district. It may require some tact on the part of the organising officers to keep this problem within bounds.

It should be borne in mind that the work of a statistician tends to look to the public to be a matter of "all take and no give". It is bad policy to leave a neighbourhood where a good

deal of the informants' time has been occupied without formally conveying the thanks of the organisation to them. Whether this is best done by a letter to the local chief or a visit from the local supervisor is for decision in the light of local conditions, but one should not just disappear without a "thank-you". There may be scope, too, for feeding back some of the results of the enquiry in carefully edited form. In many communities there will be people who can appreciate the relevance of some of the results. Even if they cannot, the recipient is likely to feel more sympathetic to the enquirer if he is evidently remembered. The local staff of a cooperating department might also value summaries of the result, perhaps comparing their neighbourhood with an adjacent one. If this is going to be done it really means choosing for use certain figures that can be obtained quickly and easily in the course of tabulation. If one waits till the complete report is published and extracts made, the local interest will almost certainly have passed.

Where no statutory authority for the enquiry exists cooperation depends wholly on good public relations and this may encourage courtesy by the organising departments. If statutory authority exists this may be sufficient to obtain cooperation for this particular enquiry but it should not be forgotten that other work may be required later for which no specific statutory authority can be produced and the public regard for a department and its work depends somewhat on the sum total of the public's relations with it.

Definitions and language

In any study a number of definitions will have to be laid down. Some of these may be of international relevance and the 1970 World Census of Agriculture could usefully be consulted as a guide. For example, formal definitions will be needed for dwelling unit (usually a private building in which people sleep); household (a group of people who eat together); holding (all land under the economic control of one farmer); farmer or farm operator (a person or persons chiefly concerned with operation of a holding–but there may be a question regarding who, in fact, makes the decisions): parcel of land (land forming part of a holding surrounded on all sides by land of other farmers, public land, or communal land); field (a continuous piece of land within a parcel covered by one category of land use, such as one crop or one crop mixture); etc; etc. A minimum size may have to be established for some of these units, for example 25 square yards has been used in some instances for a field in contrast to ¼ or 1 acre in other types of agriculture.

Units must be decided upon and quite clearly provided for in the forms issued to enumerators. Where items of measuring equipment calibrated in different units (for instance measuring wheels recording in feet and decimals of feet or feet and inches) are in use, care must be taken that the right forms go with the right instruments. Certain local units may have to be used either because this is the natural form in which the material appears (a bunch of bananas, head of millet) or because some customarily used unit of measure will arise naturally in information obtained from farmers. It is in the highest degree important that these should be avoided in all recording work where this is possible or the conversion from the local unit to those for incorporation in the final tabulation should be done *under strictly standardised conditions*, on the basis of test weighings or measurements. For example, an Ethiopian team found that a particular area measure, for which the same term was used over a considerable geographical region, in fact ranged in size from 33 to 76 hectares. When the team asked presumably knowledgeable people about the hectarage of this unit, the standard reply was "40 hectares". Perhaps the classical example of this problem is provided by cigarette tins as measures of volume in West African markets. Here, though, the volume of tins has varied both with the brand and with the degree of fill, a combined adjustment permitting rather subtle variations in pricing of produce.

A list should be provided to enumerators of all the local names for the crops which it is intended to include in the enquiry. In some livestock enquiries a similar glossary may be needed to take account of stock of different ages. A considerable range of names is often to be found for the same products but care should be exercised to make sure that, if there are varietal differences which are significant in respect of the utilisation of the crop, these are provided for separately in the list.

When it comes to tabulating the figures there should be no doubt regarding the state in which the crop, or other products, was weighed. For example, paddy may be the natural product in one stage in the process from field to mouth, and rice another. Provision should be made for the entry to be in terms of the product as it stands at the appropriate recording stage. Conversion can then be made as appropriate in the office. Other possible occasions for confusion are shelled and unshelled maize, shelled and unshelled groundnuts, fresh or partly dried root crops, (such as yams), etc.

The peculiarities of a local language may not raise any problems in large scale enquiries of a fairly simple nature where the enumerator can obtain most of his information by observation. However, as enquiries increase in scope some risks of misunderstanding on purely language grounds may arise. It might well be worthwhile if a questionnaire is written by a statistician in a local language to have it translated by someone else, preferably someone who is not a statistician, back into, say English. Similarly if it is drafted initially in English then translated into a local language, it might be valuable to get it translated back by a non-statistician.

It should be remembered that in many societies there may be only very general terms for aspects of time. An attempt to translate "minutes" or "hours" into a local language is likely to produce a meaning similar in precision to that in the expression "wait a minute!". "Time states" may be difficult to handle, too. "Being married" may in some languages imply the ceremony and not the state of being married It may be necessary to refer to "two people who were married last year" if the *state* of marriage is to be indicated. The idea of a wife, in any event, may not be entirely clear. For instance in some societies there may be differences in a wife's rights over land depending on whether a formal marriage has taken place (e.g. a bride price has been paid) or whether the marriage is temporary. But full status may be claimed even if not justified–for example if an adult woman is not married, she may be liable to tax and may fear that the enquiry will bring this to light. When one goes into less factual enquiries the problems are considerably greater. For example, in an enquiry which included questions about a family's hopes for their sons who were at school, quite a number included in their answers some reference to his being killed in a road accident or other unhappy feature. This became explicable when it was appreciated that in that particular language 'hope' and 'expect' were covered by the same word.

Documentation

A large number of documents providing information about an enquiry, providing the place in which information is recorded or providing summary sheets to which it may be transferred, may be needed in the course of an enquiry. Various report forms are needed too. Receipts and other records in connection with financial transactions, refunds on expenditure for petrol and repairs, etc., must be provided for. An indicative check list for this is provided in the Appendices. These cover the majority of the main headings under which forms will arise. One or two particular features may be worth noting separately.

Provision should be made for transmission of documents in such a way that losses are minimised and the stage of processing of each one can clearly be assessed. That is, when they move from staff member to staff member along a chain of responsibility there should be a receipt and passing-on indication somewhere in a record which does not move with the documents themselves. Losses have made regrettable gaps in a number of enquiries and if this occurs near the end of a crop season it may be impossible to re-do the work even if staff time were otherwise available.

Arrangements should be made to move the first few schedules completed by an enumerator (say, half a dozen) through to the supervisor and thence to headquarters rather quickly so that any major oddity in the carrying out of instructions can be noted and corrected straight away. At this stage travelling by headquarters staff is likely to be valuable, if it does not completely denude the headquarters of essential people to whom ultimate reference can be made. Subsequent batches can be despatched in rather larger quantity but not too many should be placed at risk in any one consignment.

District progress sheets should be displayed in district statistical offices and kept regularly

up-to-date. They should form the basis on which the weekly or fortnightly supervisors' reports to headquarters are made. It is important that records displayed in this way should be at the point where action indicated by them can be taken It is pointless to have large displays of wall charts at headquarters relating to up-country activities unless this is the point at which some action has to be initiated on each of them.

The problem of getting revised instructions to field staff, when this is occasioned by experience in the course of an enquiry is a difficult one. As has been mentioned under training, if at all possible it would be best to arrange for senior supervisory staff to take such instructions to the people who have to implement them, explaining the background and making clear the need. A stream of revisions of this kind coming out of headquarters by post is calculated to create a wholly perfunctory—if not mutinous—attitude on the part of field staff. It is not unreasonable if some of these instructions get 'lost' at a rather late stage in their distribution!

The provision of extra copies of forms to field staff needs explicit attention. They should be instructed to indent for them at least two weeks ahead of requirements (more if postal time is long); it might in fact be useful to include in a weekly report a statement of the number of unused forms still in stock. As this number runs down it should provide supervisors and, through them, headquarters with early warning of the need to prepare new supplies. Such preparations may require a new plate or stencil to be made if the old one has been worked out or if modifications are necessary.

Equipment

This can be classified in a number of groups according to its size, or the place it occupies in the work done. The group includes computers; desk calculating machines; typewriters and reproduction equipment; smaller items of equipment for central office use including planimeters, rulers, protractors, pencils, pegboards, stationery and all associated filing facilities; basic stationery required by field staff including pencils, rubbers, rulers, etc.; compasses, chains, measuring wheels, and weighing equipment for crop cutting and the necessary receptacles for carrying it; transport whether land rovers, motorcycles, or bicycles. A check list of a number of these items is given in Appendix G. Each might be the subject of some explanation but the relevance varies somewhat with the locality, and discussion here will be confined to about half a dozen points of emphasis.

In big surveys a detailed listing of the equipment required will probably have been made in the preparation of the application for the necessary vote of funds. This may not have gone into specific detail for the smaller items, however, and where a body of material has already been built up and enquiries are being paid for out of a regular vote, early routine listing of equipment requirements may not occur. It is important to provide at an early stage for a review of what is likely to be required and what is likely to be available, and get an indication of the delivery dates of anything which has to be purchased. This may be something in the nature of a reconnaissance on the ground which might point to certain critical areas and the need for a scrutiny of the importance of certain items which it might be difficult to get. A list should be maintained of the work that is in progress *and* planned so that the total demand for particular pieces of equipment at particular times is not excessive. This may mean making enquiries outside the statistical department if it is planned to use equipment from other sections who may have their own plans for it. These reviews are intended to provide "early warning" of likely difficulties.

A detailed listing should be made early on of what equipment is going to be required, how much, and when. In order to avoid omissions it may be well worthwhile to approach this from several different directions. For instance, the various field records to be made could be listed and the relevant equipment noted, tabulations to be made and documents to be produced at the end with an indication of their size (it may be that the forms required in the summarising stages cannot be produced on the equipment in the department concerned but will have to go out to other departments or commercial printers which can handle larger sheets). Again, a list should be made of the staff and their duties and the equipment which will be needed for this purpose. The position of the staff in each stage of

the enquiry should be listed too, and this may throw up matters of tents or bedding required, equipment for meals, and so on. All draft instruction sheets should also be scrutinised and the equipment required indicated. In all these processes equipment includes not only machinery of a capital demanding kind but also expendable items such as paper clips, staples and stationery.

Particularly where special purchases may be needed the resulting requirements list should be scrutinised with great care to see whether the same results could be attained more cheaply or readily (the modern term for this is 'cost effectiveness'). Even if equipment is actually available it is pointless to use it if there is a serviceable substitute procedure which will make the whole arrangement somewhat less vulnerable to competition for equipment or cash. For instance, if planimeters are envisaged for measuring the area of fields from sketch maps in the office but they are difficult to obtain, is it possible to draw out the sketches on squared paper and count the squares instead of using planimeters? Will the contribution to error be of any significance in the total error? The desired equipment could at this stage be divided into categories, e.g. what the statistics agency owns or will control absolutely, what must be drawn from another agency, either commercial or government. For this latter group the maximum limits for the period when it is wanted should be calculated and the agency concerned consulted to see whether this is going to be a feasible arrangement throughout the possible period of demand. Ideally a positive promise is required from the supplying agency but some discretion may have to be used if flexibility in the period of demand is to be allowed for. An attempt to get a cast iron promise may merely result in a flat refusal.

In respect of the department's own equipment a very careful scrutiny of its service record should be made. If equipment has not been overhauled recently then this should be done in advance of a heavy demand on it. For items like transport equipment it is important to check the past maintenance records to show what proportion of equipment is normally serviceable at any particular time. It is not necessarily impossible to work with a high proportion of the equipment out of action at any one time, provided this is allowed for in the original plans. Account should be taken of the seasonal conditions in all planning of this kind. This may seem too obvious to mention but the triumph of hope over experience has led to many enquiries having to be abandoned with inadequate samples.

Resources needed

Variations in the details of the design of the enquiry can lead to variations in the cost of carrying it out and in the precision of the results finally obtained, as is noted above under sampling. For a given sampling fraction and a given proportion of the agriculture of the country which is actually recorded costs are likely to be reduced if the units are bunched in localities, thus reducing the amount of travelling time. Obviously the detailed outcome depends on the organisation of the field staff and many other details and can only be worked out with any precision if a pilot enquiry to get the actual quantities concerned has been made. Consequently part of the preparatory work in any study should include records of the time taken to undertake certain work and to transfer staff from place to place, including injury time and time out for sickness and disruption due to local holidays, and the like.

It would be useful if one could give some sort of indication of the likely manpower needs of particular kinds of study. Not much can be done in this connection because enquiries differ so much in their content and in the conditions in which they are undertaken, but some examples can be quoted. Two survey teams of two or three people carried out 470 interviews, each lasting on average twenty minutes, into farm conditions in 150 villages in 38 days. The actual travelling time between villages was that involved in an enquiry of 20 miles or so diameter. Broadly speaking one supervisor is needed for each five enumerators or each five people in the office. In another enquiry, involving a large acreage survey, 51 teams took a little over 7,000 man-days to complete it. Of this about 600 man-days was occupied in travelling from one sample location to the next, about 1100 in interviewing chiefs, choosing samples, etc., and something over 5,000 in finding informants, identifying plots

and measuring the plots. Generally speaking, a two or three man team could measure 12 plots a day, but this could be increased to 15 under pressure or under favourable conditions.

Statistics of development schemes

The above discussion is written with largely unorganised development conditions in mind. However, there are many examples to be found of rather closely controlled developments. It is extremely important when any such schemes are planned that specific provision should be made for the collection of the statistics necessary to describe their progress, physical and financial. Evaluation should then be possible both from the overall standpoint and that of the farms and families which are concerned.

VII. ERRORS AND ERROR CONTROL

Introduction

However excellent may be the design of an enquiry the reliability of the final results depends on the accuracy of the recording and the processing of the material. No amount of statistical manipulation will correct for errors in original recording, particularly systematic errors. Errors tend to be cumulative in the sense that if some occur and are tolerated this encourages others. Such toleration may be the result either of basic slackness in the control of an enquiry or it may be the inevitable consequence of the organisation being over-stretched. It is, therefore, highly important not to start on a bigger piece of work than there is men, money and equipment to handle. This is, to some extent, a council of perfection. Nevertheless statistical work of this kind is probably not a case where it is 'better to have tried and failed than never to have tried at all'. If an enquiry is clearly proving too big for the resources available, then it should be trimmed to a scale which can be handled before a great many poor quality records have been made. For instance, if an inter-penetrating sample design has been used it can be cut to half in scope while maintaining the same rigor of sample design–such sacrifice may be needed only in certain areas. The overstretching of resources may come not only in quantitative terms but by attempting a type of enquiry that is beyond the capacity of the calibre of staff available. Generally speaking, multi-subject enquiries need much better staff than such studies as the estimation of the acreage of a few main crops.

Good staff and proper training are probably the main contributors to sound statistical material. Interested, alert and conscientious field supervisors are probably the kingpin of the whole organisation.

It is important to look at the totality of errors in an enquiry and to get some idea of which ones contribute heavily to the uncertainty attached to the final results. It is pointless to spend a lot of effort in reducing the error in a particular element of the process which contributes no more than plus or minus one per cent to the error of the final result, when there are other elements which contribute ten per cent. It is not merely that the effort might be better spent in reducing the ten per cent, but that the outlook which allows such a concentration is inimical to a sound commonsense approach to practical agricultural statistics work. Probably the most serviceable approach is first to make all efforts to avoid error, then to try and assess what remains, and finally to explore the consequences.

It would be useful if the makeup of errors in different surveys were more often available. Unfortunately, the analyses seem rarely to be published. Obviously an informant may give wrong information, or the enumerator may make a wrong measurement, but there are also errors which come in as a rather incidental consequence of the design of the enquiry. Identification of holdings account for many of the problems, for example, the main core may leave no difficulty but social practice in the community may mean that there is a substantial fringe for which it is not clear whether they should be regarded as farmers, or not. Again, where there are several return visits by enumerators in the course of an enquiry, the farmers may have changed their customary name in the interval and be difficult to locate and match with the records of previous passages. This may not be too much of a problem if the same enumerators come back each time but changes may have to be made through illness or responsibility for other enquiries, and new men may come back on each occasion. Again, a proportion of the land which ought, strictly speaking, to be included in the enquiry, may not be found in a first attempt. An experienced Central African worker reckoned that in his enquiries 90 per cent of the area in the selected sample farms was, in fact, found on the first

visit. Subsequent visits increased this somewhat, but highly intensified observations might be needed to bring this up, to, say, 98 per cent. Some interesting pointers have come from the comparison of main survey with post-enumeration test surveys in various countries. In a Middle East study the census showed 47 percent more fields per household than the post enumeration survey, but the latter showed something like 45 percent more area operated per household. It is difficult to reach any firm conclusion from this except that the different conditions of the two enquiries produced markedly different results. In some Japanese studies where the information was obtained by enumerators from the farmers, and subsequently checked by actual observation and measurement, it was found that irrigated rice was very accurately recorded but upland crops might be under-estimated by upwards of 10 or 15 percent. It was found, too, that crops, such as fruit or rape, which were mainly cash crops were more usually under-estimated than those, such as rice, which were to a substantial extent used for home consumption. There are some indications in other enquiries that irrigated crops were accurately recorded but that as the enquiry moved out into the higher land, particularly into mountain areas, field observation in a routine survey gave under-estimates largely because of the omission of fields attached to sample farmers, fields which were revealed when a closely supervised post-enumeration test survey was made. Among the most common errors, again in some Japanese studies, was the transfer of acreage figures wrongly from a cadastral record to the working tabulation, next came faulty identification and classification of land. The most important source was often the complete omission from the classification of units which strictly should have been there. No two surveys are quite alike in the problems presented but the risk of errors in transfer of figures is underlined by these results.

Error control by improved recording

The quality of recording turns on the soundness of enumerator training and the rigorousness with which the field supervisors make sure that field staff actually do what they are supposed to do, watching them carrying out the procedure, and testing from their records that they have done it in other cases. Typical examples of danger points are 'give and take' lines to designate the boundaries of irregular fields, and the actual seeking out and counting of livestock and equipment which is to be entered on returns. In overcoming problems of continuity from one passage to another the use of maps, even rough sketch maps, or sketched-in details on rather crude enlargements of small scale maps, or tracings from air survey photos, may be a considerable help. Some enquiries by their nature put a lot of responsibility on the enumerator and/or informant where details must be remembered over days or weeks. Some reference to this is made in the section on farm recording below. Generally, in sample enquiries the process of substituting another farmer for one who cannot be found, or who will not cooperate, receives rather less emphasis than it should. The mere refusal of a farmer to cooperate is not enough to initiate a substitution procedure, neither is finding a deserted farm, nor the absence of a farmer. In the latter case information can probably be obtained from someone else. Substitution should not be allowed at the discretion of the enumerator but specific permission obtained from at least one stage up the line of responsibility.

Control by check procedure

The supervisors of field staff hold a key position here. They should inspect all forms of questionnaires completed by field enumerators and should look for internal inconsistencies. A sample should also be checked by regional supervisors before the whole is sent on to the central office. Typical indications of trouble, and probable need to refer back to enumerators, are the case where areas for individual crops and for the farm as a whole are measured and entered but the total for individual crops is greater than the indicated holding. Supervisors should suspect trouble if there are many zeros or blank items in forms. They should also be able to check the transfer of figures from one point to another in a form, e.g. for further computation. These are items which could, in the main, be picked up at

headquarters level but this delays the return of the forms to enumerators for checking to a time when they may have moved to new areas. There may be various items which cannot be detected by any internal checks but which supervisors may query out of their own local knowledge. There may be a special advantage in scrutinising the data for any product which is the subject of control, e.g. where planted areas are licensed. Examples have been found where areas are entered which are appreciably smaller than the true area. Licensing may not have been enforced in a locality for some time because evasion had become too frequent to make control possible. In these circumstances, no enumerator wishes to highlight the fact of this evasion by entering a figure above the one that is nominally the correct one, whether or not he personally has any responsibility for enforcement. These opportunities for correction open to supervisors underline the need to get a few completed forms into their hands quickly so that consultations on prevalent errors can be got under way. There are often opportunities for useful experience and ideas to be passed on from district to district at this stage if senior staff can find time to tour.

Procedures for aiding control

As has been noted already, some indication of operator error can be obtained from inter-penetrating samples provided that the error is not a systematic one common to all. A call-back system can be instituted in some enquiries by which supervisors can get back to the farmers who have already provided information and, making some reasonable excuse (about lost schedules, for example) re-do part of the recording. It would probably be sufficient if it is taken to the extent of elucidating the number of fields which the farmer operates since it is more than likely that the field measurement will be accurately done, but that the enumerator's errors will arise from incomplete listing of fields. Some countries have arranged that these call-backs are made by staff who are engaged on more intensive surveys in the locality, consequently this allows rather more highly skilled staff to be used for the check procedures. It might be said in passing that the reconciliation of figures obtained by the two sets of observations may require a certain amount of tact. There may be cases where the enumerator has good reason for what he has put down and no good purpose is served by criticising him before hearing his reasons.

VIII. ESTIMATION OF CROP AREA

Introduction

This chapter is concerned with estimating areas of annual crops. Estimates concerned with general land use, including potentially cropable land and land unsuitable for agriculture, raise rather different problems and are dealt with in a separate chapter. So is the estimation of tree crop areas. The individual crop areas and the aggregated value of these, representing the total area of the farm, are amongst the most basic of agricultural statistics; they are usually the first to be collected when statistical work is begun in a country. In some ways one can regard a survey or census to collect this information as being a typical general-purpose enquiry.

There appear to be no short cuts to the estimation of field crop acreages. At first sight it might be hoped that aerial survey would yield such information without the cost of a large field staff. In practice, so far, aerial surveys are usually made only at long intervals and for purposes which rank rather higher in the priority list for funds than do agricultural statistics alone. Typically, they are undertaken as a basis for mapping in a country where maps have hitherto been inadequate or for the preparation of land utilisation maps and water control programmes. In order to obtain annual acreage estimates from aerial survey it would be necessary to repeat a photographic cover, at least in sample form, each year at a time suited to the particular crops concerned and to use a scale for the photographs substantially larger—perhaps treble—that normally used for mapping and general land-use purposes. There seem to be few, if any, cases in developing countries where this has been possible. Occasionally in certain developed countries either a very pressing need for acreage data under control programmes of one kind or another, or unusually ready access to air survey facilities, have allowed the technique to be used fairly freely. Spot checking using light planes and an incomplete coverage is, however, quite common in many situations in both developed and developing countries.

It might be noted that even if the air survey technique could be made economically feasible it is not necessarily as attractive as it might seem at first sight since it cannot be combined with any enquiry relating to the production or disposal of the crops concerned, or even their yield. For this purpose contact has to be made with the individual plot, and usually the individual farmer and, if the field organisation for this has to be maintained, it may well add little to the cost to extend its coverage to include a basic area estimation.

Occasionally, the existence of records of registered areas or licensed areas may offer a promise of a short cut. Where a crop is subject to some kind of control so that producers are required to be licensed, two possible sources of discrepancy should be watched for. One is the perfectly proper exclusion of certain categories of the crop from the registration. Sometimes a variety is excluded from the registration but would be included with the main crop in any ordinary statistical estimation procedure; sometimes small plots for home use are excluded; sometimes the production in certain localities is exempted from licensing. Usually this kind of discrepancy can be judged from the regulations governing the registration through the proportionate extent of the omission may be unknown. The second possible source of discrepancy is the evasion of the regulations to a greater or less extent. In principle, it should be possible to derive a correction factor to convert registered acreage to true acreage by undertaking a survey of the two in a selected sample of localities. This should not, however, be undertaken lightly. It may be that such a factor can be derived from the figures of a particular crop when they are gathered together with figures of various other crops in a *multi-crop* survey. The ratio obtained in one year may then be used in subsequent

years if the matter was of interest and repeat enquiries of the multi-crop scope were not possible. However, if an attempt were made to make a special survey of the actual acreage under the registered crop alone, it would be a matter of embarrassment to local enforcement administrative officials and it would be difficult to forecast what the statistical and other outcomes of such a situation would be.

Trouble has also been noted over using the land registry records for the area of individual fields or farms as an indication of the true area. Here, too, the basic reasons for the discrepancy may be part and parcel of the originating legislation but they may not be very actively in the consciousness of those concerned at any given time. Reference has been noted in the literature to such records which under-estimate true area by up to 15 per cent for irrigated rice and up to about half that margin for upland crops.

The definition of area for the present purpose is by no means straightforward. For instance, should it include the paths, boundaries and channels which are intimately associated with the portion of land on which the crop is grown? If only the portion on which it is grown is to be included, is this to be understood as including the whole of the land prepared for its reception, or only that part which is actually planted to the crop, i.e. excluding the unplanted area round the edges? Is it to relate to the area planted? Or is it to relate to the area harvested? Is it to relate to the superficial area under any of these definitions of area, or to the area projected at sea level (the latter would be that observed on a map of the land concerned)? Quite clear principles could be laid down in respect of all these points of uncertainty; the question is what is feasible under practical conditions.

As to the inclusion of boundary strips, paths, and so on, it is usually difficult to be sure what definition has been applied in practice whatever may be the instructions to field staff. There is some indication that under fairly intensive irrigated conditions the minor drains and channels might amount to from one half of one per cent up to perhaps three per cent of the total area, perhaps one and a half percent on average (this is an example from Pakistan). Some Japanese figures suggest that the margin might be seven percent for paddy and three percent for upland crops. Obviously if the estimation is under close control it may be possible to pin down these definitions but the field operations may not be entirely straightforward. The matter of including the whole of the land prepared rather than the land actually cropped may not raise grave matters where the agricultural system is a fairly workmanlike one. One statistician commented that if he were recording labour inputs then these would relate to the total area dug for the crop and it therefore seemed sensible to regard this as the area of the crop; this seems reasonable. Where the boundaries of the prepared land are much less definite, as with the less organised types of shifting cultivation, there is a real problem which is discussed somewhat more fully later. From time to time the matter of superficial area versus projected area arises for debate. It seems rarely to be a serious issue in practical working conditions though adjustments are discussed occasionally. Even in a 25 per cent slope the difference between projected area and actual area is only about 5 per cent and this is probably among the least of the uncertainties attaching to a good deal of this work in developing countries. Where estimates have to be undertaken on steeply sloping land many other problems are usually involved and to concentrate on this particular issue is likely to divert attention from more important ones.

Perhaps the matter of the minimum area to be recorded in estimation procedures might be included under the general heading of definition. This has to be settled locally in the light of the purpose of the enquiry and the nature of cropping practice. Some countries have set a minimum area of 1/40th of an acre, in other cases as small as 9 square yards for an annual crop or 81 square yards for perennial crop have been noted. Broadly speaking, it costs as much in field enumerator's time to deal with a 50 square yard plot as ½ acre plot, and where small plots are common some preliminary study of the justification for dealing with them would be valuable. The omission of some of them in enquiries where the bulk of the crop is in much larger units may be of little consequence. If they are important in a particular situation, as might be the case for herbs used in "side dishes", they are all grown on such a microscopic scale that it is very doubtful whether any useful purpose would be served by

pursuing that particular piece of information. For instance, it might be much better if an idea is required of the contribution of side dishes to nutrition to attack the problem at the consumption point.

As might be expected from what has been said before, both population and area frames have been used in sample enquiries to obtain crop area statistics. In principle either is suitable (their relative advantages and disadvantages have been discussed already). It might be said in passing that examples of upward bias in crop acreage estimation using the area frame method have been reported, apparently arising from the enumerators managing to move the position of the selected sample units to somewhat flatter ground where they would otherwise fall on hilly land. Obviously the risks of this depend on the scale of the maps available and the closeness with which it is possible to supervise the enquiry. Typically, studies have depended on finding by actual measurement the area relating to the primary population sampling units–sometimes a village or similar small group, sometimes a farm family–and deriving the national totals from these records in the usual way. In principle it would seem that there might be cases where the whole of crop A would be found in a solid block in one part of the village lands, the whole of crop B in another part, the whole of crop C in a third, and so on. Within each block individual farmers would have their own areas, but the boundaries might not be readily visible to a visitor. Under these conditions it might be economical in time not to take a sample of the farmers. To do so would require a visit to each of the crop areas in the village, the identification of each man's own parcel of crop A and its measurement. This may well be more costly than measuring the larger, but visibly defined, area, of the village's total of crop A. In fact there seem to be relatively few reports in the published material of this kind of short-cut having been used. The reasons for this may not be difficult to envisage. For instance, the crops may be in a more or less compact locality but still the boundaries may be irregular and, perhaps with belts of trees or bush breaking up the main block. Again, it is more than likely that other information will be required in the same visit to the locality, information which can only be obtained from contact with individual farmers. It will usually be valuable to be able to associate such information with the crop area data at some stage and for this purpose the area records for the individual farms will be needed. The possibility of such short cuts should be kept in mind, however, though the bulk of the rest of the discussion will be concerned with estimating procedures based ultimately on the area of the relevant crops on individual farms.

There would be few problems in acreage estimation work if the records did not have to be made under pressure of time. Sometimes this is imposed by the growth of the crop, sometimes by the needs of the statistical service to supply data urgently for operational purposes. Since the basic record required is the area assignable to particular primary sampling units (as has been noted, usually farmers,) it is essential to locate all the crop grown by the units concerned. For various reasons, when a farmer is approached he may not take the enumerator to all his plots or fields. Sometimes this problem can be overcome if the village head is cooperative and if his personal position in the village is firm. However, he may be reluctant to take a visiting official three or four miles to the other side of the village land in order to see two or three odd patches of land perhaps intended to produce food crops for a rather despised group of labourers whom the village people employ. Where local customs permit it the list of plots assignable to selected farmers can be usefully checked in a village meeting. It is quite likely that any omission, accidental or intentional, will come to light if a good tempered discussion can be sponsored. In some surveys the use of this village meeting was made compulsory on all enumerators,–as for instance the Basutoland (now Lesotho) 1949/50 agricultural census. A further type of cross-check may be obtained by listing the fields reported by the farmer according to some sort of pattern which is innate in the local system. For instance, enquiries in the forest area of West Africa have required enumerators to tabulate the farmers' plots under the year of clearance of the original bushland. In a system of husbandry where land is cleared from bush, cropped for a regular number of years, and then returned to bush, this cycle provides a reasonable check on the completeness of the list. A farmer should have an area assignable to each of the years of the cycle then current. This kind of check must obviously be keyed to the local practice.

There are four main methods of obtaining the basic area record for the primary sampling unit–

(a) by locating the individual fields and measuring with measuring equipment,
(b) by preparing a map of the areas under the specified crops,
(c) by asking the farmers for the required information,
(d) by some variety of cluster or point sampling.

These methods are described in the following paragraphs, beginning with field measurement since this seems the most common. The chapter concludes with a few miscellaneous techniques which may be of interest less for themselves than for the possibilities in particular situations which they suggest.

Field measurement

There are two main groups of considerations under this heading, one concerned with methods of estimating areas from the basic measurements and the other with the equipment used to make measurements. There is also a question of whether the calculations from the measurements should be done in the field or the office.

There are four main methods of calculating areas.

(a) For fields that have at least two sides parallel it is possible to proceed by estimating the mean length of the two parallel sides and the distance between them and taking the product of the two.
(b) If there are more than three sides to the field it can be converted to a polygon and broken down into triangles. The area of the triangles can be determined either by the half base times height to apex, or the semi-perimeter formulae. These are described in Appendix E.
(c) These polygonal fields can also be recorded by a traverse of the circumference of the field taking bearings and lengths of the several sides.
(d) A variant on this process is the use of the planetable where a map of the plot is built up by bearing and distance measurements, but not by a traverse round the circumference.

The rectangulation and triangulation methods require only measurement of distances. For this process pacing, knotted cords, tapes (steel and linen), survey chains, rods, large scale versions of 'dividers', and measuring wheels have all been used. For angular measurement the standard equipment is the prismatic compass with spirit damped card, and, rarely, the planetable.

Turning now to the working details, any arrangement for field measurement must make it possible for any observation to be repeated, if required, by the supervisor. If the farmer and the particular field on his farm are regularly identifiable, it may be sufficient to have these recorded. This would be the case if farmers are reasonably cooperative and their farms are compact or conform to some kind of pattern such as being made up of fields cleared from bush on a short annual cycle. If this possibility of rigorous checking is not provided by the mere listing of farmer and field, it may be desirable to make the field staff provide a sketch of the village land and an indication of the paths leading out from it with the location of the plots actually visited.[1] When the field has been located, the first step is to prospect the situation to see where its boundaries are and what number of sides are needed to provide reasonable approximation to its area. Of the methods noted above for getting area estimates, the use of rectangles or parallel sided figures seems now rare in area estimation work, except where fields are laid out on the square at the time of settlement, or, occasionally, where terraces of regular widths are set out on hillsides. The method for other situations has tended to leave too much discretion in the hands of the field staff in approximating the boundaries to the formalised rectangular figure. If a situation does arise where the local land system suggests this method would be serviceable, then care should be taken that where the plots are not exactly formal rectangles, parallelograms, or trapeziums, the approximation is not attempted using too few rectangles.

1. It might be said in passing that the procedure for locating plots must conform rigorously to the random sampling pattern; attempts to get mean plot areas per village or district made by telling field staff to measure a certain number of plots, have almost invariably proved to give serious bias.

The triangulation method is always serviceable; any polygon can be broken down into triangles, and the areas of these can be calculated, given either all three sides or one side and a perpendicular upon it. The difficulty here has usually been in getting the field staff to put themselves in the right position for measuring the perpendicular. It must be remembered that a line bounding a triangle which crosses a field has no very clear indication on the ground except as a line of sight. Moreover, the apex may not be clearly visible unless a fellow team member locates it and stands there. If the crop is low and the distance is small the problem may not be significant, but in high crops and irregular ground there is an obvious possibility of trouble. Perhaps the simplest practical way of getting the perpendicular is to walk along the base with an eye on the position of the apex until it is believed that the right point on the base has been reached. The enumerator then faces the apex, stretches his hands out away from his sides, pointing along the line of the base; if he then swings them round evenly to join in front of him, his fingers should be pointing directly at the apex; if they are not, he moves accordingly.

Apart from siting accurately in high crops the actual walking of the base line and using measuring equipment along it will possibly cause difficulty in growing crops. The use of the perpendicular requires even more traversing of the plot than the ordinary measurement of the three sides.

Though the triangulation method seems not often to be used now, it should be kept in mind as one which needs the minimum of equipment. If pacing or knotted cords are used for measuring it can be operated with no more than one tape in each locality for checking purposes.

The preparation of traverses of the circumferences of fields using a compass and some form of linear measuring equipment is a well tried technique and, if the field staff are well trained, it is likely to be found serviceable and not unduly demanding of resources. The usual instrument for taking bearings is the prismatic spirit damped compass; and various forms on the market all seem reasonably satisfactory. The choice of measuring equipment is a subject of some argument. The chain and steel tape are entirely serviceable, although chains in bush conditions may get strained and steel tapes in wet conditions need care. In recent years measuring wheels, (some about two feet in diameter, others about one foot) have been used. Some users have found them entirely satisfactory, some have discarded them as introducing too many errors. The chains and tapes need two staff members, or one staff member and a local helper, whereas, in principle, a wheel can be operated by one person. This is sometimes a less clear distinction than it appears because some assistance may be needed, (it may be available out of good nature, or curiosity, merely for the purpose of finding the way and getting equipment from place to place. Moreover, even with the wheel it may be necessary to make a trip to the destination corner before starting the measurement if an assistant cannot be posted there. Lines of sight in developing countries are often far from clear when doing traverses round plots. To try to crystallise the issues a little, it would probably be fair to say that if money is short and large numbers of field teams have to be operated, chains or steel tapes will probably give better results than wheels. They are also to be preferred if the boundaries to be traversed include bush stems which have to be got over or round, or ground so uneven that the axle of the wheel moves up and down by, say, 6 inches within 6 feet or so of run, or where the surface is crumbly and liable to collapse or skid under the wheels. Minor roughnesses in method are likely to be more damaging to accuracy with the wheel than with the other methods. It is important with any of this equipment to stress its careful use and to keep a very close watch on the way in which the operator does his work.

The conversion of these field records into an estimate of area raises no problems of principle. The formulae can be applied with pencil and paper or by computer—and by all kinds of calculation equipment between. The main practical question is how to avoid errors of computation and how to get errors of field observation checked most smoothly and quickly. Generally speaking, it seems to be better so to arrange communications that faulty field observations can be sent back to the enumerator readily, and to do all the calculation in the office. If, however, areas need to be calculated in the field, in practice either the

rectangulation method or the triangulation method using the half base times vertical height formula is used. The more usual method is to send the field records direct to a central office for checking and computation. Most commonly the compass and measurement traverse of the circumference is used although triangulation may be included from time to time. Typically the first process in the office is to check the closing error of the traverse; this can be done by computer which will then print out all the observations showing the discrepancy between the end of the traverse and the opening point as a percentage of the total distance round it. If these facilities are not available, as is usual at present, the practice is to draw out using a circular protractor and ruler, usually on squared paper, the various plots and to calculate the percentage error. A decision has to be reached on how much error can be tolerated and a working basis might be to pick a closing error of somewhere between five and ten per cent, which will not result in more than perhaps 15 per cent of the observations being returned to the field staff for re-checking. If in a particular case these seem too crude a tolerance limit then this is all to the good and a source of satisfaction.

It is important to keep this stage of the computation fully up to date because rechecks are usually impossible when the enumerator has moved on to other villages. It does not follow that access to a computer will necessarily make easier the quick despatch of recheck instructions. This will only be so if access is obtainable regularly at the time required. When bearings and distances consistent with a tolerable closing error are complete for each observation, the closure must be completed and the area calculated. This, too, can be done on a computer or by hand (see Appendix E); in fact with experience of plotting out differently shaped fields, a rule of thumb can be found to shorten this work. For example, in many instances little if any error is introduced if the first and the last points are joined and the failure to close ignored.

When this stage is reached the acreage can be calculated again on the computer or estimated in one of the indicated ways—for example, by counting the squares or superimposing a grid and counting from that.

There may be a tendency to strive for precision in such measurements as these to an extent which diverts attention from the overall complex of errors or discourages work which might nevertheless provide useful, even if rather rough, guidance. The possibility of undertaking area estimation should not be abandoned merely because equipment of the desired standard is not to hand or funds available to buy it. For example, though it is known that pacing can be an inaccurate method of measuring distances with errors of up to 50 per cent, it is also known that a well-trained field man who has the knack of pacing, and in conditions which suit it, can get results within plus or minus five to ten per cent without too much difficulty. This would be a considerable improvement on guessing the area in a country or even on asking farmers how much they grow. If faced with this sort of position the way ahead is to put the available staff through a careful course of training and to watch for those who have an aptitude for the work, then to reject the others and put a great deal of care into bringing the better men up to a high standard. Again, knotted cords as measuring devices have been referred to most contemptuously by some workers who argue that they shrink and stretch in changing conditions. Nonetheless, if cord can be obtained, and sufficient tapes or chains cannot, then it can be made serviceable provided a strict routine of checking is imposed, using for the purpose the few tapes or chains available. Similarly, a fairly simple compass, provided that it is damped and has a reasonable arrangement for sighting, can give serviceable results. However, if funds are very short it might be better to use triangulation methods rather than compass-traverse methods.

In working up sketches, whether at field level or in the office, it is desirable not to have too many different scales even if the size of plot varies somewhat. It might be sensible to concentrate on some measures of the order of 1: 1,000 and stick to this.

Training has been mentioned on a number of occasions in these paragraphs and cannot be over-stressed. The important thing is that the field man should do what he is supposed to do and do it methodically. If he fails to do this, then there is a grave danger of a systematic error, which no manipulation later can correct for. The level to which records can sink is indicated by a test of some 250 measures of fields in which only 44 came within ± 5

percent; 8 percent were outside the range of ± 50 percent of the true figures.

There is always the possibility that farmers will object to field staff making measurements of their plots and walking across them. Provided that there is no external source of distrust–for example anxiety about land alienation–amongst the local people, it seems that usually they do not raise much objection, even to enumerators crossing their growing crops. However, the possibility must be kept in mind and action taken if trouble arises. For instance, one report spoke of farmers being complacent about the measuring but regarded wheels going over their fields as likely to bring ill luck!

Farmers as informants on area

This possibility might reasonably be considered next since, unlike most of the others which follow, it allows acreage data to be associated with other farm business information if desired. It also permits the total area of a farm to be assessed. It seems rare for conditions to arise in which collecting information on crop area directly from farmers is satisfactory in developing countries. The local people may have their own very serviceable assessment of area but, if so, the units are usually not standard over wide areas. If they know the area because of a land registration or a licensing of individual crops the likelihood of bias already described would make the method suspect. It seems only possible to note the feasibility of the method, given farmers' cooperation and standardised units of measurement, and to indicate its relative infrequency of use.

Mapping and Grid Sampling

Where large-scale accurate maps are available, particularly if they show the actual positions of fields, and where it is not desired to have the area statistics associated with individual farms, a range of possible techniques is available. The most highly organised, used in a number of enquiries in India over many years, proceeds by using the grid sampling method (see Sampling, above). Fairly typically a square area of about 2–2½ acres is located within each half square mile of the land surface. If the plots on these maps are numbered, those falling within the grid can be identified and listed. The list of plot numbers is then sent to the field staff in the areas concerned with instructions to enter against each the crop occupying the field. It will be noted that, in the conditions assumed for this technique, the boundaries of the designated fields are quite clear and that they will be well known and clearly designated in local records. There will, therefore, be no doubt about the boundary of the fields, and if they are occupied by more than one crop then the proportion of the area, or the actual acreage under each crop can be indicated, as may be best in the light of local practice. These statistics are then returned to Head Office and the computation of totals for regional groupings proceeds in an entirely straightforward manner. If the plots are not numbered and registered in local and national records but are, nevertheless, clearly indicated in the maps, a variant of this method can be used. A section of the map containing the grid is sent to the local field worker who enters on the map the portions of the several plots which are occupied by designated crops. This uses the 'closed segment' approach in which the recording is confined to the actual boundaries of the sample grid. This need not be so, any plot which has a specified part of it (say, the S.W. corner,) within the bounds of the grid might be included and all others excluded without introducing bias. For these techniques a mapping scale of the general order of about 8 or 10 inches to the mile is desirable but they could be operated with somewhat smaller scale if necessary. There are reports of a further variant in which maps of perhaps 2 inches to the mile are available with accurate indications of the location of physical features but incomplete or non-existent records of actual fields. Enlargements of sample portions of these, with grids inserted, have been sent to the field staff with instructions to sketch in within the grids the portions of the land surface occupied by the various crops required. Grids bounded by natural features, roads, etc., have also been used. The field worker identifies the desired portion of the grid by reference to the local physical features. No doubt this method is capable of being satisfactorily worked with care. It seems worth noting, however, that these area type methods have the disadvantage of being

difficult to link with individual farm records. A variant of these techniques is where the sample locations for observation take the form of clusters of nine or sixteen points within areas of the same general order of magnitude as the grids just described. The location of the dots is indicated on maps and these are despatched to the field staff who return a statement of the crop corresponding to the point on which each dot falls. The problems of locating dots on fairly poor maps must be considerable and it seems at least possible that the risks of enumerator bias are rather greater with these than with grid area records.

Miscellaneous methods

There are a number of methods in use in various parts of the world for getting an indication of acreage which should be mentioned before going on to more general techniques. A very common method of getting a measure of acreage where the amount of seed used is known, is merely to divide the tonnage of seed used by the usual seed rate. This method is found widely and some users—for example, estimators of Irish potato acreage in the West Indies—claim considerable accuracy for their estimates. In some Middle Eastern countries the area is arrived at by multiplying the work animals and machines by their daily work capacity. Possibly where the type of organisation is non-standard and statistical control is so far slight, such methods are better than nothing. In many countries it has been customary to collect through the field officers of the Departments of Agriculture, reports of acreage planted each year quite early in the season. These records are usually very difficult to assess. Sometimes they have continued after other techniques of objective estimation have been developed. In some cases for diverse reasons, the published statistics are the average of the reported and the objectively estimated figures! There is a special problem, which will be referred to again below, of obtaining early estimates of crop area to combine with estimates of probable yield for crop forecasting purposes. It may very well be worthwhile to try to formalise a reporting system if one has been established. The essential feature here is a systematic programme of reconciliation meetings between the statisticians and the regional agricultural reporters. It is assumed that the statisticians will, after the season is over, have reasonably good estimates of the crop either in the form of the results of objective sample surveys or as records of the quantity of crops sold for export or local use. The techniques of graphical comparison of estimated and reported area, (as outlined in the section below on forecasting crop yields) can be applied here. The essential point, however, is that all successful experience of this kind of work shows that reconciliation cannot be done by correspondence; it is far better to go and sit down with the reporter and take all available records, match them against his and reach an agreed reconciliation with guidance for future operations. It is possible, too, that a reporting method of this kind might continue for rather minor crops even when objective sampling had taken over the handling of the major crop statistics.

These various methods of making an estimate of area are, in principle, applicable at any period of the crops' life from planting to harvest. (The cases of continuous cropping are dealt with below.). If there were ample staff resources it would be possible to pick when, in the cropping season, the record was to be made and, if desired, a series of records could be made to trace the sequence of planting and of harvesting. In practice, the difficulty is normally to get completely around the area even once before the crop is harvested. Different conditions require quite different approaches. A country with reasonably adequate staffing, and a clear sequence of seasons with different crops, will wish to make a series of passages through the area in months chosen to catch the various crops. This is a matter of technical decision in each locality. More generally, it is worth considering how to divide between the passages the total field work consisting of selecting sample units and making the local frames, recording crop areas, recording crop yields, compiling financial items relating to the farm business, etc., etc., depending on the statistical programme. It may well be sensible in the first passage to complete the local listing to provide frames, and to select the farmers who will be the primary sampling units and to locate their fields, and to indicate the designation of fields by a stake with a number painted on it in black paint. Only the field recording has then to be done on the next occasion when the area is visited and any other

time available on the visit can be used for collecting information directly from the farmer. Some statisticians have found this works very well, farmers having no objection to the presence of the plot designation. In other situations, it has been the practice to include the main crop area estimate in the first round and to make forecasts of the likely date of harvesting. Clearly this is easier to do if the crop is nearing maturity when the circuit is made. It may be necessary, if all areas are to be covered within the constraints of planting and harvesting dates, to work through a stretch of country, starting at the early planting end of it and finishing up at the late planting end.

Early estimates of acreage for use in production forecasting give problems to everyone. One method has been indicated above in passing. It may well be that in most developing countries the problem must be solved either in that way or by pushing ahead with the acreage estimating passage through the country as quickly as possible. There are various sampling techniques in use in developed countries for this work using systems of voluntary reports. For instance a request is sent to a more-or-less random sample of farmers before the planting time, asking for a statement of the acreage they intend to plant and the area they planted the previous year. Those who reply will not be a representative sample but, broadly speaking, the ratio of "intentions" to "last year's area" gives indications. These indications must be corrected by experience of earlier years, since intentions seem always to be biased to a degree.

Further sampling enquiries are made early in the year in which farmers are asked to state how much they have planted. Similar ratio methods are used to derive early estimates of the crop, stage by stage, as the season advances. It is possible that special occasions will occur in developing countries where this kind of technique can be used. However, it should be kept in mind that the results will usually need correction for various kinds of bias–some from the non-representativeness of the people prepared to reply to voluntary samples, or some from bias in the individual informants' replies. It may take several years to build up the necessary experience to provide correction factors to eliminate or reduce these biases.

IX. ESTIMATION OF CROP PRODUCTION

Introduction

It is desirable in descriptions to keep separate the ideas of 'rate of production', i.e. the production per unit (such as per hectare or per tree), the total production of a farm or geographical unit, and the net outgo to use off the farm. Interest in this section centres on production of national and sub-national areas, but in practice this is usually derived by combining statistics of rate of production with those of numbers of production units.

There are several groups of problems in this work. One is a matter of definition. Clearly between the formation of the grain or root (which, in due course, will be the agricultural product) and its final preparation for consumption the quantity associated with a given area of land will vary. Among the elements of this variation may be—

(a) the addition of carbohydrate. protein, etc., and water in the process of growth;
(b) loss of water with ripening of grain crops;
(c) loss of substance by bird and other damage before harvest;
(d) loss by shedding at harvest and in transfer to the store, also bird, monkey and similar pest loss from the products stacked in the field;
(e) loss in store by pests or rot;
(f) loss by drying or respiration.

It might be noted that products are drawn for consumption from crops at all stages from soon after the formation of the grain or tuber (well before maturity) right through to the end of the storage period. For example, green maize cobs may be eaten in the field before normal harvest while, at the other extreme, a product from the same crop may be used from store right up until the start of the next harvest. Consequently the sum of the weights of crop as actually withdrawn for consumption between harvest and the final using up of supplies may not match the tonnage of crop actually recorded at any point. Even in developed countries the problems are not handled very rigorously, still less are they in developing countries where statistical control of production estimation has, usually, not yet been consolidated. However, a working aim appears to be to deal in statistics of production as they would be measured when the crop was in store after harvest and before any appreciable storage losses had occurred.

A second group of items which need definition concerns the timing of the making of estimates, as distinct from the stage of the crop to which they relate. For practical purposes in this bulletin, figures prepared before harvest are referred to as "forecasts" and are dealt with in the following chapter and those made at harvest, or afterwards, are called "estimates". A confusion appears in some discussions about the usefulness of these several figures. For instance, references are still to be found to the value of production estimates for planning crop movement and for dealing with food shortages. In fact, figures must be available well before harvest if they are to be of use either for movement planning or for alleviating food shortage. Production estimates are an important, indeed usually an essential, intermediate stage for production forecasts, but it must not be forgotten that this further stage must be reached if these two particular aims are to be achieved. However, production estimates, after harvest, may be valuable as general indicators of levels of production, as elements in national income estimation, and as contributions to the study of input-output relationships. If a situation is such that forecasts alone can give the necessary guidance this will usually require a very special concentration of resources in order to push the work beyond the stage of production estimation to the one of production forecasting. Otherwise, where the urgency is less, it seems best to concentrate on estimating crops of which a large

proportion are sold. These usually provide the opportunity for cross-checking estimates and for general toning-up of the field techniques. As for subsistence consumption, though it might be the subject of ten-yearly enquiries for the purpose of contributing to the general pattern of national product studies, any nutritional interest is probably best studied by direct consumption surveys.

A third group of problems is concerned with the techniques of estimation. The main methods available are–

(a) the weighing of crops if they pass through some natural bottleneck in their utilisation;
(b) eye judgement by experienced reporters;
(c) crop sampling;
(d) utilisation table estimating.

The method of sampling crops, cutting small clearly defined areas in properly selected samples of the fields, seems now to have become fairly standard in developing countries. In most developed ones there is a long history of judgement reporting. However one cannot say that the problems of production estimating have been thoroughly solved in either. The sampling techniques give a quite reasonably low sampling error but they are subject to certain biases under the conditions normally met in developing countries. They probably represent the best line to pursue but it is wrong in principle to attempt to depend here (any more than for any other statistic) on a single source of figures. It should be the practice to arrange to cross-check each figure obtained by one technique with a figure of similar definition obtained by another. This seems the only approach that is wise in production estimating at present.

Direct weighing

A few crops, typically the non-edible industrial crop, are sold to processing plants in the producing neighbourhood. It is usually possible to arrange for a return to be made to some central authority of the tonnages handled at bottle-necks where counting is possible–for example, marketing boards, ginneries, export points, etc. Since, by their nature, these may be operated by fairly powerful agencies, it may be necessary to obtain statutory powers in order to collect this information; on the other hand such agencies may be extremely cooperative. It is absolutely essential, when using this source of statistics, to check that what is passing through the recording point corresponds to the desired definition. Examples of discrepancies include the case where the desired figures relate to total sales but good records are obtained only for citrus sold for export and little is known of the amount from the same plantations which moves into local consumption. Again, products may move into or out of the area being considered. Thus in a number of West African countries there is appreciable movement across land boundaries. Consequently the quantity marketed through particular channels may be greater or less than that grown in the country concerned for which production estimates are sought. Again, where the recording points are not under the control of a single organisation, it is possible that the same produce may be counted more than once. For instance, if the traders making the returns are acting both as primary and secondary wholesalers, their records may include some produce which has already been included in other traders' returns or will be so included later on. Stock changes may also spoil the current throughput statistics as indicators of current production, but usually if a complete harvest season is used this is not a great problem. The "utilisation table method" noted at the end of this chapter provides a framework within which data obtained by direct weighing can be used with the minimum risk of misunderstanding.

Judgement reporting

Many developed countries have built up systems under which voluntary reporters, or the field staff of the agricultural departments, send in their judgements of the average yields of selected crops in the localities for which they are responsible. In effect this requires them to form a judgement both about the absolute level of yield in the individual fields which they observe, and also about the relative proportions of good and bad fields within their area, so

that they can develop a correct weighted average for the area. Probably they do not consciously proceed in these two steps, however. In practice, they depend not only on their own judgement but also on the results of conversation with farmers and on the actual out-turn of the fields harvested early. Developing countries have also used this method, drawing estimates from agricultural officers and village accountants. Agricultural statisticians customarily condemn the technique as difficult to organise and gravely biassed. There is some truth in this view, particularly when it is remembered that the staff who are in a position to make the estimates are usually not those of the department charged with collecting the statistics. Nevertheless, the method may be written off too lightly. If no system were in use and a new one had to be set up there are good grounds for choosing the crop sampling method. However, if one is already running, even in a somewhat unorganised form, it would be better to get the most out of it than to discard it completely. In particular, it may be useful in providing cross-checks with crop sampling results and with the records of disposals through the channels described in the previous section. If 'seminars' are arranged so that those concerned with these various sources of statistics can reconcile discrepancies a good deal that is of value, in addition to agreeing on a single set of figures, can be obtained. Sometimes, too, it may be sensible to operate a reporting system for the less important crops, if some indicators for them are desired, while concentrating the field work of crop sampling on the bigger and more important crop areas. Though no explicit account of such an arrangement seems available there is no reason why the field staff who are making estimates by crop sampling should not make more general judgement reports about the yields of minor crops not included in the sampling programme.

In a system which uses agricultural department staff, reporters should be asked to report the average yield for the specific areas assigned to them for their ordinary duties. No correction of the figures from individual reporters should be allowed at regional or similar level; the original record should be sent as it stands to headquarters, though it should be open to the regional office to attach comments. Two possible correction devices for bias are available. If, as occasionally happens, records of this kind have been sent in regularly for many years as part of a routine reporting programme, diagrams can be drawn up in which the judgement report is plotted against final out-turn as judged by marketings or exports, using methods similar to those described under forecasting below. This should indicate the nature of any biases in reports and provide the means of adjusting for them. It might be sensible to make such summaries separately for the individual climatic zones or, if this is impracticable, for regions or provincial areas of a country. Since the personnel concerned are likely to change fairly often, it is pointless to make the summaries for very small areas. The second possible method can be used, given the necessary cooperation, even if back records are not available. Each reporter is asked to give his judgement of the average yield of a number of plots (e.g. "master farmer's" fields, experimental station plots, etc.) where the actual yield will subsequently be known. A comparison of "judgement" and "actual" will provide a basis for adjusting the judgement report. Of course, this takes account only of any bias in the appraisal of the individual fields; it does nothing to correct any error in weighting of the good fields and the bad over the whole area concerned. Typically it is found that reporters tend to over-estimate crops in poor years and under-estimate them in good years.

Crop sampling

Experience has shown that an estimate of yield per unit area can be obtained with tolerable accuracy by cutting small plots suitably chosen in a representative sample of the fields in a country. The procedure has become fairly standard in broad outline. A sub-sample of the fields which are selected for area recording is chosen for the estimation of yield. Typically perhaps 1/5th or 1/10th of the sample for area estimating is so chosen.

The chosen fields are visited at an appropriate time and sample 'quadrats' are sited within them using a representative method. Usually at least two quadrats are chosen in each field. At harvest the crop on these quadrats is cut, thrashed, and the produce weighed; a sample is retained for drying to give a correction for moisture content. The estimated yield per unit area for the whole territory is derived from these records.

As to quadrat size, many sizes have been used ranging from one foot square or less up to 1/10th of an acre or more. For reasons discussed below very small plots, i.e. of the order of 1 square metre, are to be avoided in developing countries and a quadrat in the general order of size between 5x5 to 10x10 metres seems sensible. Numerous instances of enquiries using this general order of magnitude are to be found. As to siting within the field, three methods seem satisfactory. One is the random coordinate method. The staff normally know the size of the field from previous recordings made to estimate area. They are required to note the distance across the field and to deduct from it the length of one side of the quadrat, then to do the same for the width, and within these limits to take two pairs of random numbers. The first of the pair gives the distance from the corner of origin along the chosen base side of the field and the second the distance at right angles into the field. By deducting the length of the side the majority of occasions where a plot might fall outside the field, or partially outside it, will be eliminated.

The second method is the use of a grid. The field-man is provided with a transparent sheet of stiff plastic, say about 8 or 10 inches square, marked out in squares by two series of parallel lines at right angles to each other. Each square is numbered. A plan of the field in which the quadrats are to be sited is drawn to scale by the field-man. He selects the longest side of the field as a base and places the base line of the grid over it. He then adjusts the position of the grid from side to side so that its left hand boundary is over the angle of the field sketch which is farthest to the left. A random number is then drawn from within the total number of grid squares. If this numbered square lies over the field sketch then it designates a sample quadrat. If it lies outside, then successive numbers are drawn until two fall within the sketch. The size of square on the grid, the desired quadrat size and the scale of sketch needs to be co-ordinated, but there are several fairly obvious methods of arranging this. A cross staff (see Appendix C) is likely to be useful for locating sites; it may be especially useful for siting small quadrats since its use should cut down the opportunity for personal bias. Quadrats falling near the edges of fields present some problems, too, since their treatment could also lead to bias. Various procedures for dealing with these quadrats might be instituted but some may be self-defeating if in practice the field staff are confused by them or ignore them. It is probably best to accept only quadrats which fall wholly within the field. If one quadrat does not, another is chosen by the standard procedure.

The third method, used in big enquiries where practice is fairly standard and the level of cultivation reasonably good, takes two points along the diagonal of the field often in more or less systematic forms, e.g. 1/3rd and 2/3rds, although two random numbers have also been used. Tests suggest that no significant bias is introduced by this method and the saving in time in locating quadrats is valuable.

The quadrats may be either square, triangular or circular but the majority of workers seem to favour the square plot. When the location has been decided by one or other of methods described, the first side is laid down using rods pushed into the ground to mark the corners. A tape is usually used for measuring the length of the side, and various methods have been used for getting the right angle. The compass is quite serviceable, so is a cross staff. Some workers have preferred a right angle square, though this might be difficult to carry. Still another method is to knot a cord to give divisions in multiples of three, four and five units. When the two ends are held together and the other knots pulled till the strings are tight a right-angle triangle is set out. The choice between these methods may be a matter of relative availability of equipment and people. The cord is the simplest and cheapest but it is difficult to operate with less than three people. It may also be difficult to use in high crops. A cross-staff really needs two people but, by back siting, and providing a peg to hold the end of the tape, a compass and tape can be used by one man. The corner points can be either stakes cut from the bush or, and this is probably better, they can be metal rods which are carried by the fieldstaff for the purpose. They are not necessarily much more precise than the bush stake but they tend to contribute to a general air of precision and they may be easier to get into the ground. One or two departments have made them from tube in sections which can be dismantled for carrying. They have also been arranged so that the same rods can form the tripod from which the weighing equipment is suspended.

When the corners of the quadrat are clearly identified the boundaries are set out with cords pulled taut and carefully dropped into the crop down to the ground. Care is needed in this process to make sure that the line is kept straight and that ears originating on stems which lie outside the boundary are excluded and vice versa. With plots of the size suggested it is not possible to have rigid boundaries. Rigid frames have been used when designating quadrats of the order of ten to thirty square feet; it is rather easier to slide the rods which form these frames under any stems which lean across the quadrat boundary.

The cutting and thrashing of the crop presents the most difficult of the processes in the crop sampling schemes; if it is a root crop much the same holds true[1]. The central point is the labour of separating the product in the form in which it is used or traded from the crop as it stands in the field. Usually this will mean for grain, thrashing, and for roots, lifting and separating from soil. The labour of thrashing by hand a number of plots, each of a tenth of an acre, is considerable; this may slow down the work to a point which endangers the programme. Conditions vary but it might be best to think of a team of two handling two to four such plots a day. It is, in principle, possible to save some of this labour by weighing the crop as it is cut, taking a known sample of this, thrashing it and weighing the produce, then calculating the weight of the produce of the quadrat by simple proportion. This is probably to be avoided since it is a stage which is rather difficult to supervise correctly – at this stage field staff must be left a good deal of discretion on timing so that they can act quickly when a farmer intends to harvest, and supervisors cannot readily keep up with them. It seems best to harvest and thrash the whole quadrat yield; further sub-sampling, as suggested, should be resorted to only if the organisation would otherwise get out of hand. Some pilot testing of the time needed in the local conditions for locating and completing the sampling procedure should be made. For grain, when it has been thrashed, it is customary to mix the crop thoroughly, to take a sample of, perhaps, two kilos and put in a loose-meshed bag which is kept under the control of the fieldstaff (or sometimes left with the farmer). If the field staff keep it, it is exposed in the sun for three days, being taken in at night, and then weighed, to give a drying-out allowance. If it is left with the farmer he is asked to keep it in his hut in an airy situation, but away from rats, hens, etc., until it is weighed when the field staff call a few weeks later.

The question of payment to the farmer for this work is a vexed one. If the arrangement can be made without payment being required, so much the better. One or two enquiries have operated without difficulty using a quite nominal payment to the farmer for the general disturbance caused by the particular crop sampling arrangement.

One of the chief problems in this work is getting the field staff to the right point when the crop is being harvested. Sometimes the headman can help by sending a message to the field staff when the harvest is likely to start. Otherwise, when the farmer is visited in the course of the area estimation work the likely date of harvesting can be discussed and a reasonable forecast made. At least one enquiry arranged to cut half of each quadrat or half the quadrats when the field staff visited the farm a few days before harvest and the rest at harvest if they were successful in getting there at the right time. This gave an insurance, and a comparison of the two, where they were available, gave a means of correcting yield records for the plots which had not been visited at the right time. No doubt a farmer could be persuaded to leave the sample quadrats untouched when he harvested, if the boundaries were marked beforehand, but this would usually be of little advantage since such scraps of crop would be the goal of every bird in the neighbourhood.

The square plots, which have been assumed in the above account, seem largely to have taken over from other shapes. In principle, triangular plots are rather easier to define since three cords and one base line position are sufficient. However, since they carry a higher edge-to-area ratio than square plots, they are probably best avoided (see below). Circular plots have been reported occasionally, and it is not entirely clear why they have not been

1. The method of harvesting to be used, in comparison with the one the farmer uses, poses problems. Where it is feasible to do so the farm method should be used – e.g. ear by ear if that is the practice. Sometimes this is impracticable (e.g. if mechanical harvesting is practiced on farms). Even if farm practice cannot be followed entirely, it should be in respect of such features as length of straw cut with the ear.

more used. Once the siting point is determined, and the stake put in, the plot could then be designated by rotating a cord or tape of the required radius about the stake as centre, (16.6 feet will give a plot of 1/50th acre). Some disadvantages are obvious but exploration of this arrangement might be worth pursuing.

Notwithstanding the wide use of crop sampling schemes they are not entirely the answer. However, probably the correct way ahead is to operate this method as rigorously as possible in the field and to direct attention to elucidating the nature of any biases to which it is subject. As reported, these biases are usually upwards (i.e. the production in a region as estimated by this method is higher than that which can be accounted for by the commercial harvest or disposals). There appear to be a number of possible sources of this. One is that the field operator may be inclined to include as within his quadrat boundary portions of crop which, strictly speaking, lie outsideit. This is a kind of personal bias which has often been noted in field experiments. Again, under field conditions it is very difficult to lay down the procedure for siting quadrats so precisely that it leaves the field man no room for choice. He may be tempted to move a small quadrat somewhat towards a better part of the crop than the correct selection procedure would designate, especially if he avoids having to cut prickly weeds in doing so. If the quadrat is large this is less easy to do. Again, there may be some confusion arising from the fact that the crop sampling procedure does not usually allow any major harvesting losses whereas the ordinary farm harvest may lead to quite substantial losses, either by shedding or by bird attack in the process of getting to store. Studies have suggested that perhaps one half of an upward bias of ten to fifteen percent was attributable to this difference in damage. Gleaning of sample areas of farm harvested crop may give an idea of shedding losses but not of bird and monkey damage. Finally, in practice there may be a bias arising from the field man sampling from a different superficial area from the one which he measured as the area of the crop earlier in the season. This will be so if the crop suffered damage in the course of growth after the area measurement was made. If the plot for cutting could be laid down at the same time as the area is measured no problem would arise but, since this is often impracticable, the area from which it is sampled at harvest may exclude this failed area. In some enquiries where the failure was of a fairly clear-cut nature, on fields which had distinct boundaries, an arrangement has been operated where an estimate is made of the amount of the field which has failed, either at a passage made for this and other enquiries, (including forecasting the likely date of harvest) a little before harvest, or at the time of harvesting. This will not, however, deal with the typical case in shifting cultivation where bush has invaded and swamped the crop at the edges.

Though there are reports of complete fields being harvested in test enquiries– particularly test enquiries for comparing sample-based estimates with commercial out-turn – there seems little evidence of their use as part of a regular sampling study for crop estimation. Clearly it would be difficult for field staff themselves to harvest complete fields but occasions may be emerging where it is possible to get a quite reasonably good estimate of the produce of a field by noting what is sold, or by noting the number of sheaves carried off and weighing test samples of sheaves. Similarly, as farm record studies expand, more and more records of this kind may emerge. There seems some advantage in giving attention to the possibility of recording production per field, or production per farm, partly as indicating commercial out-turn and partly as indicating the frequency distribution of scale of production – a matter not by any means irrelevant in commodity planning.

Utilisation table

This is used for various purposes and is, strictly speaking, a formal, and basically wholly simple, technique for reconciling statistics of various elements in a commodity supply picture. Typically for a commodity such as a grain

opening stocks + production + imports = use for food + use for animal feed + exports + waste and drying out + closing stocks.

If all these, except one, could be rigorously estimated then this final one could be calculated. It is rarely possible to achive such rigorous calculations even in developed countries, but by piecing together firm estimates from international trade figures, by assuming that stock will

not differ much from one end of crop year to another, and by inserting estimates based on small-scale enquiries for consumption, wastage, and so on, a reasonable reconciliation of figures from numerous sources can be built up. It is extremely important that cross-checks of this kind should be made whenever possible and a reasonable amount of time be spent on them. One incidental advantage of this kind of tabulation is that it provides an indication of food usage which is not available from any other source (Food Balance Sheet tabulations by F.A.O. see publications listed). Associated with this calculation in the office should be a programme of reconciling figures obtained by direct statistical estimation using the methods suggested with those reported by field staff and the trading figures of central organisations. A good deal of tact is required, both to get serious consideration of such work and to carry it through without mutual irritation, but it is well worth doing, partly because when well done it improves cooperation and partly because new light on the actual working situation for a commodity is often thrown in the process.

In the main, these paragraphs have been written in terms of grain. The basic approach is very similar for groundnuts or roots, though the detailed process may have to be varied because of the different harvesting processes.

The equipment for weighing and measuring is outlined in Appendix C.

X. CROP FORECASTING

Introduction

A forecast is a statement of the most likely yield or production of a crop on the basis of known facts at the time of making the forecast, and assuming certain conditions of weather, damage by pests and diseases, and other factors which may affect yield between the date of the forecast and final harvesting of the crop. For working purposes it is usually assumed that these factors will be about the same as the average for previous years when the state of the crop at that time of year was similar to that in the current year.

When considering crop production forecasting four broad stages are recognisable;

a) just before harvest when all the fruits (or other commercially used parts of a plant) are present and are the same size as they will be at harvest;
b) a late season stage when all the fruits, etc., have set or, if they have not, their probability of contributing to the final yield is virtually zero;
c) early season, when leaves and stems are formed but the fruits are not completely set; and
d) before planting.

The first and the last of these can be set aside for the purpose of the present discussion. The first deals with the stage at which the crop sampling scheme, described in the previous chapter, applies. It may technically be regarded as a forecast in that the estimates are actually made before harvest, but in working practice in most cases in developing countries the results are not available until the crop has moved to market. The last situation – namely the forecasting of production prior to planting – is usually in a different category from the rest as being based either on the economic relationships between prices of crops and other related variables and the area planted as estimated from statistics for a run of years, or on returns of farmers' "intentions". The two approaches are equally applicable to breeding plans for livestock or any other consequence of farmers' production decisions.

For the rest – namely the late and the early season stages –three methods of approach have been explored in various agricultural statistics departments. Of these methods, one is based on reports from observers in the field who make appraisals of the general state of the crop in their localities, the second uses historical relationships between weather and yield as a basis for deriving a forecast of yield from the weather in the course of a current season, and the third is based on objective measurements made on the crop. All three need to be combined with early estimates of the area for harvest in order to derive a total production for the geographical area. Various corrections and adjustments may be necessary to all the basic figures designed to indicate likely rates of production. It is possible, though the situation seems rare, for judgements to be arrived at not by an area estimate multiplied by a yield estimate but directly in terms of overall tonnage for a particular locality.

The possible lines along which crop forecasting might develop are indicated in the paragraphs which follow. However, some of them must be viewed as highly tentative because the amount of well-consolidated forecasting work of good precision which is going on in developing countries is rather small. Where a method has found commercial acceptance the situation is usually one where a cash crop attracts an outstanding amount of interest and in consequence it has been possible to assemble enough resources to build up a system of judgement reports. Even so, it is not improbable that a wholly rigorous assessment of the precision of early season forecasts would show only modest success. The impression conveyed is often better than this because the actual manner of operation results in regular revision of the results as the season proceeds so that, by the close, the forecasts have tapered

into estimates and thence into records of actual achievement. Where this happens the identification of the degree of forecasting success is possible only by rather disruptive postmortem examinations of successive forecasts compared with the final estimate. Some of the methods which are suggested below are based on techniques in use in developing countries, others on those from temperate countries modified by adjustment for conditions in developing areas.

Early estimate of productive area

The possible need of early estimates of the productive area has been mentioned from time to time in the course of the discussion on the techniques of crop area assessment. If the forecasting organisation is based on plant counts and similar records which are made on sub-samples of the sample used for crop area estimation it is likely that adequate early area estimates will be obtainable. Where suitable area estimates are lacking it is usually either because a significant proportion of the crop fails between the time of the 'passage' for area recording and the time of harvest or because the crop area suffers no change during growth but there are too few field staff to complete the area recording in time.

There is no easy solution, the more promising of the lines of attack require substantial resources in cash and skilled manpower, and it is essential to look at the whole matter critically to decide whether the need for forecasts justifies the foregoing of so much other work. If it does, then the resouces must be allocated. It may then be possible to speed up the regular recording work to give the area records in time. If this is not possible, the next most promising line for securing pointers to the extent of crop failure is to collect judgement reports on the subject from the field staff, from agricultural extension staff, or from village accountants. If the main problem is slowness in completing the area records it is possible that reports from local observers giving their judgement of the current year's area as a proportion of that of the previous year might be useful.

Historical comparisons of this kind are liable to suffer from various distortions – as the U.S. Department of Agriculture Crop and Livestock Reporting Service experience showed – but with the experience of several years for comparison the method might be made serviceable. It is not an attractive prospect, however, except perhaps when some agency – e.g. a marketing board or cooperative – has staff in close continuous touch with farmers.

Condition reports

In the world as a whole this method seems to be the most widely used though all countries recognise that it is capable of giving dubious results on occasion. However, the condition of a crop during growth seems to be something which observers are able to assess with some measure of precision. Naturally, however, they cannot at a given point in time forecast what will be the effect of weather and other influences between that point and the final harvest. Generally speaking, there are two bases for such reports, the commonest being an assessment of crop condition and the other a forecast of prospective yield in terms of weight per unit area. It is found in practice that experienced observers of crops have an idea in their minds of a 'full' or 'normal' crop rather than an average crop. In this context a full crop is one that is healthy and undamaged by drought, insects, disease or accidents, and which shows the state of growth which is reasonable under such favourable conditions. The fact that crops do suffer damage from drought, pests, accidents, and so on, accounts for the not uncommon remark that crops never come up to average these days! Condition reporting is usually done in terms of percentages of the full or normal crop. However, it seems unlikely that observers can identify more than a small number of sub-categories between crop failure and crop perfection. It might well be enough to report on a scale of 1–10. There have been in use from time to time reporting systems based on the number of fingers in a hand (a five finger crop is perfect, a one finger crop is bad) or coins making up some coin of larger denomination. If permanent reporters are being used there seems no particular advantage in choosing this kind of device but it might be sensible to use a scale running from 0 to 10 to

narrow somewhat the range of choice which they feel is effectively offered them. Each reporter needs to be told the precise geographical region for which he is responsible for reporting, the crops for which reports are required, and the dates on which they must be despatched. It is probably particularly worthwhile to visit reporters engaged on this kind of work and to discuss with them what is in the organiser's mind and what problems they are themselves meeting.

The reports from the individual reporters should be passed to some form of local scrutiny, but *no amendment* should be permitted at this stage. The regional officer concerned may make a comment before passing on the document, or he may pass it back and ask the field staff to look at it again. In the central office these indicators will be weighted by area of crop in the respective localities and the consolidated estimate for each main geographical region derived. It may be enough to leave these in this form, indeed unless there are historical data which will provide a guide to the usual relationship between such estimates and the final outturn, little else can be done. If such historical series are available then a graph of condition against final outturn may indicate a relationship which will allow the probable yield corresponding to the condition figure for the current year to be read off. Obviously, all such relationships may change with changing technical conditions and this must be watched for. Though in principle it may be possible to derive mathematical relationships between condition, other variables (such as pest damage or the weather) and final yield, it seems usually to have been found convenient to deal in graphical terms. Where a major deviation between the condition and the final outturn is experienced a second influencing variable may be available – for example, reports of pest damage or rainfall records for a critical stage in the growth. A useful correction may be obtained by drawing in on the first graph the line – linear or curved – which most closely fits the plotted points. Lines drawn by eye are likely to be quite serviceable for this purpose. A note is now made each year of the difference between the actual value of the final yield and the value on the line of best fit corresponding to the condition estimate for that year. These values are then graphed against the pest or weather variable and a line of best fit for that graph drawn in, too. As the condition and pest, etc., data come in each year the corresponding yield values on the lines of best fit of the two graphs are noted. The sum of these two values gives the forecast yield.

A rough and ready indication of the likely harvest level should be obtainable from this kind of procedure. It might be that one of the techniques based on objective measurement described below, may be more suited to important crops but if a condition reporting system is already in use it would be wise to continue it in addition. To push judgement reporting to much higher levels of precision would probably require some individual person to take a continuing interest in the forecasting methods for a particular crop. It might be well worth arranging this if a field organisation is available through which judgement reports of this kind can be obtained. The objective method described later offers prospects of methodical improvement but there is likely to be a long way to go before success is achieved with it; a more straightforward approach in terms which look sensible to the ordinary trader and farmer is not to be despised.

Crop-weather relationships

Since the weather noticeably affects yield, there has been for many years an interest all over the world in relating crop yields to weather factors. It seems natural to believe that if the weather in the growing season is so influential on the crop growth then from historical relationships, combined with current observations of meteorological data a good forecasting system should be derivable. In spite of the very large amount of work that has gone into this procedure, its representation in the world's forecasting schemes is slight. For a few years certain spring-sown wheat and other grain areas in North America used weather conditions and subsoil moisture conditions alone as indicators of prospective yield but this has been abandoned as a sole method. Early attempts in any particular situation are usually encouraging but beyond a certain point discouragement sets in. Part of the problem is the sensitivity of the crop to conditions in specific short periods of its growing phases and

meteorological records for these periods may not be easy to assemble. Again, important weather factors may come late in the season. For instance, root crops are very sensitive to growing conditions in the last few weeks before harvest and all grain crops are very sensitive to conditions at harvest. Any number of weather relationships and yield can be explored using regression analysis if so desired. For exploratory purposes the procedure of plotting the yield against the apparently most important variable and drawing in a line of best fit, then plotting the deviations from this line against the second factor, and so on, may be revealing.

Generally speaking, it seems best to look to weather to provide a modifying factor for a crop reporting or an objective measurement system rather than the main indicator.

Crop measurements

For many years attempts have been made to find single measurements of a crop which will provide useful indicators of the final outturn. For instance, the height of cinchona trees and the diameter of their branches seem to give a useful indication of their yield of bark. Not many useful results of this work seem to have appeared. Of late the approach has been by way of analysing the progress of particular fruits through the season in relation to the stage of maturity which they have reached at the time of observation.

In the opening paragraphs of this chapter two situations were noted as applicable to forecasting, one of these relates to the later part of the season after the fruit or other commercial product is set. Here the important feature is the probability of survival of these fruits up to the time at which they are harvested. Earlier in the season, before the individual fruits are set, reliable indicators of outturn are much harder to obtain.

For the later stages, basically there are three elements in the picture,

a) how many fruits there are,

b) how many will survive to be harvested,

c) what will they weigh.

The way ahead in this work appears to be to pick an important crop and to aim at quantifying these various elements by concentrating on this crop all the available staff in an effort to get as many relevant observations as possible.

The approach has usually been to classify the fruits observed at any particular time of observation into maturity classes. Then the number of fruits surviving at harvest from the ith maturity class can be regarded as given by Pi Ni, where Ni is the number of fruits in the maturity class i at the time of observation and Pi is the probability that fruits in that class will survive to harvest. The total number of fruits harvested will be given by summing this product for all the maturity classes.

To quantify these relationships two sets of observations are needed. One is a count of the number of fruits in the various maturity classes; this requires a definition of maturity class. This matter of definition has produced some difficulties and some short cuts. As noted below, they are fairly easy to define for cotton; they have proved largely irrelevant for oranges; but for maize they have been very difficult to define in a form which is easy for field staff to operate. For a particular crop in a particular locality the consultation between agricultural botanists, husbandry staff, and statistical staff is highly desirable. The estimation of survival probability is made by noting the survival of tagged fruit on sample plots.

There remains the question of weight of fruit per unit at harvest. This may be obtainable from historical averages – average weight of a field crop, ear of grain, cob, etc., may have been recorded for various purposes. If this source cannot be relied on it may be necessary to work out growth relationships from actual observations and sampling of fruits with a view to defining the final weight of fruits which have a known weight at a given stage in the season. Obviously this particular element could be the subject of elaborate enquiries, but in the season or two while exploratory work is going on, there should be little difficulty in arranging to weigh some commercially harvested crops and get a working estimate of average fruit weight. Refinements could then be pushed ahead as opportunity offered.

It is likely that any department undertaking objective plant measurements as the basis of

forecasting will already be engaged on fairly large scale sampling enquiries directed to crop production estimation.[1] Given such field enquiries, the sensible approach would be to add measurements of the crop in the course of growth to the routine observations for acreage and final yield. This extra work can, initially, best be put on teams with slightly easier working conditions or somewhat more competent staff. Using these facilities the basic relationships, probabilities, and mean weights can be explored. Further, some attention should be given to the best way of using available time. Where the work has been done most extensively (in North America) much smaller plots have been used for crop sampling than are recommended here for conditions in developing countries. Consequently it is likely to be better to count the plants per quadrat of the size recommended (plant counts tend to be easy to do) and then to take a small representative sample from the total quadrat, perhaps 20 stems or 2 square metres, for a count of fruits per plant.

The way ahead towards results of substantial interest seems clear. However, it would be rash to be over-optimistic; there are many variables and each technical change will make a change in the relationships. It seems highly important that the need for forecast should be weighed with great thoroughness and if at all possible, the work concentrated on no more than two crops until some light is seen into the best working techniques.

1. This is not absolutely certain because if it is believed to be the main contribution to crop variability, it may be desirable to get some indication of year-to-year variation in the yield at a stage in the evolution of the statistical service where crop sampling for yield estimation may be possible only at five-yearly censuses.

XI STATISTICS OF PERENNIAL CROPS

Introduction

Perennial crops set rather different problems from the annual crops which have been the subject of the preceding chapters. In some ways the situation is simpler since the crops are relatively stable in productive area, so that rapid or frequent surveys for the purpose of estimating area are not required. On the other hand, a number of them present special problems either of identification, or recording of produce – for example some yield fairly steadily all around the year which means that there is never a single occasion when the season's products can be assessed.

For convenience this chapter is concerned with all kinds of crops which occupy the ground for, say, two years or more and with the estimation both of plant numbers (or area) and production.

Perennial crops can be divided for statistical purposes into about four categories.

Estate production

A number of perennial crops,– rubber and coffee are examples in certain countries – are grown in solid blocks on a regular plan under highly organised management systems. The same species may be grown by quite different methods in the same areas, also. Statistics of acreage, production or forecasts of production, of the estate element in a country present no basic difficulty since the office records of the controlling agencies customarily contain such information. The information may not be forthcoming unless statutory requirements are imposed but from the statistician's standpoint the problem is straightforward. It is highly desirable when collecting statistics from this type of producer to make sure that the scope of the enquiries matches the prevailing range of interest in the changes occurring in the sector. For instance it may be less interesting to know details of the production of, say, rubber, than to know whether the enterprise is changing over from rubber to other products, or whether it is ceasing to expand in rubber, mainly because it has begun to channel its expansion on newly cleared land into some other product. This is not, of course, likely to happen with an alert statistical service but there is some risk that, out of custom, the statistics will have concentrated so entirely on the former key crop that the initial indicators of a changed investment policy may pass not, indeed, unnoticed, but unquantified.

Commercial tree crops owned by smallholders.

These may be of the same varieties as those to be found on plantations or they may be of varieties which customarily are not used in estate production. Many coconut groves are operated by small family operators rather than by joint stock companies. There is no basic difficulty in extending the type of sampling programme outlined for crop area and production estimation for annual crops to embrace permanent crops. For example, if the scheme provides for farmers as primary sampling units, then the area of trees of the required crop operated by the farmer can be recorded as if they were annual crops. If the trees are irregularly planted it may make better sense to abandon attempts at area estimation and to count the number of individual trees.

There is a certain range of marginal problems attaching to tree crop estimation by sampling techniques, however. For example the custom in certain countries may provide for tree crops to be held under different systems of tenure from that applying to other uses of the land on which they grow. Consequently a farmer may be chosen as a primary sample

unit and indicate to the enumerator his farm land, but an entirely different set of enquiries might be needed in order to elucidate which trees he has rights in. Moreover, it is at least possible that he may have moved his home but retained rights in trees which he planted in his original home area, or if he may have gone to some distance to find land on which to plant permanent crops (as has been the case of cocoa in some countries in West Africa). These circumstances may result in the field staff having to travel much farther than was expected when the programme was worked out. Perhaps more dangerous to the success of the enquiry is the probability that farmers will be reluctant to admit to having trees at a distance if this would mean a trek across country at a time of year when they do not want to leave home. It is quite possible under those conditions for the crop to be under-estimated in the sampling studies.

Wild economic trees.

In some parts of the world it is customary for the local people to harvest the produce of trees without having any personal and continuing rights in them. In these circumstances it may be of interest to have an estimate of the total number of trees in a locality as a general indication of the production potential. It is likely under these conditions that the proportion of the total fruit which is harvested will vary from season to season (as for instance 'wild' oil palm production in West Africa). Some approach to estimating total numbers in restricted localities can be achieved by walking along bush paths, treating them, in effect, as transects and counting the trees within, say, fifty feet of the path on either side, (the width estimated by eye but checked from time to time by measurement). Other approaches have used linear transects of a chain in width at regular intervals across the area. Typically, a road is used as the baseline, and from points chosen along it transects are traced on a compass bearing. The trees are counted, either completely or in sections at intervals, depending on the resources available and the size of the country to be covered. This approach is really better treated as part of a general economic survey of the commodity concerned than as an element in statistical work. Unless trees are firmly within the economic sector is is probably not worth spending many statistical resources on them.

Any perennial crop lends itself rather better than annual crops to obtaining information from aerial photographs. Solid blocks of planted trees show up clearly on aerial photographs, unless they have been planted under taller shade trees, and estimates can be taken by outlining the block concerned on the prints in chinagraph pencil, and then measuring the area covered by a grid or planimeter. Counts of individual trees scattered through mixed vegetation, if they can be identified without difficulty, are probably best made on a random selection of prints from the whole mosaic. If the trees are close together so that a standard print would involve a very large amount of counting, then randomly sited quadrats or transects seem equally applicable though no very ample literature on the subject appears to have been published for agricultural crops (though there is an extensive literature on forest estimation, of course).

In all records of area or numbers of economic trees, it is essential to distinguish significant age categories. The particular ones relevant to the crop and the locality can be determined in consultation with the technical staff, but, broadly, it is important to distinguish trees that are different in respect of their bearing capacity rather than to distiguish age for age's sake. Those too young to bear should be distinguished from the others, trees which have begun to bear but which are not yet in full bearing, and full bearing trees. It may also be necessary to distinguish those that are, in fact, broken down and scarcely fruiting. These distinctions are essential if reasonable production estimating is to be obtained.

Production estimation

Basic principles of estimating production do not differ from those described for annual crops. In some situations judgement reports are reasonably satisfactory and various techniques of crop sampling have been tested. It is only fair to state that it has proved very difficult to handle some tree crop estimating and forecasting procedures satisfactorily.

A basic system would proceed somewhat as follows. Starting with the farmer as primary sampling unit, his plantation would be surveyed and two quadrats, each of an area which on average would contain about ten trees, sited. The trees are then recorded one by one with an assessment of age, a strip of bark or palm frond being tied round each as it is counted to avoid double counting. Two trees in each quadrat might then be selected at random and the primary branches of each counted and recorded. One branch from each tree is then selected at random and the fruit on that branch picked and weighed. The successive sampling fractions are then applied to the records of fruit weight for the individual branches to derive an estimate of the total production for the geographical unit desired.

A very similar process can be used in the early part of the year after the fruit has formed, to obtain a count of the number of fruits on the branch. If they are small (for example, coffee) a feasible procedure is to strip the fruits into a container and weigh them; a portion of two or three kilos is weighed and the fruits counted. The ratio from the weighed sample provides a conversion factor from fruit weight to fruit numbers which can be applied to the tree yield. These records might need to be made separately for fruits of each maturity stage as described in the previous chapter. If it can be assumed that they will all survive to harvest then all that is required is an estimate of the mean fruit weight at harvest which might be obtained from tests made for experimental purposes in earlier years. Such results should be re-checked by observations on fruit from sample farms as soon as possible.

It may not always be feasible to destroy the produce of whole branches of a producer's fruit harvest in the early part of the season without loss of goodwill, even if he is paid for his loss. If an attempt is made to avoid this by counting fruits per branch without picking them a very rigorous programme of testing should be built into the enquiry. Even with quite good working conditions and large fruits it has been found very difficult to get consistency between observers in counting the same fruit on the same branch; this appears to be a line of work where there is much still to be discovered.

Suggestions for useful approaches to the handling of crop estimation in smallholdings may be obtainable from a group of enquiries made in recent years in U.A.R. The details there varied with the crop, and they will no doubt vary from place to place for the same crop, so great detail of description seems unnecessary. Broadly, the enquiries started with records of plantations containing the economic trees concerned with an indication of the number of trees, the varieties and the ages. However, the circumstances in which these details were compiled meant that they could not be regarded as accurate. Hence this group of studies also sought to provide an improved estimate of the number of trees as well as of the yield per tree. The location of the plantations was ascertained from local records or with local help and they were listed in sequence according to the lie of the land. They were then grouped to give primary sampling units of rather similar size—the sizes of the individual plantations varied considerably. In one study such units included 300 date palms, in another about 2,000 citrus, and in another about 800 banana hills. (If there is any degree of standardisation in planting it might be reasonable to use a more-or-less standard area of trees, in place of numbers of trees, for some kinds of basic record.) Five to seven percent of these primary sampling units were then chosen at random for detailed recording. The trees on each were recorded on a map of the unit concerned, the position of each tree being indicated, and the characteristics, such as variety, age, and status (in the case of trees thrown up as suckers from other trees) were indicated. This information provided the basis for an accurate assessment of the numbers of each variety and age group, using the sampling fraction for raising. The numbers of trees in the several selected units were then totalled cumulatively and 5 trees were chosen at random within each primary sampling unit. Associated with each of these were the two adjacent trees, making a total of 15 trees for the unit (i.e. 5 groups of 3). These groups were then visited at harvest and the crop weighed. Variants of this method were necessary if, as in the case of bananas, the product is not harvested at a specific time but may be cropped more or less evenly through the year. In this case sufficient time was spent at the sites concerned on a sufficiency of occasions to note the total number of bunches which had already reached maturity in the course of that season (as judged by the cut-off stalks), as well as those which were in course of growth or were being harvested at

the time of the recorder's visit. Where possible the field observers tried to visit the plantation on the days when the merchants arrived to harvest the crop.

The fact that the field records consisted of sketch maps showing individual trees made such sequenced recording feasible. However, the system may be difficult to operate successfully without fairly ample labour for field work.

In studies of coconuts various approaches have been used, depending on the data available to form a frame and the size of the holdings. When units of fairly small size can be identified for use as primary sampling units (with necessary data for raising) all the trees on the units (say, smallholdings) can be numbered serially and a random sample taken from the whole list. About four trees per smallholding has seemed useful for yield observations. Some gain in convenience without appreciable loss in precision may come from choosing only one tree per holding and associating with it the next three in serial order.

For large blocks of planting a similar system of sampling from a serially numbered list of trees might be developed provided an intermediate stage were inserted in which units of reasonably manageable size were selected from within the total block. If the scale of production precluded this then a means of estimating the density of trees per acre would be needed to allow records of the areas under trees to be converted to numbers of trees. Sample quadrats of 40 yds x 40 yds appear to be suitable for this.

If planting is in regular rows, a tree can be chosen by reference to a random choice of row and of tree within the row. Such a tree can then be made a defined corner (e.g. the S. West) of the 40 yds x 40 yds quadrat. For yield records this designated tree and the three others next nearest within the plot may be used.

For cocoa, techniques seem to have been worked out in the main areas but, broadly speaking, the practices already outlined above are applicable under such conditions. The plot size used appears to be about 18 trees per farmer. Complete pod count is usually made of all the pods at intervals through the season–in September, November, April and May, for instance, under categories of "fermentable" and "unfermentable". Later, the pods actually harvested are recorded.

Which techniques will be most suitable and what sampling fraction to use at the various stages depend so greatly on local conditions and the availability of background data that each design must be matched to the particular case. Arrangements must usually be made to provide for recording yields over a period of several months since the harvest may not be concentrated at one point in the year. It is usually necessary also to arrange for records to be made of annual crop planting amongst tree crops, and also of grubbing and re-planting activities on the trees. These are probably best handled by pursuing the enquiries for a sub-sample of the chosen primary sampling units. Perhaps the greatest comfort to the statistician in dealing with tree crops is that the productive area–except following boom planting times–tends to change rather slowly, so his records, once made, stay up to date for some while.

A possible technique for forecasting tree crop production before harvest where the outturn is ultimately known depends on this rather slow change in the productive area. Where there is a body of observant and intelligent field staff (normally, men with the basic training and experience with views of their own) they can be asked to state when the crop is set what proportion of a normal crop they think is standing that year. Some systems refer to a 'par' crop and ask the reporters for an assessment based on a 'par' crop as 100. If the consensus is 120 compared with "par = 100" and if it later becomes known that 6000 tons was marketed one might reasonably conclude that a par crop was $6000 \div 120 \times 100 = 5000$ tons. If the reporting is well founded and no abrupt changes in planting or grubbing occur, then the estimates of a 'par' crop from a series of years should show some consistency–possibly tending upwards by a few percentage points each year in reflection of tree growth.

If such consistency is found it provides the basis for a forecasting system; if it is not found then it should clearly indicate to the field staff that their judgement is of doubtful value and clear the ground for an attack on the problem using objective methods.

(For detailed work on tree crops see F.A.O. publications listed, also U.S.D.A. Crop and Livestock Reporting Service Handbook.)

XII STATISTICS OF MIXED CROPS & OTHER SPECIAL CASES

Introduction

So far, various problems of defining boundaries of crop areas have been referred to but it has been tacitly assumed that there is never any doubt about the definition of the crops which lie within the defined boundaries. However, in many countries this is far from being the case; local practices in the growing of food crops and even of cash crops may include mixing them on the same land surface. This may be found in a variety of forms, some of them representing crops coincidental in time, some successional.

The simplest situation is that where a parcel of land is planted in a number of portions, each under a different pure crop. Thus, one field might be divided into four–one part planted with maize, one with millet, one with groundnuts, and one with cassava. Often, however, the mixing is done by interplanting the various crops either by alternating rows or by a scattering of the plants of the one crop among the plants of another. Sometimes these mixtures consist of one crop which is quite clearly dominant with a few scattered plants of another. For example, in some places women have the established right of planting a thin scatter of cotton plants in the main cropped area, but the density per unit area is by custom kept very low. Other examples are fairly simple mixtures of two or three crops which are part of the regular and established pattern of cropping in the neighbourhood. The individual plants are scattered but the relative importance of the various crops making up the mixture is more or less constant from farmer to farmer and field to field. There are situations, too, where there is a great number of mixtures but experts can yet recognise them as having some individual identity. Finally there are cases where the whole pattern of cropping has broken down and odd attempts at planting various crops are scattered throughout the cropped land in more or less complete confusion.

One might add to this the various kinds of successional cropping which are discussed below. These may bring in not only mixtures of annual crops but mixtures of annuals and perennials.

Statistics of mixed crops is probably the most discussed subject in this whole field of agricultural statistics in developing countries, but no satisfactory solution has emerged that is both technically satisfactory and practically feasible.

The existence of the problem indicates that there is uncertainty as to exactly what is to be measured by statistics of crop area. Tacitly it is assumed that in the various respects in which users are interested one hectare of crop A in one locality is sufficiently like one hectare of crop A in another locality for comparisons of the two or aggregates of figures from the two places to have some meaning. Clearly in the present context there is a question whether meaningful results can be obtained by aggregating area statistics from pure stands and from mixed crops.

Approach to measurement

The various types of crops briefly noted above present different problems to the agricultural statistician. First, the situation where a parcel of land is divided into portions, each of which is planted to pure stand, presents very little difficulty beyond the extra time needed in the field to complete the more detailed records. There is some possibility of doubt about the correct way of drawing the boundary between crops of rather different growth and cropping patterns, but this is not usually a matter of very great significance. For example, grains might be planted broadcast at intervals of a few inches, whereas maize might be planted in rows at distances of two or three feet. It is usually possible to regard the part

plot under maize, to take this example, as ending at a distance outside the last row equal to half the inter-row distance. If it is not planted in regular rows then half the interplanting distance beyond the last plants might be a reasonable limit at which to draw the assumed sub-plot boundary.

The cases where there is a single main crop with a very small scattering of secondary crop can probably, for practical purposes, be regarded as a pure stand of the principal crop and be entered accordingly.

The rest, namely those where the individual plants of the several species concerned are scattered between each other, admit no such straightforward solution. One possible approach might be to regard each component of the crop as occupying the whole area. In effect this is saying that a mixture of maize, groundnuts, and cassava is not fundamentally different from a mixture of maize and weeds from the standpoint of statistical reporting. The maize and weed crop would be regarded without compunction for statistical purposes as being "maize", notwithstanding the competition of the weeds with the crop. Groundnuts and cassava could be regarded as merely weed-like competitors for maize, and similarly for the others. The total area under crop, if the statistics are handled in this way, would naturally be treble the total superficial area of the land, but this effect must always be expected where more than one crop is grown in the course of a season, whether they follow one after another, whether they partially overlap each other, or whether they are mixed, planted and harvested more or less at the same time. Again, an attempt might be made to divide the area under the mixed crop between the individual crops concerned on some concept of the "proportion of the area occupied" by the respective crops. This demands rather complex perception on the part of the observer. What he can assess on the ground is likely to be either the numbers of plants of the respective crops per unit area, or the proportion of the ground surface covered by the plants of the respective crops. In some instances, the proportionate coverage is something that might, with practice, be judged without too much difficulty. However, the plants in some mixtures differ considerably in height and the same piece of ground surface may be covered by two or more sets of foliage. If we remember also that the observer may well try to take into account the appearance of pure stands of the respective crops in making his mental calculations for sub-dividing the superficial area between the various components of the mixed crop, it is clear that the principles on which such a division can be made are likely to be very difficult to standardise.

Another approach might be to convert the plants on any given superficial area of the mixed crop to their equivalent in pure crop. This again reguires an appraisal of density of plants in a reference pure stand and in the mixed stand. Moreover, it requires some tacit allowance for the possibility of competition between the various components. For instance, if the same number of plants of crop A appear in a mixture as in a pure crop, the plants of A presumably are no more competitive with each other in the mixture than in the pure stand and the same would appear to hold for B. But to what extent does A compete with B in the mixture and hence to what extent must correction be applied to the acreage of the mixture in order to convert it to the "equivalent area" of crop A? Some reporting services have met this problem by not probing too deeply into the conceptual difficulties noted here, leaving their field staff with instructions to sub-divide the land by judgement. Others have subdivided the mixed acreage between the components, or calculated pure stand equivalent, on the basis of seed rates.

All these methods appear to be building a reporting service with dubious bases of assessment. This is so even if in each locality there is a pure stand of each crop which can form a basis of comparison – in many areas, one would need to travel a long distance before finding anyone planting certain crops in pure stand, so local reporters could have no experience of this. It is arguable that it should possible to standardise some of the elements of judgement criticised here by using sampling techniques to form estimates of plant density. This is probably true, but assessment of area and of yield per unit area places a heavier strain on field services than they can bear in most countries, and the addition of a density measuring stage would increase the strain still further.

The most satisfactory way ahead for these mixtures where the individual plants are

scattered would seem to be to record the areas concerned as being under the various specified crop mixtures. It may be possible to describe these in terms of principal crops and secondary crops or it may be possible merely to refer to them as including the several crops. This should give some indication of the cropped land-use which has bearing on the description of agricultural practice and, for this purpose, is likely to be more revealing than any of the derived estimates of "area equivalent" which are referred to above. The straightforward cases where a crop is in virtually pure stand, or where the plot is broken into sub-plots with pure crops, would be handled in the manner suggested above.

It would then be at the production stage that an attempt would be made to supply, on the one hand, estimates of the area of the crop grown in mixtures which could be associated with the part grown in pure stand where this occurred, and on the other estimates of yield per unit of superficial area. The estimation of yield per unit area for mixed crops is, in principle, practicable using the crops sampling techniques described here, but there appears to be no experience of continued successful handling by judgement methods. The approach using sampling requires the normal procedure of selection in one or more stages to give a final list of primary sampling units, such as farmers. The area of each field operated by selected farmers is recorded at a suitable point in the growing season and these field records stratified on the basis of the crop mixtures indicated therein. Sampling for crop cutting then proceeds within these strata, but otherwise in the normal way. At harvest the plots are cut either crop by crop or the ground is cleared and the material then sorted, the economic parts separated and weighed, and provision made for drying samples as required. It has to be recognised, however, that the presence of a number of crops on the same piece of land greatly complicates the already awkward step of getting the enumerator on the spot at the right time. It may well be that enumerators must be brought back to the field as many times as there are crops. If we bear in mind that some crops (e.g. cotton) must individually be harvested on more than one occasion, the burden of this kind of work is clear. Nonetheless, there appears to be no short cut and where the need is pressing appropriate resources will have to be made available.

This seems to be a case, as for livestock, where considerable care should be applied in assessing the exact needs of statistical material. Some of this work would be easier to handle by resident reporters conducting farm record enquiries. It might be best to arrange for a country-wide sampling scheme to get a pattern of land use for mixed crops and to depend on records from a much smaller, and perhaps scarcely representative set of closely recorded farms to provide interpretative yield data to amplify the area record.

In conclusion, it might be noted that if it is desired to have available some area equivalent for the crops grown in mixed stand, a rational one could be derived from estimates of yield of pure stand and by dividing the production of a crop in all kinds of mixtures by the average yield, in pure stand, a "pure stand equivalent" of the mixture can be derived.

Successional Cropping.

It is common practice where the climate permits rapid growth for more than one crop to be produced within twelve months. If the practice is simply to plant one crop, wait for its maturity, harvest it, then plant another crop and waits for its maturity, no specific statistical problem arises. It is necessary to arrange sufficient visits to the chosen sample areas to allow the necessary records to be made but, this apart, the work is not particularly difficult. Other examples of successional cropping may present their own problems. For instance, where a piece of land is cropped with annual crops but has, for example, bananas and citrus plantings added, it is likely that it will remain as an area of interest for the farmer in respect of these "tree" crops even if the area has reverted more or less to bush as far as annual cropping is concerned. Here, too, the principle may cause no trouble although the cost in time of keeping track of this kind of marginal production activity may be heavy. This problem may reach its gravest form in some tropical areas without very much variation in growing season, where the agricultural practice provides for a more or less amoeboid growth of cropping. Thus Ugandan practice in some parts might begin with a planting of bananas which was split,

one part of the plot a few weeks later being planted to beans, maize, potatoes, cassava, and the rest left in bananas only. The banana portion might then be combined with a portion cleared somewhat later and interplanted with a mixture similar to the first. A little later an additional plot of land might be cleared adjacent to another of these parcels, and a further crop planted extending across both of them As the months pass, crops will be harvested when they come to maturity and the land replanted with an annual if the spacing of the bananas permits it, or it might be left as a pure banana garden.

One can only say that under conditions such as these it is extremely difficult even with intensive concentration of field staff to keep a track of the progress of cropping and the extent of cropped area over a twelve-months season. It is inconceivable that any economical method of handling the problem could be devised, and the best approach appears to be to seek from larger scale enquiries an estimate of the total cropped area in use by the farmers of the localities concerned and then, using intensive surveys of small samples, to build up a picture of the detailed cropped land use and the consequences in terms of output.

Incompletely harvested crops.

Crops in all types of agriculture may remain unharvested, sometimes for technical reasons such as the land being too wet to allow access, sometimes for economic reasons such as failure of the market for the produce and the consequent lack of incentive to harvest it. What might be regarded as a special case of this "economic" cause of incomplete harvesting is found when more of a reserve crop is planted than farmers find they need for subsistance use. The classic case of this is cassava in West Africa.

Cassava is grown in some situations as a cash crop and the ground is completely cleared when it reaches maturity, which may be from nine months after planting onwards. However, it may also be planted as a reserve crop – it is under shifting cultivation often the last crop in a cycle before the land is allowed to go back to bush. Under these conditions the usual practice is to harvest what is required, choosing the younger plants which are likely to be more palatable, and allow the rest to go to waste. Somewhat the same situation may hold for some tree crops – for example, plantains.

There are, of course, two different possible definitions of production in such circumstances. Actual production, as represented by the product harvested and used, and potential production as represented by the total crop whether harvested or not. It should be noted that, whereas the former is a resonably clear-cut concept the latter is much less so since the weight of produce which is regarded as being available will depend on the date at which the assessment is made. As one passes from the early days of the growth of the tuber the weight available will increase until a stage is reached where parts must be regarded as inedible even under conditions of grave food shortage. After a certain point these must be written off for food purposes and the quantity effectively available will begin, perhaps, to fall unless new growth has taken over by then.

This "cassava problem" is another which has attracted a great deal of discussion and cannot be said to have been solved. Some progress in any particular situation would be represented by agreement on which of these definitions it is desirable to use. Production for market for which the crop is completely cleared presents no more difficulty than the production of any other cash crop. If potential production is required then no special difficulty, other than the work involved, is presented by a sampling scheme undertaken at a suitable period after the planting of the crop. This might be towards the end of the twelve month period in the case of cassava in many situations but varieties vary in the periods in which they come to maturity. If actual production is required the problem may be of a different order. Usually it is not possible to estimate harvested area because users pick a root here and a root there from amongst the growing crop as seems best to them. Under these conditions the only reasonable approach is to include the production of the cassava within the recording programme for a farm output survey of the kind outlined in the following chapter.

Though in the context of developing countries cassava provides the commonest example of incomplete harvesting it may also be found in cash-crops when market conditions are too

discouraging to justify further harvesting towards the end of the season. The remaining crop may be ploughed in, fed to livestock or otherwise disposed of. If this kind of incomplete harvesting is a significant feature estimates can probably be provided by reporters who are in touch with growers. Reports are usually made in terms of "proportion of the crop unharvested" or "proportion of the area unharvested" as may seem appropriate in view of harvesting custom. In practice, only a rather broad indicator of abandonment may be possible. One set of problems arises from uncertainties about definitions when growers pick over the best from their crop and leave some crop on all plants. It must be expected, too, that there will be a tendency to exaggerate abandonment of crop in the atmosphere of disappointment of a bad season.

XIII LIVESTOCK NUMBERS AND PRODUCTION

Introduction

Collection of livestock statistics in developing countries is usually either a reasonably straightforward process or an intensely difficult one. Among the few reasonably straightforward situations are first, those where there is a body of fairly large farmers, with well organised and managed herds, who can supply information as required from their own records, and second, those cases where livestock are an integrated part of small-farm production systems where a sample survey of farms can be designed to provide the required material. However, in many countries the majority of livestock is to be found in the nomadic sector, often where the economic use is secondary to the social value of the stock, or in small herds associated to some degree with farms but often grazed at a distance from the farming headquarters. This being so, it seems necessary to proceed with livestock statistical work with a good deal of circumspection; probably the following three stages provide the best way ahead.

The various requests for statistics of livestock should be scrutinized with great care to assess the possibility of meeting the threshhold level of completeness and precision necessary to meet the need. This may require long and patient consultation with potential users; veterinary, agricultural and statistical departments may all be concerned. This process should include the drafting of the specific questions which the statistics are expected to answer and preparing an explicit statement on whether the data in prospect would, in fact, supply this guidance if they were available. For example, if their value for assessing overstocking problems was the basis of a request for data on livestock numbers, would a knowledge of current livestock numbers alone advance the understanding of evolving conditions? Is the need perhaps, for an indication of the *trend* in livestock numbers, rather than for an estimate of their absolute numbers? Again, would statistics of population be much guide to potential market supply if the annual output of stock for sale is no more than 5–10 per cent of the population? Or again, is it possible to get any guidance about the significance of the contribution of livestock products to human nutrition without a good indication of the distribution of the product between sectors of the population, as well as an idea of the total supply? It would seem that only a nutritional survey would throw much light on the nutritional aspects of meat supply in most instances. If the answers to questions like these are specifically 'yes', then some attempt along the lines indicated below should be made to operate a sample enquiry to obtain the necessary information.

Low cost methods

If is found that the information is either not urgently needed or if available, it is unlikely to provide the kind of guidance which is sought, then it seems best to get together the best guidance which is to be obtained from available statistical material supplemented by relatively cheap exploratory enquiries which give qualitative guidance. There is usually a certain body of data already to hand in most countries including regular reports from the Veterinary Departments which are compiled in the course of field veterinary control work; data from taxation records where livestock taxes are in use: the results of various *ad hoc* enquiries: the statistics of imports and exports of stock across land boundaries or through ports, or their movement over recognised routes; etc., etc. Attempts should be made to list all the markets through which livestock are sold, and, for as many as possible, to estimate the numbers passing through at different seasons of the year. If numbers and quantitative data are impossible to obtain the main pattern of seasonal variations should be sought

together with reports of the origin of the stock, their destinations, some idea of their weight, and of the prices at which they change hands. Some idea is also desirable of the organisation of the trade, the number of traders involved, their sizes, relationships and so on. In the large towns there are usually slaughterhouses whose operations are closely controlled on hygenic grounds, and numbers slaughtered here may be available. It may also be found that in the regional offices of the civil administration, the agricultural department, or the veterinary department, there are appreciably larger numbers of useful statistics than are to be found at head office. However, it must be said that many of the available livestock statistics appear in great detail but, when checked are incomplete to a substantial degree or conflict with each other. No doubt in every country there are exceptions to this but it is a situation which it would be wise to expect. Consequently, such staff as can be spared for specific enquiries under these conditions might well be put on to operating check surveys in a selection of situations which are planned to throw light on the degree of deviation of the departmental statistics from the true situation. The villages or herds which might be covered in such enquiries might well not be a proper representative sample of the whole but the results should be a guide to the likely errors. Local opinions about such statistics should be treated with reserve until objective comparisons can be made. Usually, too, information which will allow a reasonably accurate indication of livestock output to be derived from the outline statistics available is also lacking. Typically, there is a tendency to impute to the herds and flocks of an undeveloped system a level of production a good deal higher than is likely to be found in practice. Consequently it is desirable to get an indication of the main features of herd structure (for example the proportion of cows to stock of various ages, etc.) and also of the number of births per hundred head of breeding stock and the death rate in the first year of life. Cattle are mainly in the main in mind here, but comparable information is relevant for sheep and pigs also. It may not be necessary to undertake a time-consuming enquiry to elucidate the main levels of these several items if available experience is tapped and the indications cross checked.

When assembled, this material should add up to a serviceable indication of the main features of the livestock industry and may be sufficient for many purposes. To improve greatly upon it is likely to demand very heavy expenditure of statistical resources. It is at least possible and, some would say likely, that when the country's livestock owners who have previously had little contact with an economic system move into the market, then the prospect of collecting statistics will improve. This improvement may come in part from greater interest on the part of the producers and in part from greater preparedness of governments to vote funds for the study of a sector which is currently of increasing interest.

Sampling enquiries

So much for low cost assembly of statistical material. Where it is decided that some direct and positive collection process should be undertaken, the way ahead which promises to be most likely to be successful is outlined below. The history of work in this field suggests that in any but straightforward cases, where the herds are part of the regular system of a settled agriculture, the propsects of initial success are small.

The frames and sampling units available for formal sampling enquiries depend on the administrative and special enquiry records already available. For livestock which are part of a mixed farming system any of the frames already discussed as serviceable for locating farms should be suitable for livestock enquiries. In this connection, it seems necessary to say that while there is no reason in principle why livestock statistics should not be collected as part of a multi-purpose subject enquiry, it is highly probable that this will not be successful. The sort of attention which is required to obtain good livestock statistics is somewhat different from that required for crops. For example, the livestock, although associated with the farm quite closely, may be grazed away from the farm and the enumerator must be prepared to persist until he actually sees them. If they are grazed at a distance of several miles it is unlikely that his persistance will reach that far. Possibly, too, if he has had no personal acquaintance with livestock he may look on this part of his job with little enthusiasm anyway. Consequently though the frame may be the same as the one used for such crop

enquiries as have been described already, it is preferable to mount a separate study (possibly with selected enumerators) – or at least a separate passage – for the purpose of collecting livestock statistics. Since there is a very tempting argument for using the same travelling time to collect as much information as possible on each visit, these problems of livestock recording must be stressed if they are to be given the right weight when enquiries are being planned.

In this kind of enquiry rather special care is needed to watch for "landless holdings". This may have different specific meanings in different places, but quite often there are producers who are holders of livestock but do not crop land and the need to include them in the frame either together with the crop producers or in a separate section of it must be stressed. Frames for specifically livestock enquiries are often derived from one or other of the veterinary records. These may list villages with the livestock in each, kraals or their equivalent, or individual stock holders. A typical enquiry might proceed by stratifying the villages by size as indicated by recorded numbers of stock; within villages herds might be stratified by size. Some enquiries have then included about ten percent of the stock in the final tabulation. It is possible that numerous variants on the most common pattern of livestock holder will be found on enquiry. Thus there may be regular stock owners located in villages, all with more or less modest herds, possibly other stockholders scattered over the countryside with similar sized herds, and still others with massive herds. In the latter case, it might be desirable to make as complete an enumeration as possible of this large herd sector while sampling for the rest.

It has been found on many occasions that initial enquiries by enumerators collecting livestock statistics are only partially successful, less so even than for crop statistics. Consequently, there is much to be said for so organising the enquiry that the stock owners have a number of opportunities of contact with the enumerator so that, gradually, the full total connected with their holdings may become apparent. It might be possible to arrange this either by including the livestock work amongst the responsibilities of a resident enumerator engaged on a range of statistical work – though being careful to guard against the risks already noted in multi-subject enquiries which have already been noted – or by alloting enumerators to livestock work for a period and getting them to collect a fair range of inter-related information. This latter method is probably preferable if the circumstances and staff make it feasible. In this case it might be arranged that they visit each holder once a month, recording the livestock he holds at each date. The enumerators would move on to other villages in the interval. They should be instructed to see any livestock that are to be reported, though this may be, initially, a council of perfection. Additions to the herd, births, purchases, amalgamations of herds, etc. and also numbers leaving the herd by slaughter, sale, or death should be recorded. Particular attention should be given to the death rate for young stock.

It must be recognised that, while a monthly visit may be as frequent as available staffing, and the patience of the livestock owner, permits it probably is too infrequent for reliable working. If a two week cycle could be arranged it would be preferable in many instances; circumstances differ but it is usually found that a month is rather a long time for accurate memory. Even in developed countries where farmers may be vividly conscious of the identity of individual days and dates in the month, it has been found that enquiries which require them to remember the dates on which young stock were born or sows farrowed up to a month previously are subject to substantial "recall" error.

An intelligent enumerator with capacity to get on with livestock owners is likely to be able to progressively bring his records up to improved states of completeness. By noting any discrepancies between the totals he had on the previous note and the reported ingoings and outgoings, it should be possible to frame a line of conversation (rather than specific cross-questioning) which will elucidate where the explanation lies. Each type of information then acts as a check on the other and both together promote completeness. Of course it must be remembered that livestock owners tend to be independent people, even if they are not superstituous about counting their stock, and may be reluctant cooperators in statistical studies.

The categorisation of stock for these enquiries may need to be adjusted somewhat to match the standard statistical returns required by national agencies under their commitments to international statistical services, and the local needs. Generally, it is desirable to distinguish between unimproved stock and improved stock, and amongst the improved, between pure stock and graded-up stock. Usually in any particular situation the difference between improved and unimproved is well recognised locally. It will also usually be desirable to provide for separate figures to be collected about elements in the total population which are directly indicative of output. For example, where milk is the main product the number of lactating cows at different times of the year should be separately indicated. For meat stock, which are fairly completely within the economic sector, the age classification is usually as sound an indicator as is likely to be obtained. Given the age categories, the survival rate of young born, and the numbers of stock sold or otherwise disposed of, a population flow chart can be derived which gives valuable insight into the overall working of the system, and provides cross-checks statistics one against another, and provides serviceable pointers to the level of output.

Nomadic stock

The above enquiries point the way to compiling serviceable statistics, which one may hope will progressively improve with experience of this work on the part of both the producers and the statisticians. However, these methods have no practical application to truly nomadic livestock. In principle, it might be possible to deal with them by a sampling procedure, or an enumeration procedure, based on water holes, dipping gates, etc., but in practice schemes of this kind that have extended to more than a small fraction of the country, have not provided reliable results. Usually the veterinary department has certain broad indications of the numbers of stock concerned, and for most purposes this is probably all that it is worthwhile seeking to handle. There is the possibility of obtaining estimates of livestock by aerial counts, either by photographing or by personal counting. This technique has been used fairly extensively by ecologists interested in the large herbivores. For instance, a slow aircraft operating at, perhaps, 80 knots can be used to count livestock in selected areas, or in transects across larger portions of the country. It appears that the actual counting does not present great problems unless very large herds are concerned and here they can be photographed and counted at leisure. In principle, such photographs could be used for deriving some age categorisation given the necessary conversion factors and from linear dimensions on the photograph to age of beast. Clearly this would be a great deal easier to do if the stock were distributed more or less evenly across the country, but any substantial concentration at water holes, or under solid blocks of trees, might make reliable counting much more difficult. It would seem from reports of ecological studies using this technique that the method is applicable; the question is whether in any particular situation the interest in the statistics is acute enough; if it is,then it may be possible to secure the funds necessary to contract with aircraft operators for the work.

The above discussion concerns livestock numbers with only incidental references to production of live animals ready for slaughter, or milk, eggs, etc., for market. It must be recognised that even in highly organised agricultures in developed countries, the estimation of livestock and livestock product production is the least elegantly organised of agriculture statistics. Usually, the information cannot be obtained by methods parallel to those used for crop production estimation, and where satisfactory data are available, they are almost invariably obtained by returns from the purchasers of stock at bottlenecks such as markets, slaughterhouses, or milk processing plants. Since the importance of centralised services of this kind for livestock in developing countries is usually slight, the proportion of the total output which can be counted in this way is small. Typically, statistics of numbers of stocks slaughtered in urban slaughter-houses are available under hygiene and sanitary regulations. The milk handled in milk processing plants can usually be estimated from the records of plants concerned; because of the cost of the equipment they tend to be reasonably organised. With the arrival of "industrial" systems of poultry meat and egg production, using hybrid stock, it might be possible to obtain indications of the usage of birds for this

purpose by records of the numbers of chicks imported or hatched locally for broiler and egg production respectively. The numbers of hides have customarily been used as an indication of the output of meat and this practice is still to be found. It is, perhaps, better than nothing but hides can move a long distance and some little research on the hide trade, marketing, local usage, etc., may be needed before any calculations are based upon them with even mild assurance. Some care should also be given to the possibility that hides may be recovered from beasts which are not used for food either because they are too far gone when found to be used for humanfood, or because of religious or social prohibitions. The derivation of estimates of numbers of stocks leaving herds, has been referred to already under the classification of "information to be collected". It remains here only to stress the need to approach this with as many cross-checks as possible. Some methods are extremely sensitive to the precision with which a single conversion factor is known – for instance if the cattle output is calculated as either a percentage of the total herd or on a basis of the numbers of calves born alive per hundred cows, the reliability of the result depends entirely on the reliability of these conversion factors. Often experienced observers have a personal view of what these should be but often when the total figures are known they depart from the expert view by a substantial amount. One further conversion factor is usually needed before population figures can be converted to meat production and that is the average yield of meat per beast. Enquiry and weighing at a central slaughter-house may give some useful guidance on this but two points should be kept in mind. One is that the stock slaughtered outside the main towns may be of different sizes and different varieties from those in the centres of population, and their weight may differ accordingly. The second point is that practices may vary considerably in respect of the parts of the body which are used for human use. This may vary in several directions – parts might be used, even specially valued, which are rejected in, say, European practice. On the other hand, some parts may be rejected for local reasons which are regularly used elsewhere in the world. Presumably the interest in most cases is in the supply of edible material for the local population, although on occasion the possible supply of flesh as defined internationally may be the main interest.

These are not very satisfying proposals for handling livestock statistics but to judge from reports of statistical work in this field in various developing countries, satisfaction is rare. One point might be made in conclusion. Where specific development plans are underway for livestock production, it is very desirable that the provision of adequate records should be built into the development programme – and for this purpose the European style statistics would often be a useful pattern.

XIV FARM OPERATION AND BUDGETARY RECORDING

Introduction

In recent years it has come to be recognised that certain data which are essential for general economic planning, for evaluation of projects, as well as for the work of advisers and instructors seeking to help farmers, can only be obtained by rather intensive recording of operations on as large a sample of farms as can be accommodated. In particular, information is needed about the relationship between resource inputs (for example, inputs of labour and fertiliser) and product outputs. Involved in these assessments are clear and reasonably quantitative descriptions of farm operations in terms of the work actually done, limitations which farmers meet in their management, the results which they obtain, etc. There is no reason, in principle, why this kind of information cannot be obtained using a rigorously selected sample of farms, though, in practice, the cost in enumerator time may prevent the sample being a large one and the need for some consistancy of cooperation may make some selection other than of a representative nature unavoidable. It is probable that rather different information will be needed for national planning, regional planning, and individual farm management purposes, but the following notes take a general-purpose line embracing something of all of them.

Frames

The normal frame such as is used for enquiries into crop acreage, or livestock numbers, is suitable for this work provided that farms can be identified. Typically, farms selected for intensive enquiries of this kind might be a subsample of the sample drawn for acreage estimation purposes. Where a major cash crop is part of the farming system, it has been found useful to stratify the farms which are included in the sample by reference to their area of cash crops, and to draw the sample for the farm records from within these strata. If cash crop area were not available for a large sample of farms (e.g. if a sample were drawn specifically for a "management type" enquiry) the necessary data might be obtained from a pre-enumeration survey of limited scope.

Scope

The detailed content of the enquiry in a particular case may reflect the active interests of the potential users, but in many instances the information sought will include the following, – basic areas of the main crops both cash and subsistence; numbers of livestock, both on the farm and held elsewhere; land held but not cropped and land to which access for grazing is permitted; inputs of farm resources, including labour, fertilisers, sprays and any other inputs that are relevant in the situation; output of farm products, not only for cash but for household use in the farm family and to provide food and drink for labour; receipts in cash from the sale of farm products and for activities outside the farm and remittances from members of the family at a distance; expenditures on farm and household purchases; savings; equipment owned and capital formation generally. It is to be expected that for a number of these items records may be scanty but workers in a number of situations have found that with patience useful guidance can be obtained on all of them.

Approach to recording and farmer contact.

Labour is used on farms on most days of the year, or at least over a substantial part of the season, and there may be numerous occasions when products are sold and purchases are

made. Consequently reliable records of this kind of information cannot usually be obtained by enquiry on a single occasion in the course of the year. Frequent recording is needed, possibly more or less continuously. It cannot be said that there is clear agreement between workers on the most effective approach to this work. There are reports of enquirers who have sought to collect data relating to the whole of a year's use of labour on the farm by questioning the farmer on a single occasion; others have visited the farm once a month; others once a week; still others insist that only by detailed day-by-day, diary-type, recording is it possible to be sure that the data are reliable. In part, these differences no doubt reflect differences in the operating situation, for example if a farmer sells only on a single occasion in the year and then to a cooperative society, it is quite possible that he has documentary evidence to show how much his receipts were for that year. Even if he has no record, with his permission the society is likely to be able to provide one. No doubt there is a certain amount of difference between farmers in different localities in their ability to remember or in their possession of records which will provide the basic information needed by the enquirer. Nonetheless, those who are coming to this kind of work for the first time should give great weight to the view of those who say that only by careful recording will true data emerge. For example, in obtaining information about how much time was spent by workers, paid or family, in producing a particular crop, a complete record of what everyone was doing who had any concern with that farm, each working day, is desirable. It has been found by some workers that it is not sufficient to record what time the farmer or his helpers said they spent on work on a particular field. If the enquiry is pushed on to ask how else they spent their day, the figures which would otherwise have been entered for the farm operations are likely to be changed. Only when the farm data are available as part of a complete record of the use of time will a reasonably true figure for labour input in the farming operations be obtained. Having said this it must be recognised that the survey director is likely to find himself forced away from this ideal by the objection of the farmers to this degree of cross-questioning. Nonetheless, taking one country's experience with another, it seems highly desirable that at least a selection of farmers on a selection of days should be treated in this way so that the enumerators and the enquiry directors learn the kind of uncertainties about labour recording that are peculiar to their particular area.

Given that the approach and documentation has been settled, before starting in each area a formal visit should be made to the chosen villages to explain the purpose of the enquiry to ensure that there is no misunderstanding about the use to be made of the information collected – so far as it is ever possible to ensure that there is no misunderstanding – and about what the cooperating farmers are going to get out of it. Generally speaking, there should be no promise of any kind made to them except, perhaps, that they will be given a certificate saying that they have cooperated. It is a good idea to arrange a village or community meeting for these explanations and for a senior officer in the unit directing the enquiry to be present to give it status and to explain authoritatively what is intended.

The demands of this kind of work put very special responsibilities on the field staff compared with the demands on those in area estimation and similar 'passage' work. They are going to be in contact with the community, even if not with the same individuals, over a period of perhaps a year or more and there is a very high premium on their tact and innate decency. The degree of personal contact between the enumerator and the village people could be considerable but it is desirable that contact should be confined to the actual working sessions and that he should keep away at other times. The problems have been noted already under Staff Control. Staff of dubious reliability under these various headings are a grave risk, particularly if they are recruited locally.

The ability is useful too, of promoting the kind of general conversation which will bring out the required formation for the record without the use of rigorous questioning.

Staffing.

The amount of staff required for farm recording obviously varies somewhat with the amount of work to be done and the amount of travelling. A rough indication might be given rather as follows. If fairly continuous daily or alternate day recording of inputs, farm

operations, receipts, and expenditure, is required it is unlikely that one enumerator can handle more than four or five farm families. If this work is to be kept under close supervision and the data moved forward for tabulation on a strict timetable about one supervisor per five enumerators will be needed. In a big enquiry one can reckon on needing probably one senior economist-statistician in charge of a survey which involves fifteen or twenty field staff. Obviously this may not be possible on occasion but, unless the work has already been established as a routine, there are grave risks if some such concentration of expert guidance is not available at least for some months after the start of an enquiry. The statistician would, however, be concerned also with the office tabulation of the data and for this purpose it might be wise to reckon on about one clerical officer on office tabulation for every five enumerators.

Obviously there will be variation from these sorts of figures in the light of local circumstances, but some such pattern as this is likely to emerge. Where it is not possible to attain this concentration then it is likely to be safer to concentrate on getting a small number of properly organised and controlled detailed farm records, and to collect more broadly from the rest of the sample some indication of available resources, inputs, and so on. This is likely to be more successful than attempting to cover detailed recording on a wider range of farms and fail through incompleteness of supervision and lateness of tabulation. If some rigorously supervised data on, for example, input/output relationships are available from a small sample of farms it might be possible to cut costs and yet extend awareness of the working conditions on the farms concerned by collecting data from a rather wider sample by questioning farmers as intensively as the goodwill permits.

It is possible that where area estimation and similar enquiries are in progress the call-back checks and general post-enumeration control for that enquiry might be carried out by staff on intensive farm enquiries.

Recording techniques and problems

It is impossible to foresee all the problems that are likely to arise in work of this kind. They run the full gamut of technical and human origins. However, there are a number of points which it might be useful to note as pointers to the kind of things needing attention in particular cases.

If one of the objects of the enquiry is the determination of input/output relationships from crops it is necessary, or at least highly desirable, to be able to associate the input of each item with the field or crop on which it was used. Consequently, field records must have a place for the individual plot identification to be entered and this must be readily available in the field. A considerable aid to accuracy in recording is a good sketch map of the farm concerned. If enumerators are well trained in this they can provide their own, but if they are hesitant it would be preferable to pick the best of them at the work and get him to prepare the maps for all his colleagues within reasonable range. To avoid errors in recording details for a specific field, a stake with the number of the field is a considerable help and often survey organisers have found that farmers have no objection to this.

Sales off the farm usually present relatively little difficulty if they go to centralised buying points. Even if they go in small quantities locally, there may not be much of a problem provided that the recording programme provides for entries every couple of days, or so. Consumption on the farm is likely to be less straightforward than might be hoped initially. For instance, some crops are eaten on the farm before they are fit for harvest. This is always an extremely difficult feature to deal with and the best that seems possible is to ask the farmer or worker who is being recorded to drop the remains of the food he has eaten somewhere within sight and for the enumerator to go along and note them at intervals. Obviously this is not a possibility for regular reporting but some insight into the amount of green maize, for instance, eaten on the farms, might come from counting the cobs thrown away. Crops brought into the farm house for consumption also present problems because only a part of them may come straight from the field to consumption without staying for a while in store. The harvest of one day may be mixed with products brought in on other days before the food is prepared thus losing their identity. If weighing is possible, well and good,

otherwise measuring may be possible with subsequent conversion, or the enumerator must note the approximate size of the items concerned, match them with other material and weigh these. The enumerator should be instructed to be present when produce is being prepared or, if there is a daily pattern, when it is being brought in from the fields. If they miss, they should ask the farm family to show them about the size of basket or size of roots concerned.

It may be difficult to transfer quantities into values for subsistence items. Provision may need to be made for the purchase of samples of similar products in the markets of neighbouring villages in order to get a price. This may be better done by the supervisor than by the enumerator.

It is risky to make any assumptions about the available labour on farms, or to ask farmers what labour they have "available". It has been noted on many occasions that farmers are operating on a scale which they could not possibly handle with their own family labour and thus, therefore, must be hiring or otherwise controlling outside labour which they do not mention in reply to an enquiry. In handling labour records it is important to keep a note of the names of workers associated with a farm, if these are at all stable, and check the stated labour use against these for consistancy. One must remember that time may mean very little to people in some situations. "Minutes" and "hours" as units of time may have no equivalent either in local language or local thinking. A hasty enumerator will get a figure which reflects much more the kind of question he put to the informant than the actual amount of time that the man spent on the particular job.

Whether recording labour or receipts or expenditure at short intervals, but not daily (for instance weekly or twice a week), the record sheet should provide a space for an entry on *each* day back to the one on which the last record was taken. It has been shown repeatedly that people do not remember precisely what has occurred during a stated recent period. They are inclined either to omit a day or to include with the total requested for the period "since I was here last" an item for at least one day included in the previous records.

It may be necessary to make fairly detailed records of what people do in their non-farming time and to reach some decisions on which of these activities, if any, ought to be associated with the farm task. For example, if a worker goes to the dispensary in the middle of a working day, should that be included as part of the labour use on that day? Probably it should. Again, it has been found that in certain resettlement schemes relatives, who previously could have made ready contact with each other in their old community, now find themselves separated and the women take time off deliberately to visit each other for social contact. This may not be thrown up without reasonably detailed time records. Is it part of the productive time or not? Probably it is, if those procedures are going to be customary. The cost of labour occasionally introduces a difficulty. If there is a daily payment settled regularly this may not be a particular problem. However, on occasion there seems to be a considerable interval between the time that the work is done and the time that the payment is made and this both confuses the record and leaves some doubt as to whether this alleged circumstance is concealing some other transaction. Again, the payment for group labour may not be in cash but in food or drink. Some care may be needed to make sure that purchases of material for group entertainment, or withdrawals from the home crop for this purpose, are entered as a labour cost and not as family consumption.

The amount of recording of cash receipts and expenditure and the number of people who are concerned may vary greatly with the situation. Sometimes the number of sales is very small and it is easy to locate what happened to the funds. At other times, sales are going on all the time and by more than one member of the family, perhaps there may be non-farm activities bringing in money, and migrant workers from the family may be sending remittances. It is at least possible that some of these will not be reported to the enumerator in a proportion of the families. It may be possible to cross-check some of the transactions by asking the farmer what he has sold, how much he got for it, and then what he did with the receipts. If he then mentions purchases which add up to less than the stated receipts he is questioned on the discrepancy. If he mentions items totalling more than the alleged sales brought in, this may throw light on other sales which he had forgotten to mention. This sort

of cross-checking may be impracticable either because the necessary goodwill is lacking, or simply because of the number of items concerned. Moreover the receipts may be coming to the farmer and the expenditure may be largely the concern of the farmer's wife or wives and two or more different people have to be questioned – and each may have certain transactions he wishes to keep secret from the other.

A net excess of receipts over expenditure might, naively, be regarded as representing savings. It is highly unlikely that at least in the earlier stages of an enquiry, any useful information on savings will be obtained by direct questioning or even by comparisons of this kind. The most that can generally be hoped for is that a few farmers will be cooperative enough to give fairly complete information and thus give an insight into the way the farm/family finances are likely to be run. More than one enquiry has suggested heavy drains on savings throughout the period of the study but this probably merely represents under-recording of receipts. Some of the cross-checking is spoilt by very long delays between the transfer of goods and the subsequent settlement of the cash account. This may mean that long after a farmer has purchased something he in fact pays out the money for it, or it may mean that long after he has supplied someone with some farm product he receives payment for it.

In many instances, in the course of an enquiry lasting some weeks or months, new and better information may be gathered as farmers gain confidence. It is desirable to have on the daily record sheet spaces in which corrected information of this kind can be entered with a note of explanation so that the main tabulations can be adjusted accordingly.

Studies for assessing fertiliser response on a field scale under normal agricultural practice can follow exactly the same pattern as those for estimating yield per unit area in general statistical enquiries, with one exception. Normally a design for this purpose would provide for two sample units per crop within each sample village (or possibly within each sample farm) and no particular attention would be given to their agricultural treatment. In order to be able to assess efficiently the consequence of applying fertiliser (or any other identifiable treatment) it is necessary that one of each pair be treated and the other untreated. If a pre-enumeration survey can be arranged, then the fields can be stratified according to their treatment and selection carried out within strata. Otherwise, extra fields will have to be drawn for the fertiliser assessment study until "with" and "without" plots are available. The results of such studies are usually very valuable modifiers of the guidance obtained from field experiments. Two comments might be made, however. In the assessment of response on farms it is usually only possible to compare "with" and "without"; it is not practicable to distinguish different levels of "with". In many instances this covers the main area of interest. Secondly, it may be necessary to draw the samples for comparison from different farmers, within villages, because the farmers either use fertiliser on all their fields or they use it on none. If this is the situation, the difference between the "with" and the "without" will reflect a combination of fertiliser effect and "farmer competence" since some types of farmer are more likely to use fertilisers than others. Something of the same effect may occur even if both samples are drawn from within one farm because farmers are unlikely to choose at random the fields on which they will use fertiliser.

Useful guidance on the state of an agricultural sector and its course of adjustment can sometimes be obtained from the answers to questions seeking qualitative information. Comparable guidance by way of analytical statistical studies might require very considerable bodies of data. Such questions might concern–

- whether improved varieties or practices are in use.
- presence or absence of specific diseases.
- rotation length (in shifting rotations).
- how many months the family can live on their own staple food (indicating degree or self-sufficiency)
- what they would buy if they had a windfall profit or bonus.
- what they can buy now that they could not 10 years ago.

No doubt some of the records proposed here should ideally be compiled daily round the year. Very few normal individuals will stand such intrusion for more than a few weeks. The

overall design must, therefore, provide for relaxing the questioning or ending it before crisis tension is reached. As some guide it might be well to continue daily questioning on labour use, budgets, etc., and assessment of quantities of produce brought into the household for no more than one month, moving then to other households in the sample. Twice weekly visits without interference with household practice might be possible through a year. Of course it may be possible to get away with heavier loads on the informants than this, but it is important when approaching overload levels to give some thought to the prospects for the *next* piece of statistical work attempted in the neighbourhood.

XV PRICE STATISTICS

Introduction

Statistics of prices connected with the agricultural industry are extremely important to an understanding of the working of the economy. It is more than likely that more insight into the state of an agricultural industry could be obtained per unit of expenditure by enquiries into prices than enquiries into many of the subjects discussed so far. Some degree of price reporting is to be found in most countries, but it is often oddly superficial. Among the deficiencies and elements of confusion are the following:- commodities are not fully defined; the quantity units are uncertain; farmers may sell at the farm and the local market and in some distant locality to which they have transported their produce; they may sell effectively at retail, at wholesale and at a mixed-stage in between; there may be statutory fixed prices which are not, in fact, adhered to. The quantities passing through particular markets fluctuate sharply from one time of the year to another, which upsets reporting procedure. Prices as quoted may be largely nominal because growers are tied to buyers by debts incurred for goods supplied and consequently they do not get the market price. Amidst this confustion it may not be clear for what purpose the statistics are wanted and hence what criteria should be used in their choice. Furthermore, it is easy to get together a body of material in the price field and unless some external agency demands critical scrutiny, these price statistics will pass into the record with little comment. It is in the face of these kinds of problems that improvement in price reporting has to be carried on.

There are usually a number of possible points along a distribution chain at which prices might be assessed, always a large number of farm products which might be included and usually a number of production requisites (such as fertilisers) which should be included, too. For the purpose of this chapter, emphasis is placed on farm product prices at the first stage where a reasonably stable and broad market is to be found. The matter of prices along a distribution chain is dealt with in the following chapter on marketing. In other words, this chapter is concerned with a regular routine of price reporting. However, though many of the suggestions for reporting apply equally to retail prices these are not specifically dealt with in either chapter because interest in retail prices in developing countries tends to centre on labour relations and wage fixing in urban areas which is outside the main agricultural area of interest. Index numbers derived from wholesale or retail prices are not included either, though the price series prepared from the reports outlined in this chapter would be suitable for the preparation of agricultural price index numbers if desired. With these many sources of uncertainty and error attaching to price reporting it seems better to concentrate on getting one series or group of series in a reasonably reliable form than to try to bring a whole range of possible series up to a modest level of reliability. Suggestions are made below for getting a general indication of relationships between different series for use where such amplification of the basic series is essential.

A list of the main uses of price series seems a useful background to further discussion of this subject. Among the uses for which statistics of prices are compiled might be included

a) To provide comparisons over time, place and, occasionally, form. Governments are usually very much more concerned with trends in prices and relative trends in prices than they are with the absolute levels.

b) To provide the basis for converting physical quantities produced or marketed to valuation in money terms for the purposes of regional or national accounts. Occasionally they may also be needed to supplement farm records in deriving estimates of farm income. For these purposes prices of both farm products and farm requisites are required.

c) For market guidance in current trading operations. Thus traders are usually interested in price reports to give some general indication of the state of the market in places to which they have no direct access.

d) To give an indication whether the price which a farmer obtained from a dealer in his recent transaction was in fact a fair one in the light of general market conditions at the time.

e) As the basis for contracts where it is customary for the price of a commodity sold under contract to be linked to current market prices.

These various needs represent very different demands on the statistical service. Thus a) requires great consistency in definition and other details of reporting if the comparisons are to be valid. On the other hand, the general level of the prices reported need not be precise nor need the quotations be frequent. Even monthly prices are often quite suitable for giving inter-country comparisons or time series for general assessment purposes. Often the same applies to b) though in principle it might be desirable to have the figures at the right level. However, there are usually so many conventions in accounting work of this kind that adjustments may be incorporated without too much difficulty. c), d) and e) all demand that the prices as quoted shall correspond to the actual commercial transactions in the markets and be recognisable as realistic by those trading as buyers or sellers. Moreover, where trading is continuous, and consequently prices may be of interest at any time, at least daily reports are usually necessary. These two groups of figures are orientated in such different directions that a good case can be made for organising their collection by different staff. This is not to say that the two are not handled in many countries by the same agencies but, nonetheless, the outlook required in order to be thoroughly successful in the one is rather different from that required for success in the other. Those collecting statistics for a) and b) are serving the statistical department; those collecting statistics for c) and d) are serving those operating in a marketing field. In big centres, where it is possible to separate these functions, there is much to be said for doing so, even if both sets of staff are employed by a statistics department. In some ways there might be little loss if those providing daily reports were members of a marketing department. However, if there is this dichotomy, a regular practice should be instituted of checking the indications from a selection of the daily reports against the indications of the reports for a) and b) and generally holding a postmortem on the comparison.

The main ways of obtaining statistics of prices of farm products and/or farm requisites are

a) from farm recording studies usually by dividing the values involved by the quantities rather than by direct noting of prices at the time;

b) from reports obtained from farmers about prevailing prices in the course of general statistical passages for physical recording work;

c) by observation and discussion at markets or with traders;

d) from schedules of fixed buying or selling prices of some kind of statutory organisation;

e) from private trading price lists;

f) from marketing boards and other semi-statutory organisations which keep their own statistics.

The bulk of such work uses methods c) to f) and consequently usually requires staff in quite different places from those where the main recording of physical statistics takes place. Market reporting tends to be concentrated in the big towns. It may be possible to mix price reporting with other kinds of statistical work but it is usually better to assign staff to specialising in price work, perhaps combining it with office work on general statistics.

Reporting procedure

Proceeding now to the outline of a reporting procedure, it is probably best to review the main characteristics of the commodity field which has to be covered and the markets through which the produce is disposed of. The first need is to select first-hand selling points which are representative of a substantial proportion of the trade. Where it is customary for farmers to take their produce to a market centre and there sell in fair sized quantities to first-hand buyers, there may be little difficulty in making a choice. Where it is customary for

producers to sell to travelling buyers who act as collectors for larger buyers in market centres, a difficulty is introduced, since the small itinerant buyers are probably not good sources of price reports. Under these conditions the best approach may be to split the reporting. One group of staff should concentrate on the normal reporting at the main market centres, while the other does as good a job as is feasible in collecting reports of the prices at which they have sold by direct enquiry from farmers. Reports of this kind should also indicate the kind of trader or other buyer to whom the sale is made.

A standard day for the report should also be chosen. This is usually better as a day of the week rather than as dates in the month. The activity of many markets varies quite markedly with the day of the week and consequently there is some risk that if the chosen date falls on a different day of the week, fluctuations in reports will appear which have no useful indicative value.

Amongst the markets which correspond to the stage of first-hand selling required, a choice should be made of those most suited for the purpose in hand. This choice should depend on such features as the commodities on offer and the variety, and also on the size of the market and its consistency over time. It is also important to define precisely what is to be regarded as the boundaries of the market for reporting purposes. Sometimes in a city there are a number of markets with slightly different characteristics, and the ones to be used for reporting should be clearly defined. Even within a market of this kind, however, and even where a physical market area is provided, there may be firms on the fringe who handle a substantial quantity of the produce. If these are to be included then the fact should be built-in to the reporting system.

A practical approach to this may be to determine, in the light of local information, whether the trader, or traders, concerned handles the same kind of produce from the same kinds of producer as the main market place. If there is no clear separation of trade then it might be better to confine reporting to the market place unless the firm is notably cooperative. It is usually easier to maintain consistency in reports from a big market than from un-cooperative traders, treated more-or-less as individuals.

Markets customarily vary in their activity and price level over the day. Some can be regarded as opening at a price related to the previous day's trading and closing at a price relating to the supply and demand situation for that day. However, in many up-country markets business may trail off during the day so that no clearly defined closing price can be discerned. In other markets, trading may go on virtually round-the-clock. In still others, each transaction is a separate transaction and its level depends very considerably on the haggling skill of the parties. Generally speaking, it seems best to review current practice and to decide on the time of the day when the peak trading occurs. This may vary from season to season, and should be locally known. Reports are best quoted as the *most frequent* price for trading within the peak trading period.

Since the aim is to secure comparability of price reporting, the need to report price of "the same thing" on each occasion is clear. Unfortunately, it is not by any means obvious what is to be regarded as "the same thing". The same variety of a crop may not be available all the year around; or even if the same variety is available all year around it may represent a very different position in the hierarchy of users' preference scale at one season than at another. Though real enough, this latter aspect is difficult to reflect and the best approach is one directed to using as nearly as possible the same variety throughout the year. For this purpose it is desirable to consult the technical experts before reporting is begun to find out in which periods of the year different varieties can be expected to be on offer. One that is available for most of the year should be chosen as the basis for reporting, but one or more other varieties should be selected to cover the periods when the first is not available. The reports should be collected for the standard variety for each day that reasonable trade is carried on, but for periods overlapping the end and beginning of the main season by two to four weeks reports should also be obtained for the secondary varieties. In all instances the quotation should refer to "fair average quality". Some care may be needed to get this idea into the minds of the reporters and it may be necessary to bring together a selection of produce from various points and indicate the type of goods corresponding to the desired standard of quality.

Units to be used as a basis for reporting cause many troubles. Generally speaking, reports should be in terms of the unit as sold—baskets, hundredweights, bunches, etc. Provision should be made for converting these to standard units e.g. kilos, etc., at some point, not necessarily in the field. This is very much less straightforward than might seem to be the case and the particular troubles may emerge only with local exploration. First, the conversion should not be made with nominal conversion factors but by ones actually determined by weighing representative samples of the units as used in the market. On occasion this weighing may produce no particular difficulty but there may be cases where the particular conversion rates vary from time to time. Probably the classic case is the quotation of retail prices per cigarette tin where the money value of a tin-full is unchanged for a while as prices rise but sellers change to a brand in which the tin is smaller. Differences in fill of receptacle may be relevant at all stages in the marketing chain, corresponding to measures from cups to kerosene tins. Getting the correct conversion factor may also be confused by customary practices about "allowances". For instance there may be a practice under which first-hand buyers quote on a per hundredweight basis, but expect to get anything up to one hundredweight and a quarter in actual product at the farm for the price of a nominal hundredweight. In such cases the price per unit at farm, as quoted, may be similar to the quotation at wholesale or even retail. This may be a fairly straightforward point to pick up if the produce is handled altogether, but if the extra above the nominal quantity is supplied as a perquisite, so to speak, it may be difficult to identify.

Finally there may be straightforward cheating over the weighing in some places. If the scales are faulty, some indication can be obtained on a single occasion by the observer weighing himself on bulk weighing scales, having already a good idea of his weight to the nearest pound or two. Obviously this is not a trick that can be used very often in the same place. Again, effective cheating may occur at some point in the distribution chain by moistening the crop before weighing. On the other hand, it may be necessary to check that foreign matter in the sample is correctly treated for the definition desired (e.g. "fair average quality" corresponds to no more than a certain level of such adulteration).

There is usually a fairly clear cut season for the more perishable crops but stored crops may survive beyond the start of the next season. In this case provision should be made for separate reporting of new and old crops.

The above should cover the majority of the general features of definition needed to establish a working system of reporting for farm products sold through regular market channels. Where some form of guaranteed prices is operating the price can be obtained, nominally at least, from the published price list concerned. Some care should be taken to make sure that these prices do operate as they appear to operate. Thus a nominal price may be subject to variation by counting the same crop as differing in quality depending on the state of the market. Again it may be found that part of the produce is sold in this way and part in some other way. Where departures from the legal working of a fixed price system are known to exist, the statistician is usually in a difficult position since it is probably not possible to report prices other than as conforming to law. However, there is no need to convey this fiction into derived statistical work and a correct definition should at least be available for the purposes of computing regional accounts and the like. Where farmers sell the bulk of their produce through cooperative societies it is usually possible to get useful data on prevailing prices from the societies. It should be noted, however, that sometimes this kind of source can more readily offer average values rather than prices. This means that it may not be possible to give the price for specific varieties but only the weighted average of all purchases.

Regarding statistics of purchases by farmers, a clear distinction must be drawn between purchases for farm use and purchases for family use. Often this presents no problem but sometimes it is a matter of difficulty. Generally speaking, statistics of prices of such inputs as fertilisers, insecticides, fuel, tools, etc., can best be obtained from sellers. There are, however, certain points which should be checked in using this source if misleading results are to be avoided, particularly in connection with income estimation. For example, it may be that there are various programmes under which inputs can be obtained by the farmer apart

from ordinary private sale. For example, fertilisers may be available under a special price scheme to producers of certain commodities through the purchasing organisation for these products. In addition, fertilisers may be available through the ordinary channels of trade. It may be quite unrealistic to take the figures from either of these sources alone, and some study is necessary in order to get any figures on which reliance can be placed. Again, the whole of the supply may come out through certain channels where prices are well known. However, if the product is highly desired, it may find its way to certain sectors of producers through "black market" channels at a price appreciably higher than the formal price. This may be well known in the locality, or some exploration may be needed to bring it to light. It is possible that prices of this kind may vary substantially from the originating warehouses to the end of the transport system up country.

Again, when dealing with products which are storable and which have been the subject of policy measures of one kind or another, it is possible that supplies have reached a distribution point from consignments which were priced at one price but have not been disposed of before the selling prices have been changed. Just how this will work in practice can only be ascertained on enquiry. Such occasions may not greatly affect a routine reporting system but they may be fundamental in explaining the behaviour of producers as buyers.

In conclusion, a few details concerned with operational arrangements of a more general kind might be useful. It is important to stress the need for supervision in this reporting work. There is a great deal to be said for not embarking on new reporting procedures – or attempting to improve ones which are already operating – until provision can be made for adequate supervision. This means not merely a checking of documents in the office, but a steady continuing interest on the part of the supervisory staff in the marketing practice, the reporting conditions, and the user requirements.

The form of report on which the market reporter is to make the return should be standardised. It should provide for a clear statement of the product and the variety, the price and the units in which it is quoted, and for spaces in which the equivalent in standard quantities can be entered either by the reporter or by clerks in the receiving office. Reporters should be strictly instructed to obtain their information in respect of markets by actual personal visit, and observation of market conditions in consultation with the trade. They should not attempt to get information by sending out a runner for the purpose nor, except when dealing with large firms under a procedure agreed with the statistics authorities, should they collect by telephone.

Instructions should also require that any notable features about the return should be explained. For example, if no figure is entered, an explanation such as "no trading" should be required. Again, very high figures, or very low figures, should attract an explanatory comment. A column on the report form should provide for this.

The emphasis in this outline on the consistent reporting of prices at a single point in the distribution chain makes it desirable to build up certain associated information. Much of the market data referred to in the following chapter falls under this heading, but special mention might be made here of the need to record, as separate background enquiries, data indicating the normal price differentials between varieties at different times of the year and inter-market differences for the same variety.

The reporting routine outlined here assumes that it is possible to obtain information about the prices in which transactions take place by asking one or other of the parties. In a number of situations, perhaps more concerned with retail and micro-level transactions than with bulk farmer-to-first-hand-buyer sales, it may be very difficult to obtain such information except by making an actual purchase. Probably there is no wholly successful method of handling this, but various departments have approached the matter by finding staff members who engage in such transactions in their ordinary life and asking them to report their price experience. As has been indicated, from time to time it may be important to build into any intensive surveys being undertaken a procedure for collecting their experience of prices from farmers in the sample units. It may also be possible to arrange for field staff who are carrying out area estimation studies and other work which takes them

into villages to make specific enquiries along the same lines. This may not be as straightforward as it seems, particularly if the staff are not particularly good. For one thing, the kind of work on which the field staff may be engaged varies from season to season, and consistency of contact with farmers may be difficult to achieve. Moreover, the more procedures that are added to the work of field staff, the more skills they need to display and the less easy it is to keep to high standards of operation.

All the discussions so far have concerned the *collection* of statistical material. It is very important that regular provision should be made for discussing the result of such work with potential users. It may need some persuasion to bring the parties together; busy men are inclined to say that they have already made all the comments which occur to them. Sometimes there is a tendency to leave contact between statistical department and user to very junior staff who are in no position to take a firm line. However, if it is worth spending a substantial amount of staff time, week-in and week-out, throughout the year in collecting this kind of information, and if it is worth the users' manipulating it in the course of their own work, then it is worth getting together to make sure that the arrangements and the needs match as closely as is reasonably feasible. This may all seem so obvious as to need no stressing yet constructive joint action of this kind is rarer than might be expected. Needless to say, users outside Government service should be sought and included in such examinations.

XVI MARKETING STATISTICS

Introduction

It might be difficult to obtain general agreement on what statistics should be included under this heading. Statistics of the quantities of a cash crop marketed through a major marketing channel would come under this heading and should certainly do so; so should the prices at which the produce changes hands. However, when one leaves this group of statistics and moves to data on the structure of the marketing sector and of price patterns along the distribution chain and over space, some workers might argue that the statistical field has been left behind for one which is better regarded as economic research. Certainly when passing from dealing with statistics which are by-products of the records of ordinary highly organised commercial distribution to those in other fields the difficulty of the work increases enormously.

For the present purposes, however, three broad headings are recognised –

a) statistics of quantities marketed and the prices realized which are obtainable as by-products of the operation of monopoly buying organisations (e.g. marketing boards), or at export points, or by agreement, or under a statutory requirement applying to large scale operators in the distribution field,

b) routine price statistics designed to give comparisons over time,

c) statistics relevant to marketing efficiency or the structure and organisation of the marketing system.

The material available under a) has already been noted as highly important in providing cross-checks of direct methods of estimating production of agricultural products (see Contents). As was noted there, it is usually extremely important, if the best is to be obtained from such figures, to have ancillary information about the part which the channel concerned occupies of the total distribution system for those products. For instance, there may be very good statistics of the quantities of citrus sold to a citrus marketing board. But without some knowledge of the quantities sold from the same holdings and perhaps the same trees outside this channel to local users, it is difficult to cross-check production estimates for the citrus sector as a whole with the marketing statistics – highly accurate though they may be for the part of the field which they cover.

By-product statistics

The sources of such figures are usually well known in a particular country and tabulations can usually be found in the reports of any semi-statutory agencies concerned, or in the annual reports of the Chambers of Commerce, if not in the regular national statistics publications. Two or three points might, however, be usefully stressed in connection with this work.

Though they are often relatively straightforward, the definitions of such statistics are often less clear than they seem at first sight. It is extremely important that the precise coverage, in respect of quality, variety and location, should be recorded against the statistics in any formal tabulation by a statistics organisation. The kind of change which may arise is the opening of new buying points by the agency concerned so that over a period when the geographical scope covered by the records increases, from the agency's viewpoint it is quite reasonable to count the total quantity handled since its interest is focused on its own activities. Again, under a programme of quality improvement tonnages of the lower qualities may be progressively dropped from the records. Where questions of this kind are concerned,

the recording definition should include at least two stages. First, it should set out the formal change concerned – e.g. that this year includes records from 62 buying points whereas last year it included records from only 60. But it should also be put on record whether this change is as significant as it might seem at first sight. For example, the addition of a new buying point may merely increase the convenience of sale for the suppliers concerned; no new area of supply may be tapped. There are obvious risks in expressing views of this kind, since expectation may be entered as fact, but nevertheless some reference of the sort is valuable.

It is extraordinary how often, when a visiting adviser or an academic research worker comes to ask questions about apparently very straightforward series of this kind, the precise definitions are the subject of argument. One important way of guarding against some of these risks is to make it a rule that the statistics are scrutinised each year by senior staff who are not concerned with the day-to-day tabulation. Some of the difficulties arise through delegating this kind of "routine" work to clerical staff "who make the contact required every Thursday morning, and enter the figure in the appropriate place in the office tabulation sheet and set the summary procedure in motion". Unless a formal assignment to review is built into the statisticians' calendar, this kind of uncertainty is almost certain to get through. It can usefully be built in to a discussion with users of figures, a discussion which is likely in any event to throw up oddities of definition.

Sectors of the marketing field for which there are statistics tend to attract undue attention in marketing work. It is worth stressing that a substantial amount of business is done in other directions and the significance of these other sectors may become greater if measures to substitute home production for imports, now built in to many of the policies in developing countries, come to fruition. Of course, some programmes provide for the extra supplies to be channelled through semi-official agencies, but a successful policy is likely to result in expansion over a wide front. However, unless some bottleneck presents itself, the compilation of data bearing on these other channels must usually depend on the use of individual farm records (such as are described later) or semi-qualitative recording of movements through markets and along transport routes. Rigorous recording of so many channels is usually out of the question.

"Market information"

The markets in many developing areas, at least up country, traditionally are imperfect in the economic sense. In other words, some of those engaged in transactions are not well informed about conditions elsewhere in the market or in neighbouring localities. Typically, prices may be appreciably different in villages not very far apart, without any administrative or physical barrier between them. It is helpful to make available to those concerned in the marketing business information about the prices which have recently been ruling in that market and in others – the details naturally vary with the communications systems and the market pattern of the country. Thus, in parts of Africa producers valued the posting up in the market each week of statistics of the average prices ruling in the market as a guide to them on what they could expect to get in the near future for their product, and how the price which they had actually obtained in the past few days compared with the general average. In conditions of considerable chaos this may well be valuable and it may be possible to contribute notably to the efficiency of a marketing system by the provision of adequate market news of this kind. However, it should not be thought of as a particularly easy or straightforward piece of work. For example, buyers of produce from farmers are the most readily accessible source of routine reports but these are also likely to be only partially enthusiastic over any scheme that makes the other party to their transactions better informed; this process is likely to cut their opportunities for profit. As is noted elsewhere (Price reporting) for market news purposes it is essential that the prices quoted should be at the correct level and should not merely move from day to day with the right trend. It is likely, too, that much more detailed information by varieties and so on will be needed, in order to give local growers the information on their particular product of interest. In some instances this reporting procedure may be needed only at harvest time. After a short period

all the product for first hand selling will have changed hands and no further reporting is either possible or sensible. On the other hand, with increased sophistication farmers may begin to store products in order to avoid the low prices at the congested harvest season, and marketing statisticians may need to extend their period of price reporting accordingly. This is particularly likely to be the case where a development programme is providing credit so that growers are no longer forced by lack of cash to sell so early. Appreciable ranges between top and bottom prices must be expected in both primary and retail markets.

Price spreads or distribution margins

The increase in price along the road from the farm gate to the retail point for farm products, and from the primary selling point through various distribution centres to the up-country sale point for requisites attracts a good deal of attention on some occasions and much less than it deserves on others. There is a tendency in some instances to quote comparative prices at different distribution points as evidence of the inefficiency of the distribution system. Often the picture would be appreciably different if more care were taken to obtain the appropriate definitions. If large price increases along the chain are still found it would be better first to look at the structure of the trade in these localities to see whether reasonable competition can be expected and reasonably free entry of new operators (see below).

In any event, information of this kind can often be tabulated fairly readily in the course of other enquiries, or as a by-product of other enquiries, and is well worth having. Obviously statisticians and marketing staff would seek opportunities for joint consideration of information of this kind.

Where there are sectors in the whole distribution picture for which quantitative information is almost totally lacking it may be well worthwhile building up purely qualitative data on the parts of the country from which produce has come to a market, and the destinations of lorries going outwards from markets. It might be worth putting staff on lorries to see them through to their destination from time to time, and to note the general spread of deliveries along the routes. It is useful at various points along a route to talk to individuals and to find out what produce leaves the neighbourhood, for what destination and by what means and what comes in, where from and by what means. This kind of background material would be essential if formal sampling enquiries have to be mounted at any time to obtain information of a more specific character. Such work may need to be rationed since it can be extremely time consuming, but some part of it is essential if the cross-checking and other secondary statistical work, whose value has been stressed in this bulletin, is to be developed to the full. Often it is revealing to ask informants what changes in sources and destinations of produce have taken place over the last, say, ten years.

Finally, throughout work in the marketing field, it is desirable that possible frames should be sought which might be valuable for special enquiries in the production field. For example, any source of lists of growers of particular produce should be recorded; so should markets where growers are the main sellers and, consequently, where an interview with those bringing produce to market should allow a reasonably serviceable frame of producers to be prepared.

XVII STATISTICS OF LAND USE.

Introduction

What is in mind in this chapter is much wider than the use of land for agricultural purposes and its sub-division between crops. The emphasis is rather on broad categorisation, either between ecological types or between agricultural land, potential agricultural land, unusable land, etc. There is no theoretical reason why enquiries directed towards obtaining this kind of information should not be integral parts of the kind of crop assessment work already outlined, but in practice two special features of "land use" data tend to make the two fields of work largely separate. One is the fact that broad land use information embraces, in some countries at least, a good deal larger superficial area than the crop land. Further, whereas it is fairly economical to locate the crop land for observation by approaching it through a frame of individual humans or communities, a similar approach may not lead effectively to much of the land which is not used for agricultural purposes. The second feature is the probability that the *location* of different categories of land are as interesting as the statistics of their superficial area. It is therefore usually as important to have a map of the various categories of broad land use as it is to have a tabulated statement. Though the enquiries that have been the subject of the rest of this bulletin can yield data from which a certain kind of land use map can be derived, they obviously will not throw light on those parts of the country in which cropping is not carried on. Sampling enquiries are not specifically useful for preparation of maps. Consequently, the principal approaches to land use enquiries have tended to be in the general direction of mapping work, and any statistics that may be required have been derived from assessment of the areas on the maps. This takes the studies rather out of the field of this manual and no comprehensive consideration is given here. A few remarks may, however, be in order.

Mapping

The main features of land use over quite serviceable areas for which no formal record is yet available may still be obtainable at little cost if energetic attempts are made to collect together on one set of maps all the information that is available in the personal knowledge of staff who have field experience of the areas. Typically, the arrangement might be to transfer to maps all the available information that is in published form, noting particularly any areas of the country for which information appears to be conflicting or where it appears to be lacking. The next step might be to arrange to discuss these maps with people who have experience in the various sections of the country, particularly those areas where uncertainties have been noted. In seeking this personal experience it is desirable to divide the country into clear cut sections by, preferably, topographical features, and to work through these systematically. A general sort of interview between, say, a collator of land use information and a man from an organisation of a quite different character, who has worked in the areas concerned, is not likely to be fruitful unless a clear discipline is applied by the parties to their discussions. When all the available information has been brought together, written and personal, the maps should be reviewed to see where the gaps now are, and to decide which of these gaps have real significance. The means of filling them may vary with the circumstances. One method is to arrange for visits to be paid to these specific areas, making a journey criss-crossing the uncertain or blank parts so that the details can be filled in. On such a trip any incidental information which can be obtained by local conversation would naturally be included. It may be, however, that there are people with private light aircraft who would be prepared to cooperate in a check on such doubtful areas by flying the

observer over them, even perhaps landing him in places if the aircraft and ground conditions are suitable. Enquiries of several kinds have been made by such semi-formal collection of material. Sometimes it has been found useful to mark identifying features on the map and to take a route from one to the next, making of a running commentary of what is observed on a tape recorder as the flight proceeds.

It may seem improbable that there would be scope for this kind of rough-and-ready enquiry when many countries have had complete air surveys, but surprisingly often features of special interest have cropped up and have been covered by local ingenuity.

In principle it might be thought that transect surveying across country might be a suitable means of recording land use information. For example, a road might be taken as base and transects at regular intervals made from the road across country to another line of reasonable access. The observers would work their way on a compass bearing along the designated transect lines, entering the information about land use as they go, and making special enquiries at regular intervals. This kind of technique has been used on a number of occasions on fairly small areas. The main difficulty is the time required and the labour of forcing a passage through certain parts of the land which is met on the transects – possibly thick bush, possibly swamp land. It would now certainly be worthwhile exploring intensively the possibility of getting a light aircraft survey before committing field staff to a routine transect enquiry. It is, of course, true that such an informal aerial survey will not give details of soil conditions, but it should be possible from it to mark in the main features of the observed situation and to select specific points where ground examination is essential. These can then be approached by the most convenient routes (not by following a pre-determined arbitary compass bearing), and the necessary observations made.

The bulk of the broad land use work in recent years has been based on aerial survey by specialist operators whose organisation prepares maps with whatever detail is provided for in their contracts. This has been noted elsewhere, and considerable expertise under many headings is needed for this work, with ground checking to ensure that interpretation is consistently accurate. This represents a body of techniques which lies outside the statistical field of this manual.

XVIII STATISTICS OF FOOD CONSUMPTION

Review

Aspects of food supply and consumption and its valuation have been touched on from time to time in the preceding pages. In fact many of the larger elements in the picture have been dealt with, nutritional aspects apart. Nevertheless, the food supply situation is surprisingly ill-covered by statistical material and what is available as a guide is often confused by uncertainty about definitions. There is a great deal of work to be done in this subject, more than can readily be outlined here, but many countries will have to manage with very crude indicators of their food situation for a long while. Perhaps the most useful point to emphasise in this brief chapter is the need, in dealing with agricultural statistics which have a bearing on the food situation – and most of them have – to be rigorously precise in the terminology used. Often it is impossible to be comprehensive in assessing a sector of the food economy but it is extremely important that the nature and scope of the sector actually dealt with should be quite clear to the users of the statistics. When the tabulated material is rigorously defined there is often a good deal of useful work which can be done in the office, as staff time is available, to reconcile estimates and to synthesise others.

As has been noted elsewhere, a wide range of products may contribute to the agricultural activities and to the food supply of people in developing countries; the agricultural activities and the food supply activities do not, of course, necessarily equate. Agricultural output may include non-edible products and the food supply may include products from agricultural industries in other countries, from ocean fishing, from local fish culture, from the local natural fishing, from wild plant and animal sources of various kinds, and the pasture industry. (On a point of terminology, it might be well to bear in mind that some countries do not include the products of herbivorous livestock within their "agricultural" statistics; they are to be found tabulated separately under pastoral or grazing industries).

The very wide range in the products of the agricultural industries in many developing countries add to this formidable list of potential food items. Furthermore, the amount of change which occurs between the point at which the product is regarded as "produced" (when viewed from the farming end) may be very different in different parts of the country, or for different dishes, or in different seasons. It has to be recognised that the chances of deriving a reasonable estimate of the nutrient supply situation of a group of people from statistics of their food supply, using that term quite generally, are not good. For the present purpose one might dismiss the nutritional angle by saying that any information of a reliable character must be obtained by special enquiries in which the observer is in daily, even hourly, contact with the food preparation point of the household. Only in this way will he be able to take account of the varying intakes of different kinds of foods, their different states of preparation, and the destinations of food where it is moved from the house where it is prepared to others where family connections may be living. Even this kind of enquiry, as has been noted elsewhere, will not pick up problems of consumption away from the home (see crop yield estimation).

Broadly speaking, there are at least the following kinds of statistics which consequently need to be recognised as having a bearing on the food situation. a) The "total potential supply": This would, in many instances, be scarcely open to definition since if there are within the customary food supply wild plants, wild animals, and crops which are partially harvested, no reasonable bounds could be put on the potential. More commonly the potential will consist of the total production of the regularly harvested products, the normal off-take of any livestock owned, together with the total supply at the best time for

harvesting of any crops which are normally partially harvested-for example, cassava. Some general indication of this figure may be obtained from a reasonably comprehensive census, but it has to be recognised that with so many crops grown, often in very small patches, no census enquiry that can be contemplated in most developing countries is likely to record them with any precision. It might, however, be worth bearing in mind that if basic carbohydrate containing foodstuffs are the main concern, then the production of many crops when expressed as nutrients per unit area after preparation for eating may not be very different, at least in the terms of the standards of uncertainty with which one is normally working. b) The actual food supply; this would need to be defined as between the farm gate, wholesale and import point, retail, and the kitchen stage. This has some degree of precision in meaning for a community whose food passes through formal marketing channels but is of doubtful definition for many communities in developing countries. Nonetheless, in principle, it should be possible to assess the main items of agricultural output at harvest and, given the necessary organisation, before harvest. The products from wild sources and from partially harvested crops can only be assessed in terms of their actual use. This normally goes for livestock products, too. Populations are doubtful indications of output for there is usually, even in highly organised systems, a certain range of age within which it is economically and technically possible to slaughter livestock. This may be a matter of a few days or weeks but it might extend over months or years if there are two kinds of products for which a particular unit of livestock is used (e.g. veal and full grown beef). c) The marketed supply of food. Even in developing countries whose statistical sources are not yet very complete there are usually reasonable statistics of the main food supplies for urban areas, and for all areas in so far as supplies come from outside the country. As is noted under marketing statistics, local movement of internally marketed crops is extremely important in building up a picture of the food supply. d) Foods purchased. It may be possible to derive estimates of the purchases of more or less standardised foodstuffs from statistics of wholesale supplies, but in most instances any information which is of much significance, particularly information relating to food consumption at different income levels, can only be obtained from specific food consumption and budgetary enquiries, for which techniques have been developed over many years. Two points might be made here. One is that it is important to have both the weight and the value of foods purchased by sample families in household budget enquiries; these figures will allow an indication of the change in the kind as well as quantity of food purchased as incomes change. It is also worth bearing in mind that interesting pointers to current and potential changes in the food demand picture may be obtainable from rather simple enquiries of a "presence or absence" nature into the foodstuffs which families actually use. For example, as a part of some other enquiry, or even through enquiries at schools or gatherings which bring groups of people together under conditions in which individuals can be interviewed, it may be worth asking them to say which of a list of foods they have, in fact, used during the previous week. These might include staple foods together with others that they might be expected to drop from their diet as they become more prosperous and others that they might be expected to begin to use. It might also be useful to ask which foods they would regard as a 'treat' if they had a windfall of cash. These may give pointers to the directions in which demand may move, from information much less comprehensive than would be necessary for formal analytical studies. e) The supply of food actually prepared for consumption. It has already been noted that this is a specialist field and is not pursued here.

The statistics which might be assembled under these several definitions are likely, in the main, to be in terms of physical quantities, and for many purposes these may be essential. It is also useful to be able to convert these to value terms, but this may present a problem in all subsistence and barter conditions. As has been noted elsewhere, (Price statistics) the only way may be to buy samples at neighbouring markets; if some foods are not traded at all it may be necessary to synthesise prices. It is extremely important that the bases of any price data which are integrated into food statistical series should be explicitly specified.

In conclusion, it might be worth restating the possibilities of getting forewarning of food shortages. First, it must be recognised that an estimate of total supply of foods from all

sources is unobtainable in virtually all developing countries until some while after the end of the year concerned – if then. It is consequently only possible to work with the supply statistics of staple products. Secondly to be able to get an idea of the acuteness of impending food shortage from suitable statistics of the staple crops supply it is necessary to know what part the staple plays in the total supply, taking into account the availability of potential supplies of wild or normally only partially harvested crops. Finally, given that "suitable statistics" of the staple would in the light of the general background of food supplies give useful guidance, then what might be "suitable" statistics? This depends somewhat on the origin of the dangerous situtaion. If, for example, the problem arises principally from a progressive tendency for supplies to be short compared with requirements, then the statistics in any particular year may be less relevant than an indication from a comparison of population and production over a period, or, perhaps even more sensitively, from a comparison of prices of the staple compared with wage levels. Such general statistics may point to an evolving situation which is leading to danger. This situation apart, production estimates for a staple are not usually sufficient to give effective forewarning of food shortage, since by the time they are available at, or soon after, harvest the market has reacted, supplies have gone into hoards, etc. Consequently, if irregularity of production is the key problem, the average over a run of years being more or less satisfactory, then only a forecast of supplies well before harvest will be satisfactory. Obviously this could be belied up to actual harvest but this problem is discussed under the techniques of forecasting.

XIX PUBLICATION OF RESULTS.

The publication of the results of enquiries for the collection of quantitative information about agriculture is of concern to most of those connected with statistical work at one level or other, though they may not consciously appreciate this. Publication is as important for people who are unconnected with Government but need information for their business, studies, etc., as it is for official collectors of statistics. A certain amount of public goodwill has been used up, even where the money for an enquiry is obtained from elsewhere, and there is much to be said for making the results available in some form that is meaningful to at least some sections of the communities concerned. It may be that in a research study the most sensible course might be to get out a brief qualitative indication, of the main results illustrated by a few figures, and to send it to the principal officers of the organisations who helped in any way in the course of the work.

Certainly where data have been collected at public expense, it is desirable that it should be published, and published promptly. There is no commodity that is more perishable than statistics.

If at all possible, it is important to include along with any tables setting out the figures comment on any associations indicated and any warnings against unwarranted conclusions. A statistician in charge of an enquiry is usually better qualified than anyone else to know the nature of the figures and the errors attaching to them The standard errors of the various estimates should, of course, be quoted but care should be taken to emphasise that observational errors may be large, also. Further there may be pitfalls in interpretation which the statistician can appreciate but the ordinary user may not.

Unfortunately for the senior staff connected with statistical enquiries, the writing of the reports which measure up to these requirements is a time-consuming occupation since it involves not only careful selection of material but lengthy thought to try and place oneself in the position of users. Nonetheless, as has been indicated under the public relations section, this is in the long run well worth doing. It is also important to make sure that information is available in a form which will be valuable to the potential active and positive users rather than those who might be generally interested. They will often be businesses and marketing organisations, etc. It may be worthwhile consulting them specifically on this material. Of course, there may be situations where the release of information is a delicate matter but the sum total of embarrassment over a run of years is not likely to be greater with prior consultation than without it.

Last but by no means least, well prepared reports, including clear statements of the techniques used and the difficulties met are the most certain means of developing a steadily growing body of knowledge. Without them, workers in these fields will be learning by their own mistakes when they might have learnt much less painfully by their predecessors'.

APPENDIX A

BACKGROUND INFORMATION STORE

A. Introduction

1. The aims here are:
   (a) to ensure that any data which would be useful in planning new statistical studies—e.g. lists suitable for frames—are looked after and made accessible to those who might want them,
   (b) to ensure that records of the way statistical studies are carried out are made at the time in a form clear and accurate enough to give to those who read them later a true picture of the methods used;
   (c) to arrange that data from small-scale enquiries and other sources, which might be useful to supplement orthodox statistical studies, are brought together for the convenience of users;
   (d) to arrange that, notwithstanding extensive developments in formal statistical work, all possible sources of supplementary material are used to the full.

2. It might be assumed that whereas such aims might conceivably have been relevant 20 years ago suitable organisations designed to serve these aims would have now been introduced. Often they have—but their results are often less than perfect, usually because pressure of work prevents the full and consistent carrying out of the arrangements and also because the designer of each statistical enquiry is innately reluctant to believe that his results would be improved by admixtures from other less well designed sources.

3. Perhaps the most pressing need is so to plan the records that they *are immune to staff changes.* A big statistical enquiry represents a substantial capital investment in building a complex tool. As far as possible its functioning should be immune to any staff member crashing his car or taking a better job on the other side of the country. This is possible only if documentation is adequate in volume and properly organised.

4. The physical location and handling of the relevant data could be discussed at length, but the differences from country to country probably make this unrewarding. What is important is (a) that each group of data should have an acknowledged place and (b) that users and potential users should know what is there and where it is housed.

5. Some workers would argue—and there is an element of truth in the contention—that 'archival' work of this kind undertaken without a quite specific and individual enquiry in view becomes a tedious and wasteful chore. Insofar as some degree of focussing of interest is necessary to maintain thoroughness, this argument underlines the value of getting users and suppliers of data in contact with each other as often as possible. It does not justify making no early systematic arrangements to collect information and depending on the information that can be scraped together when an issue arises.

6. It would be well worthwhile to arrange an annual seminar for holders and users of agricultural statistical data where each opens a discussion by a statement of his viewpoint. A mixture of newly appointed and long-standing staff members should be included. The former will not only learn something but should provide jolts if the others are complacent.

Undoubtedly the time for this will be begrudged but it will save over the year much more than it uses.

7. Computerised data banks will be urged by some as the answer to all these matters. Be that as it may, it is assumed here that 'old fashioned' methods will have to be used for some while yet. There are many demands on available computer time and classification of a considerable amount of agricultural economic material for computerised retrieval is far from straightforward.

B. Information

1. *General Information and background*

The following indicates the kind of data which are likely to be needed at some stage of statistical enquiries. Where it is feasible they should be available in the headquarters office, if they cannot be centralised then their location should be known accurately including the position of local data, if these are housed separately from the national and regional summaries.

As many as is practicable of the following should be collected in the headquarters office. For material which it would be unreasonable to keep there, a precise statement of where they can be found and the precise form in which they exist (e.g. scale for maps, smallest unit quoted for statistical lines, etc.) should be noted.

(a) Records of population censuses;

(b) Lists of villages with the population, or other statistics indicative of population (e.g. tax lists);

(c) A set of the largest scale maps available for each section of the territory. Indications of the degree of reliability of the detail are valuable, e.g. it is known that in some territories the detail on the 1 inch to the mile maps, though sparse, is accurate; in others, it is known to give only a general idea of the locality;

(d) Air photographs, both serial and isolated. It is essential that the latter should be clearly identified and dated;

(e) vegetation maps and descriptive surveys;

(f) soil maps;

(g) geological surveys;

(h) maps of tribal areas, with accounts of any characteristics of their agriculture, published or collected from staff with local experience;

(i) meteorological data, with as full seasonal and district data as possible;

(j) data on quantities of produce marketed internally to the country and international movements, with location of main markets;

(k) a farmer's calendar for each area;

(l) maps of the country showing agricultural stations, observation farms, best check points on roads, important markets and bases for agricultural department staff, and marked with indications of the extent to which each area is under surveillance by agricultural department staff;

(m) a list of the staff available for field work with a statement of the place at which each is based and the area normally toured;

(n) villages with enlightened and co-operative members likely to assist in statistical enquiries noted on reference maps;

(o) a record stating specifications, and where they can be found, of all chains, compasses, rangefinders, weighing apparatus (spring balance, etc.), available to agricultural department staff and all weighing facilities at public markets;

(p) lists of farms, 'test plots', etc., within the responsibility of the agricultural department which might provide quantitative, even if biased, data on seasonality of production, crop yields, dates of various operations, etc.

2. *Frames*

(a) In many cases the records of the last population census will provide the most comprehensive and useful frame for sampling purposes. Special care should be taken to

ensure that they are complete and that no relevant records are destroyed.

(b) Where records have been used as frames in other enquiries a carefully annotated record of any pecularities and deficiences should be prepared.

(c) Associated data likely to be of use in stratification (size of units, location in agricultural zones, etc.) should be tabulated.

(d) The main agricultural sectors should be reviewed and the potential frames for enquries in each explored for short distance and their character noted. For instance, do coffee handling co-operatives have lists of producers? Does the agricultural department have lists of those to whom it has issued improved tea plants? Apart from any other value, such contacts can open up liaison, may shorten the preliminaries if a snap enquiry is needed later, and may prevent potential valuable records being lost by a more or less casual decision. Perhaps more important, certain pecularities of such data for statistical purposes may only emerge after several occasions of contact.

3. *Conduct of enquiries*

Care should be taken to provide a definitive record of the conduct of all major enquiries. This should include a memorandum outlining the purpose of the enquiry; any modification of substance; the resources of men, equipment, etc., used; the time taken for the several sections; and the cost. Often the total cost may not be computed because of the use of departmental staff or facilities supported on other sources of finance. If so, this should be stated precisely to give guidance to others planning similar studies later, but the manpower used should be clear. Copies of instructions to supervisors and field staff, and of all relevant forms and record sheets should be included, too.

4. *Results of quantitative studies*

(a) *Published.* It can be confidently assumed that the published reports of major statistical censuses of substantial geographical coverage will be well known and readily accessible in the country concerned. Once one looks beyond these it is unwise to be confident that those who might be interested in results of a piece of work will know of its existence.

It is well worthwhile arranging for the preparation of an annoted bibliography of all quantitative agricultural studies (e.g. agricultural and population censuses, sample surveys, enquiries by marketing boards, bankers, universities, etc., including food consumption, family budget and farm management subjects). A draft bibliography should be circulated to several different interests–e.g. university library, government agricultural statistics department, chamber of commerce–in an effort to make it comprehensive. Some degree of personal contact may be needed to stimulate the necessary attention. The place where each study can be consulted should be indicated.

(b) *Unpublished data.* Large quantities of potentially valuable tabulations may not be published. Commonly they arise as follows:–

(i) Only summaries may be published from big enquiries and many details may be left in typed form.

(ii) Special tabulations, and computations of derived figures, may be made for particular purposes which could be useful for others if their existence were known.

(iii) The results–official and otherwise–may not reach beyond typed form or the duplication of a few copies because of shortage of funds.

Typically, material in these categories are scattered in different files and different offices and, after a few years, no one may be able to locate it. It is very much to be desired that a pooling system be arranged so that a spare copy of all such material is provided as a routine for the pool. Whether a government department, a government library or a university keeps it may not matter greatly if the principle of access is accepted. The report of an enquiry should state where such supplementary material is filed.

(c) The archive might usefully also contain some indication of the form (e.g. punched cards, tabulation sheets, etc.) in which the original records are available for the use of other workers and, if so, where.

5. *Weights, measures and other data*

A reference list of all weights and measures customarily used in the locality, whether recognised for statistical purposes or not, should be compiled with their equivalence in standard official measures. The source of these figures should be noted carefully and in particular conventional figures (for instance, these where a reference tends to take the form of "this unit is taken to be x pounds." Where actual observation would show that it varies from district to district, and is probably in no case very close to the figure customarily "taken").

6. *Cataloguing and Classification*

There is likely to be a very wide range of different kinds of material in a body of information of the kind referred to here, and it may never come together physically in one place. Three points only might be made here. First, it is usually worthwhile consulting professional librarians to get some idea of the basis of classification, even though modifications may be needed to handle the particular detail of information that the specialised field embraces. Secondly, there are available a number of methods of rapid location of sources of information which do not depend on expensive equipment but yet can be highly effective. Two which suggest themselves are the edge-punched cards (well known to most statisticians) and "feature" cards. For those not familiar with them, the forme come in various sizes, and they can be printed to order. Holes are punched about ¼ inch from the edge at regular intervals all the way round. Information is coded and assigned to particular holes. Thus one batch of holes might relate to crops, another batch to types of statistics and a third to geographical regions. A card which contained a reference to information on cassava acreage in Region A would have the margin outside the cassava hole, the acreage hole, and the region A hole, clipped out. A batch of such cards can quickly be sorted in such a way as to bring out all the cards concerned with cassava acreage, for example, merely by placing a slim rod through the "cassava" hole, and lifting it when all the unpunched cards will come up with the rod leaving the cassava ones behind. Those relating to cassava acreage can then be selected by sorting on the acreage hole in a similar way.

A less well known device, which is extremely effective where the situation suits it, is that of the "features card". In this system the cards are ready printed with a sequence of numbers from 1 to 1,000. Each of the sources of information in which one might be interested (for instance, reports) is listed and a number assigned to each. One of these printed cards is assigned to each "feature" in which the user is interested, for instance one card might be assigned to cassava, one to maize, and so on, and again a card may be assigned to acreages, another to production, etc., etc. The various sources of information are studied and if they contain information about cassava the number relating to the information source is ringed on the cassava card; if the source contains any information about acreages its number on the area card is marked, and so on. These marked numbers are then punched out with a special punch for the purpose, leaving, on each feature card, the numbers relating to references containing information about that "feature" as holes scattered across the card. If one now wants to know in which sources one can find information about cassava acreages, the cassava and the acreage cards are selected, superimposed upon each other and looked at against the light where the references required are indicated by holes right through the two cards. Any combination of features can be dealt with in this way, giving a very rapid location of reference material. This method uses much less space and fewer cards than the previous one, but naturally it is necessary to go back to the original reference to get the actual information.

APPENDIX B

PREPARATION OF FORMS, SCHEDULES AND QUESTIONNAIRES

1. *General Instructions*

Forms are used for the recording of definite, concrete, usually quantitative and objective data in a formal rather than an arbitrary manner. They ensure uniformity in the material collected and reduce the risk of recording only the obvious features of the subject. They are a valuable means of limiting the scope of enquiries. The agricultural economist and statistician use forms both in the collecting of data from private persons, special recorders or commercial agencies, and in the collation and analysis of records in the office.

The drafting of forms usually receives much less attention than it should. The structure of the form, and even the precise form of words used, largely determines the nature of the entries made by the informant: the mere drafting of a form and the instructions accompanying it ensures that there is keen scrutiny of the scope of the study of which it is part.

2. *Guiding Principles*

(a) If the information lends itself to collection in a standard pattern it is usually better to use a form than to try to edit the uniform versions from informants' letters or from transcripts of verbal statements, etc. A form also acts as a check list of the items required. Nevertheless, the number of forms should be kept down and new ones should be closely linked with those already in use. Headings and questions common to two or more forms should have precisely the same wording and be arranged in the same sequence.

(b) The sizes used should be rigidly standardized—6 x 4 inch and 8 x 5 inch (i.e. standard index card size) and quarto and foolscap usually give sufficient variety. A form larger than foolscap should be used only when it has quite specific advantages over the use of two smaller forms. If metric paper sizes are being brought into use in the country the choice of one of these sizes should be considered.

(c) The use of carbon copying, the clear naming of forms, the careful placing of totals and other key data. and the use of different coloured paper for similar forms with different uses, (e.g. where the same layout is used for recording the cotton acreage statistics for the several farms in a village and for the various villages in a district) are simple ways of saving time and increasing accuracy.

(d) Where possible, entries should be limited to single marks, words, or figures, or the deletion of unwanted items.

(e) A good quality paper and good quality printing should be used.

(f) There is much to be said for associating with the form a name known to the informant. In some instance it may be possible to arrange for a common heading on the form and on a Paramount Chief's circular about the enquiry. Forms to be completed voluntarily by informants (i.e. crop reports) might carry the name of the Minister of Agriculture. If there are no obvious names it may still be worth putting the *same* name on all forms so that in due course it becomes familiar.

(g) Where forms are sent by post or messenger to an informant, it is often helpful to include with them something he is glad to receive. Voluntary reporters might find that suitably printed summaries of the results of previous reports are interesting. Survey teams might welcome a brief summary of the doings of the various teams on the job, personal news, etc., especially if each is working in isolation.

3. *Sources of information*

(a) For forms on which information obtained by personal enquiry is entered it is usually best to start with simple questions and to leave awkward ones until the end.

(b) The "natural" order of entries for the informant should be used. Where more than one person contributes to its completion (e.g. where certain entries are copied from business records but others require expressions of executives' judgements) the questions for each should be kept together.

(c) For summaries requiring information to be transferred from another form, both the order of the entries and their wording should be identical on original and summary forms.

(c) Vertical working is to be preferred to horizontal for addition and subtraction. Cross casting is easier if items are staggered diagonally, e.g.

£
  s
    d.

(d) Allowance should be made for the longer entry in the total space than in those above, either by staggering diagonally or by extending the entry under the "description" columns.

(e) Items to be added from several forms should be arranged at the edge; then, by arranging a bunch of forms to overlap, like tiles on a roof, continuous lines or columns for summation are formed (see Appendices E & F).

(f) It is often helpful to leave a column blank in which coding data can be entered, particularly when tabulation is to be by mechanical accounting equipment (see below).

4. *Entries*

(a) For hand entries ¼ inch per line vertical and 1/8 inch per character horizontal spacing should be allowed for fully literate informants and rather more for others. A rough paper is preferable for pencil and a calender for ink entries.

(b) For typewriter entry 1/6 inch per line and 1/10 inch per character is usually required. Vertical spacing should be maintained in multiples of 1/6 inch irrespective of intervening matter. This may be important for forms to be completed by large firms.

5. *Copying*

(a) It is important to test samples of proposed forms and carbon papers under the actual conditions of use: copies may be unsatisfactory or the carbon may soil copies, particularly in a hot humid climate.

(b) "Master" entries relating to a particular sector of an enquiry can be duplicated on to standard forms, provided sufficient space is allowed in each entry position to allow for slight inaccuracies in registration. Thus one family record form might be drawn up to cover an enquiry in ten villages but reference data for each of the ten villages might be duplicated on sufficient copies to cover the number of families in the village. Spirit duplicators may be useful for this work but the scope of available types of copying and duplication facilities of all kinds should be explored.

6. *Adding and calculation*

(a) Fine lines or lines in pale ink help to keep columns of figures in order for addition and subtractions. Entries for deduction can be more easily identified if the space is stippled.

(b) Separate columns should be used for figures in different units, e.g. the use of alternate lines in the same column for *quantity* and *value* of sales of various items should be avoided;

7. *Filing*

(a) A margin should be arranged on the left side of the front and the right of the back of a form to avoid its contents being obscured when filed or bound. If further entries are needed after binding this margin should be sufficient to allow the parts of the form on which entries must be made to lie flat.

(b) Form names and instructions for filling in the form can be printed in this binding margin.

(c) If forms are likely to be referred to frequently over a long period important entries should not be placed within ½ inch of the "outside" edge or in the top and bottom corner or they may be obscured by "dog-earing".

8. *Checking design*

Before a form is put into use it should be checked by:–

(a) writing detailed instructions for those who have to use it;

(b) trying out the form and instructions on some of the kind of people who have to use it;

(c) making a pilot study under the actual conditions under which the form will be used;

(d) testing the convenience of entry spaces for typewritten, and manuscript entries as may be appropriate.

9. *Wording*

In principle, record forms on which field staff enter their observations, questionnaires which are completed during interviews between enumerators and informants, and questionnaires which are sent to members of the public for completion voluntarily, present different cases. Thus it is the enumerators' job to complete the record forms and they can be drilled in the procedure, whereas no one can compel an informant to co-operate. Nonetheless certain basic principles are best borne in mind for them all.

(a) The language should be understandable by the *least* intelligent of those who are likely to use it, not by the average man. This may demand repeated and patient scrutiny and editing of all forms and instructions by two or three people after pilot trials.

(b) Questionnaires to be used by enumerators should specify the actual form of the question to be asked since minor differences in the form may affect the answers given. (Cases of very frequent visits where open or mildly steered discussion are used to elicit the required data are exceptions–they carry risks but rigid questioning would carry greater).

(c) Questions which the informant cannot answer are best avoided, otherwise he will be encouraged to guess and/or the enumerator tempted to use leading questions. For example, in collecting data in conditions where management is casual it is pointless to key the questionnaire to age categories which the owner may not recognise. It would probably be better to ask for a total and then see if figures can be elicited of how many of these are breeding cows, how many calves.

(d) Leading questions must at all costs be avoided e.g. "How many bags did you harvest from this field?" "Oh a poorish crop". "Well, how many?" "Oh, bad, pretty bad". "Come on, you must know how many you sold–would it be 8 bags?" "Yes, about that".

(e)If cross questioning is expected in the enquiry, for example in (d) if the reply on how much is sold can be followed up with "how much did you get for it" . . . "What did you buy with that?" . . . "That comes to ten shillings more, where did the extra come from?" (and some enquiries have used this successfully) then space is needed for the notes.

10. *Coding*

Coding comprises many devices e.g. the use of different coloured paper for forms to be used at different stages of a study is one. So are arrangements of data to get the best out of the tabulation facilities available. Coding for this latter purpose requires an intimate knowledge of the technical features of the equipment. It is essential, if mechanical or electronic equipment is to be used for tabulation, that the experts in this equipment should be consulted in the early stages of designing the questionnaires.

11. *Instructions*

Detailed check lists should be provided as part of the instructions for completing schedules which are given to recorders. The preparation of these check lists should be the personal responsibility of the organiser of the study. Their nature will vary with the entry, e.g. a full list of crops grown in the region should be provided with a schedule about crop production, a list of dealers for an enquiry about cash crop disposal. Check lists are of the utmost importance since an informant will usually say whether he has grown a specified crop but may not mention it in response to the question "What crops did you grow?"

12. *Internal design*

(a) Space should be provided for precise description of unit, recorder and informants concerned, e.g.

Name of person ..... whose particulars are given

Household no. .....

Village .....

District .....

Map reference .....

Name and no. of recorder .....

Rank .....

Date of record .....

Period covered .....

Person supplying information (if other than person concerned or if household or higher unit concerned) .....

Checked by ..... Date .....

Comments .....

(b) Either precise instructions should be given on the form as to its transmission, or spaces provided in which the recorder enters to whom the form is to be sent and the date of dispatch.

(c) A space should be provided in which the person scrutinizing the form can enter comments, or instructions for a check to be made, and finally, his agreement, with signature and date.

(d) It is a sound rule in scientific experimental work to book readings precisely as they are made (e.g. if weighing machines read in lb. and ozs. and a reading in ozs. is wanted, book in lb. and ozs. and convert later). Similarly in surveys, enough entry spaces should be provided for each item of information supplied. Thus if crop area is being recorded, the informant should be asked about his holding in each of the village's blocks of cleared land and an entry of his individual answers made. The informant's own units of measurement (or the recorder's actual measurements) should be used and any necessary conversions made later.

(e) At least on summary forms, in addition to a space for totals, it is frequently useful to provide spaces for averages and for ratios, particularly when numerator and denominator are widely separated. Two entry spaces for such divisions are useful. In the first an overseer

enters the denominator, the form is then passed to the computor to do the division and enter the result in the second space.

13. *Scrutiny and checking*

(a) The form should be internally self-checking as far as possible, particularly by
(i) approaching each item from at least two viewpoints, e.g. by including a sale of grain among the items of utilization of the crop and as item in income,
(ii) recording both individual items and their sum (e.g. crop harvested from all the individual fields together equals the reported quantity stored plus immediate sales),
(iii) inserting a general question the answer to which immediately indicates gross unreliability. These must be suited to local conditions, e.g. in regions where every wife has a specific kind of "garden", a question might be inserted on the household record form asking how many such gardens were made. A quick check of this answer against the household composition shows up grossly 'cooked' records. A question about extent of weediness can give similar pointers when compared with the recorders' comments on crop yields or the standard of husbandry.

(b) Completed schedules should be checked by supervisory staff at the earliest possible time after completion, preferably the evening of the same day. Queries can then be followed up with much better chance of success.

(c) Arrangements should be made to check that the number of forms received for tabulation corresponds to the number which should be present. Once a bundle of forms has been checked as complete there is much to be said for binding them in some way. If they can be punched a filing clip is a cheap binding device. Latex adhesive brushed along the "spine" edges will bind successfully if they can be pressed together for a few minutes.

APPENDIX C

MEASURING, WEIGHING AND TRANSPORT EQUIPMENT

1. It should be kept in mind when considering equipment for measuring and weighing in the field that there are many sources of error in this work. For example, two careful observers may differ appreciably in respect of the position at which they draw the boundary of an irregularly shaped plot planted to a crop. Under these conditions the inaccuracies of various measuring instruments would be rather minor contributions to the total inaccuracies of an enquiry. This is not to say that inaccuracies should be lightly condoned, but rather to warn the user against straining at a gnat and swallowing an elephant!

2. *Field Measurements.*

a) Surveyors chain. This has been for years the standard field measure for general purposes and has much to commend it. It is not upset by a certain amount of mud or moisture, is reasonably portable and is robust. It should be handled with reasonable care, being picked up and moved on as required, not just hauled by one end through the scrub, and should be checked regularly for distortion.

b) Measuring tapes. Linen tapes no doubt change their length somewhat with changing conditions but it is their lack of robustness rather than this sort of variation which makes them unsuitable for work in many developing countries.

Steel Tapes are much favoured by some workers, and there seems good reason to use them provided that conditions are not too muddy. If mud is wound up on them trouble can usually be expected; they need reasonably careful treatment.

c) Measuring wheels. There are a number of versions on the market, differing in the size of the wheel, the link between wheel and counter, and the scaling of the counters. Instructions are available with each of them. Unless the land is reasonably smooth there seems some doubt about their reliability (see Area measurement), but, provided wheels are acceptable at all, the small ones which are dismountable for carrying have much to commend them.

d) Pacing. This method of measurement has been widely used from time to time; statisticians regard for it varies greatly. It is fairly clear that pacing is unreliable over terrain with greatly varying surface, through bush, and on steeply sloping ground. The method should therefore be used with a formal calibration arrangement. Different users have different views of this. It is clear that asking a field pacer to pace under observation is likely to produce doubtful results. It is probably better for the supervisor or team leader to check with a chain or tape a proportion of the ordinary work of the pacers after the pacing records have been entered, and to use the conversion factors so calculated for transferring all the paced records to metres or yards. This arrangement has the twofold advantage of providing actual practical conversion factors and of taking the "test" element out of this work. It should be possible to regard it as merely part of the machinery of preparing the final estimates. There is, however, probably little actual advantage in this closeness of control. Most of the cost of these adjustments comes not from doing the tests but from having to apply different conversion test factors to different peoples' records. It seems at least possible that a test would show that a mean conversion factor for the whole of a team determined soon after they start work in the field would be very close to the true factor right through the whole of their working season. It might be rash to lay this down as being so without test but it may well be worth checking its truth at a fairly early stage in an enquiry, and introducing the simplifying process if it seems justified.

e) Hand-made versions of measuring devices using ropes, cords, or wires. Knotted cords and similar home-made devices may be quite serviceable for many purposes if it is necessary to economise on equipment in this way. It is true that they will stretch or shrink with varying

conditions, but if they are checked from time to time, and suitable adjustments made, quite good service can be obtained from them. They would certainly be better than pacing under difficult field conditions.

f) Pedometers. These have been used from time to time in land use work covering distances of a score of miles or so; they seem rarely to be referred to in agricultural statistical work in recent years.

3. *Weighing Equipment*

Generally three kinds of equipment are potentially available for field work, namely the spring balance, scales using separate weights, and the steelyard. Different users have strong views about these various items. The advantages of spring balances are that they are portable and readily obtainable in various sizes. Their disadvantages are that they are not a precision instrument, and that they are capable of giving considerable errors, particularly if ill-used. The scales with weights are not usually chosen for crop yield type estimation, but rather for household consumption studies. For this purpose—which is in the main on the margin of the subject of this manual—they are probably the most satisfactory equipment but they are not particularly easy to transport. Steelyards are regarded as the ideal equipment for crop weighing since they are not subject to lengthening of the spring and consequent inaccuracy with use, and their life is long. However, they may not be as easy to obtain as the spring balance and they are often a good deal less easy to transport unless some form of motor transport is available. The first approach in this appears to be to consider the possibility of obtaining the various steelyards, and if they can be obtained in the numbers and the delivery date, if these are acceptable, then they probably represent a good investment. If not, then spring balances should not be despised but should be treated carefully, checked regularly, and ruthlessly culled if they are found to be unsatisfactory. Obviously the order of magnitude of the quantities to be weighed will need to be noted and balances with a suitable weight range chosen. Those which will handle the larger quantities are not likely to be calibrated in very narrow divisions. Various special devices are available on certain balances, adjustments for error, and so on, but, as a general rule, it is better to have a simple, easily obtainable type than a special version which can only be replaced by special order from a distance. Since much of the weighing must be done in the field some support for the weighing equipment is needed. In forest country it is not difficult to cut three poles and wire them together to form a tripod. In intensively cropped country or in Savannah areas this may be difficult and the field staff must be provided with equipment which they can carry. The poles that are used for marking the corner of plots, provided that they are fairly thick, can be used quite readily by drilling a hole through each, near one end, and putting through it a bolt of suitable length and slightly bent retaining three together with a nut. This arrangement will allow them to be used as a tripod and when required for ranging purposes the nut is merely unscrewed and replaced. Some arrangement for replacing the nut or for keying it to the bolthead so that it is not lost might be an advantage. Similar arrangements could be devised using tubes instead of poles. Another device developed in some areas uses a stout metal pipe about 5/8 to ¾ inch diameter. The individual rods should be 7 or 8 feet long with one or two joints, the ends being threaded accordingly.

A plate of hardwood or metal (depending on workshop facilities) about 9 inches in diameter can be provided to carry a hook for suspending the weighing equipment. The connection of the rods to the plate will again depend on the workshop facilities. The neatest provides for 3 hinged mounts on the top of the plate with connecting rods bent over the edge to connect with legs in the usual tripod position. This leaves ample free space for hanging the weighing unit.

Some thought needs to be given to the conditions under which this equipment is going to be carried, not only between headquarters and villages and between villages and farms but right down to the spot in the field where the weighing has to be carried out. Two points must be considered, the advantage of having very light equipment and the need, in that case, for suitable protection during travel. It is not very helpful to save two pounds by redesigning the weighing, measuring and tripod equipment, and to add five in providing a heavy wooden

case to carry it all. A heavy case may not be much of a disadvantage over the distances where motor transport is used, but there should be a lighter device, (for instance, a canvas holdall) sufficient to give reasonable protection from the end of the road to the field point.

4. *Compass*

There are two features of a compass which are of interest in this work; the precision with which it can be read and the speed with which it settles down so that a reading can be taken. The standard type of spirit-damped prismatic compass, which is widely used for this kind of work, seems entirely satisfactory. Occasional users may make an error in reading arising from the fact that usually only one actual number (indicating degrees) is visible in the sight at one time, and it is possible to misread 282, for example, for 278. Some makes have a convention in which the lines indicating scales divisions on the upper side of a number may be short and those on the lower side long, and this may be of help to some users, though probably not of significant advantage to the kind of field worker discussed here. If the cost of the standard model is a problem in particular enquiries, it might be worth exploring for lower priced ones. Variations come onto the market which have been tested in various situations and have proved quite reasonably successful. Probably any that have spirit damping would be found serviceable enough for area measurements of fields and sketches of villages.

5. *Cross-staffs*

These are used to locate a direction at rightangles to a given bearing. Optical instruments for the purpose are on the market. Home made equipment has been used successfully, however. For example, one has consisted of a piece of good quality wood about 1 foot square and 1 inch thick. It is rather precisely squared. The opposite corners are connected by diagonal lines and a nail driven in on these lines at about 1½ inches from each corner. Some means of attaching this wood to the top of a rod is needed so that it can be set up in the ground as required. One way may be simply to bore a hole through the centre of the wood–say 1 inch diameter and taper the end of the rod accordingly. Another version has consisted of an aluminium alloy disc about six inches diameter with some form of attachment to a suitable rod–usually, in fact, a tube. Grooves at right angles are cut in the face of the disc passing through the centre. In the wooden model sighting is along the pairs of nails, in the metal one along the grooves.

6. *Volume measurements*

There are occasions when it would be very tedious to handle initial measurements by weighing on the available scales, for example, where large volumes of a grain or of a fruit are harvested, perhaps from a complete plot or complete tree. On occasion, rain gauge cylinders have been used, and often the ubiquitous kerosene drums. It is important to standardise the fill when they are being used for formal recording. This means that the way the material is put in should as far as possible be the same for all measurements; this may be more important for irregularly shaped objects (maize cobs, for instance) than for grain (which naturally settles down). The measure should be shaken or jolted a standard number of times and the top swept off with a piece of wood. When used in this way a random sample of the contents would be weighed in order to convert to weight terms.

Incidentally, if records have to be made where very little equipment is available it may still be worthwhile counting units (for instance, yam tubers, maize cobs, etc.) and taking a quite small quantity away for measurement under reasonable conditions. When this is done it is desirable to wrap the product in some sort of waterproof covering in case a notable drying occurs between the time of count and the time when weighing can be carried out. Polythene film is so widely available now that this, or waterproof bags, are likely to be available. This note is really part of a repeated plea in this bulletin not to pass over an opportunity to obtain serviceable and useful quantitative information simply because the *most* suitable and desirable techniques for such a purpose are not attainable at the time.

7. *Travelling equipment*

The availability of transport varies greatly not only from enquiry to enquiry but over the distance from headquarters to the field and, sometimes, from time to time. One general point must be stressed, namely that it is not sufficient to note the existence of transport available on the record; it is absolutely essential to ensure that it is usable, or if it cannot be made usable, that the fact is known early. Repeatedly, plans for large scale enquiries or enquiries at a distance from the headquarters have been badly disrupted because transport was not available at the time that it was expected. One might consider four main kinds–bicycle, motorcycle, landrover, and hired lorry transport. Dealing with the last first, the main difficulty, providing that such equipment is there to be hired and that it is reasonably reliable, is that it is often difficult to foresee when a particular job will be finished or will have reached the stage when movement of personnel has to be arranged. It is important to match flexibility of transport with uncertainty of timing and, if they will not match, to arrange some buffering device, such as subsidiary enquiries to be undertaken by fieldstaff while they are waiting for the date when transport can come for them.

Bicycles have considerable advantages, being usable on most bush paths and on roads. They can carry very considerable loads–tripods, ranging rods, and some types of weighing equipment can often be broken down into units which will fit along the bar or on the carrier. Sometimes staff can be expected to provide their own bicycles, being given ån allowance accordingly; some enquiries, which have used temporary staff with whom a continued contact is certain, have made bicycles available, or funds to buy bicycles, on condition that at the end of the job the bicycle would be the property of the field workers as part of the wages. Obviously this is only a feasible proposition if the staff are known and continued contact can be ensured (this has been used mainly for students). It is worth bearing in mind that the bicycle wheel gives quite a serviceable measuring device if the bicycle is pushed along and the number of times the valve comes round is counted.

Motorcyles have been used in a number of enquiries, mainly for field supervisors. With suitable panniers they can carry a substantial amount of materials. This is, however, a case where penny wise is certainly a pound foolish, and those who have tried to manage with 98 or 150 cc. machines have met trouble. If motorcycles are used 250 cc. machines should be standard.

Landrovers and similar items are so well known that no comment is sensible here, except to emphasise again the need to check on availability.

APPENDIX D

FIELD WORK ON MAPPING, SURVEYING AND MEASURING PLOTS[1]

1. The following notes are mainly concerned with simple techniques of land survey which might be used by intelligent, but not specially highly trained, field staff. Only very simple equipment is assumed and the main work in mind is the preparation of simple sketch maps of villages and the estimation of the areas of fields or plots of land. It is envisaged that considerable numbers of these basic records will contribute to the final estimates of totals and means in sampling studies and that, in consequence, freedom from bias is more important than the highest order of precision. Doubtless high precision would be desirable but to get it would greatly increase the time needed to do the work, both because of the increased care needed and it might reduce the number of staff effectively available, as some who could handle simple equipment would be unable to master more intricate methods. Furthermore, if heavier items of equipment were needed—a plane table instead of a compass, for instance—it would mean re-organisation of the transport arrangements of the teams. Other equipment may be difficult to use under field conditions, for example chains might be preferable to pacing for the measurement of distance were it not for the difficulty of working chains through bush. Finally, there is often such uncertainty about where the boundaries of a plot should properly be drawn that it would not be justifiable to try to make measurements with an accuracy greater than, perhaps, plus or minus 1 or 2 yards.

2. *Field surveying and mapping*

(a) The most frequent need is for a sketch of a village and its neighbourhood, using pacing or measuring wheel and compass. The work entails choosing a base point, taking bearings and distance outwards along the paths to the boundary booking them in a systematic manner and preparing a rough sketch on the way. This may need to be repeated along the actual boundary though it may be possible to sketch this in by linking the ends of the paths. The idea is straightforward enough but the following notes list some of the details.

(b) The work consists of marking points, taking bearings, measuring distances, recording, and labourers' work. Usually two or three men per team are better than either more or less but the exact organisation varies a little with terrain and the capacity and experience of the men.

(c) It is usually convenient to have one man with a conspicuous pole to act as marker. He stands at the end of each 'leg' of the route and the compass is sighted on him. Often a long pole, much longer than the ordinary surveyors' ranging rods, with something conspicous tied to the top will allow sighting over intermediate bushand permit longer 'legs' and fewer sights to be used.

(d) Though some workers sight forward, sending the man with the marker pole ahead until ordered to stop, the smoothest working usually comes from back-sighting. Two men can do this if no cutting of paths is needed. The one in charge takes the compass and makes the records. He goes ahead, chooses his station, and sights back on the marker who is told, when the bearing has been obtained, to come up to his leader, measuring as he comes. The leader then moves ahead to the next station. If there were three in the team, marker and compass man could move simultaneously to their next stations. However conspicuous may be his pole, there is some advantage in the marker wearing a light coloured shirt.

(e) Each station should be noted by a letter on the sketch map. The best routine of booking of distances and bearings depends rather on the field conditions and the programme for working-up the data. Sometimes it may be feasible to enter all the necessary details on the sketch itself. Unless the outlines are simple, however, some formal manner is essential for

without it records become a hopeless muddle. This is especially noticeable if field records are sent into the office for mapping or computation. Even if the field staff do their own mapping, such formality is worthwhile, for a record which is not perfectly clear to a second person will probably not be clear after two or three days to the man who made it. Field measurements can be booked on a surveyor's field book, or in the same form on separate sheets of paper so that all the documents for one village can be kept together. A simple alternative form, favoured by many agricultural officers, consists of a rough sketch of the area with the main stations lettered, and the details of the measurements recorded in three columns:

Station	Bearing	Distances
A from B	247°	240 yds.

Such a record could be used for a succession of short 'legs' along a winding bush patch where it would be very difficult to enter the figures directly on to a plan.

(f) Advantage should naturally be taken of any natural landmarks along the paths but experience shows the advantage of driving in numbered marker pegs at suitable points as base points for locating sample plots, etc. Care must be taken that these pegs do not call forth suspicion amongst the local people about compulsory buying of land, and the like.

(g) In open grassland country where long 'legs' are possible no special difficulty in taking bearings may emerge but in forest country the bush is often left untouched adjacent to paths so that any slight bend in the path obscures the view along it. In consequence, compass-pacing traverses proceed in legs of a score or two of yards. A long marker pole may help in some situations. In others, it may be found that the kinks in the path are all minor ones so that a path plotted in 50 yard legs in fact goes virtually straight for 2 or 3 miles. If so, with practice, it may be possible to pace outwards, watching the compass and using no marker stations except natural landmarks, reference pegs and points where the path distinctly changes its direction.

(h) If none of the methods of the previous paragraph is successful the sound survey technique may help. In this the compass man goes ahead, the marker calls at intervals and the bearing is taken on the sound. This can be surprising successful, even in thick bush.

(i) Some of the details of preparing a map from such a field record are discussed in Appendix D. Usually it would be drawn at the end of the day when all the teams are back at the village for the night, or after a meal break. If the map is to be used as a basis for sampling (e.g. p. 33) there is obvious advantage in getting it done soon and the question presents itself—why not make the map from the measurements as they are made? This can be done, particularly if the drawing board can be conveniently carried but prevailing opinion is that it slows down the work because protractor, ruler and large sheets of paper have to manipulated in addition to the compass, marker and booking sheets which must be carried in any event. Moreover it makes the work much more vulnerable to weather. Against this,, it probably does reduce the risk of silly errors in booking bearings.

(j) Though it may not be worthwhile to attempt to draw reasonably accurate sketch maps while taking bearings, there may be situations where rather rougher maps might be useful. With practice useful rough sketches can be drawn by eye on wide-lined writing paper, or squared paper, from compass and pacing data. Angles are guessed in relation to ½ and ¼ of a right angle and a scale is chosen using the width between the lines of the writing paper to represent, say, 100 yards. It is preferable to complete traverses back to the base point to provide a cross check.

3. *Counting of trees, etc.*

Among aids to mental counting are the tally counters used by some tally clerks and the record on paper entered in fives—four upright strokes and the fifth as a cross stroke, Usually the difficulty is not the actual counting but making sure that the tally is complete

without double counting. One useful device is to issue each counter with a known number of palm fronds, pieces of climbing plant stem, etc. one of which is to be tied round each tree. The team leader checks that all trees in the specified area have one and counts those fronds which remain unused.

4. *Estimating area of plots and fields*

It is desirable for the field staff on coming to the plot to appraise its shape and to make a freehand sketch as a routine. This will enforce some preliminary planning and tends to prevent silly errors.

Where field boundaries are irregular it is extremely important to impress on field staff the need to keep in mind not only the conditions at the corner itself but *also* those on the line from there to the adjacent corners. Thus, suppose the field boundary changes direction 5 times and so would be shown as a 5 sided figure A.B.C.D.E., starting from A point B should be chosen not only to approximate to the point where the side changes direction but such that the line AB leaves as much area under crop outside it as it leaves scrub inside. Some eye to the line to C may be needed, too. The economy of a team of 1 rather than 2 may be more apparent than real under difficult conditions of this kind.

A serviceable 2-man drill might be as follows, using three poles; start at A, leader going forward with one pole, chooses B with an eye to the general position of the corner, and the give and take lines which will designate sides AB and BC and puts in his pole, then takes back bearing on pole held by assistant at A. Assistant leaves one pole in ground at A and, if wheel or pacing is used moves up to B, measuring as he goes, and carrying one pole which he hands to the leader who moves on to C, and so on, to the final leg bringing the pair with their poles back to A.

If a chain or tape is used then when the leader reaches B, puts up his rod and takes his back bearing he must return to A to take up the chain, etc., and measure through to the rod at B. At the other extreme from this drill, is another in which one man with a compass and measuring wheel and no rods can start at one corner of a field, A, decide on a piece of vegetation which is at the point he regards as B, take a forward bearing on it, measure with the wheel up to it, repeat for C. D. E. and back to A.

Why be complicated when one can be simple? The answer is that if the fields are clear cut, with straight sides, obvious corners and if the staff are experienced and conscientious the simple method will work. However, in more difficult cases, e.g. in shifting cultivation systems, under this arrangement the fieldman is unlikely to strike the "right give and take line", if he is diverted by bushes he may mistake the point he had selected for the corner, and he may not return accurately to his starting point. Moreover, it is virtually impossible for a supervisor to repeat the measurements as check, which he has a good chance of doing by the first method.

If one-man teams are essential then the field man should be required to cut marker poles for the angles and set these out *before* measurements are made.

Readings can either be booked in a field book

A from B	60 ft	230°	
B .. C	80 ft	93°	etc.

or on the sketch of the plot. There are supporters of each If closing errors and areas are being calculated on a computer it is probably best to book simply the figures.

If the plot is sketched it should be approximately in the right orientation—i.e. the sides should be in about the right direction compared to a N at the top of the paper. When the bearing is entered on the sketch it should immediately be obvious—to the supervisor if not to the recorder— if a bad error has been made. Further, failure to enter figures for each side shows up vividly. This should help to prevent returns of bearings and distances which quite obviously cannot close.

Much of the above holds true, with fairly obvious modifications, where it is decided to break all areas into triangles and to measure the lengths of sides only (or one side and the perpendicular). Data from this method are usually better returned to the office on sketches.

5. *Recording by sketching on maps*

Under certain conditions field records can usefully be made by entries on outline maps. It is difficult to describe typical arrangements since each tends to be a special case because of its combination of details required, quality of field staff available, area of country to be covered, scale and detailed precision of maps available and facilities for map reproduction and enlargement. Two extreme cases might illustrate possibilities:

(a) Maps of 6–10 inches to the mile available and copying facilities; areas planted to various crops required. If parcels of land are registered and registration numbers are on both maps and on local field records, the crop planted on any specified area can be obtained by sending appropriate parcel numbers to the field recorder with instructions which cover the case where one parcel is divided with a different crop on each. If they are not registered, copies of the maps are sent to the field staff with a shading and designation code and he is instructed to sketch in the boundaries of the various crops on the maps before return.

(b) The distribution of various types of vegetation is required on an area of many square miles where air survey photos are not available.

Districts should be grouped under the charge of a senior officer who is responsible for allocating work, maintaining staff and checking the results. The area under his charge should preferably cover one or more complete survey sheets.

Maps are issued to senior officers who have a tracing of their area including paths, streams, etc. prepared from each. The map area is then divided into compact sections using streams, paths and other natural features as boundaries, for one of which each field group (e.g. one agricultural instructor and two labourers) will be responsible. These section boundaries are marked on the tracing and each section of the tracing cut out and given to the field officer concerned. When he gets to his section he makes a quick reconnaissance to get his bearings and identifies important points, e.g. stream junctions. From such convenient points he puts in pegs at standard intervals, e.g. 220 or 440 yards, as a guide for pacing. On this basis he completes the tracing for all major items, paths, rock outcrops, etc. marking in at the same time the position of crops, vegetation, etc. to be noted. As each section tracing is completed it is returned to the senior officer who compares adjacent sections and builds into a full cover for his area. All field staff must be taught careful pacing and their work checked periodically.

Such methods can also be used when large scale maps are not available, but accurate smaller scale maps, say, 1 inch to the mile are provided that such information as it recorded on the maps is substantially accurate, i.e. any deficiencies in topographical survey are in absence of detail rather than low accuracy, and facilities are available for making enlargements for known magnification.

6. *Plane table survey*

Occasionally a plane table technique can be usefully fitted into a field organisation but it now seems to be rare. It seems most likely to occur where one organisation borrows staff from another where the plane table is a regular tool. It usually means more plotting across planted fields than a compass-tape type of survey (with the consequent difficulty where crops are high) and the equipment is less portable than the compass and less flexible for other uses e.g. location from maps.

APPENDIX E

OFFICE WORK ON PLOT AND MAPPING MEASUREMENTS

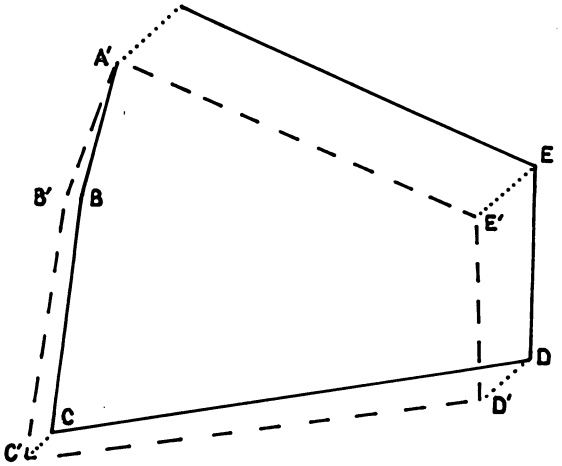
1. (a) Drawing scale sketches from length measurements only. Draw out side AB of triangle ABC with ruler to scale. Locate C by the intersection of arcs drawn with a pair of compasses with radius AC from centre A and BC from centre B.

(b) Drawing scale sketches from length and bearing records. It is convenient to use squared paper. Suppose that AB of a 4-sided field ABCD is 80 feet long and that the bearing of A from B is 290 . (This is the typical reading from back sighting). The bearing of B from A is 290 - 180 = 110 . (The reciprocal bearing). Mark the starting point at A and with a circular protractor centred at A mark 110 . Remove the protractor, place ruler against pencil point held vertically at A, rotate ruler till it touches pencil held at the 110 mark; recheck position of ruler at A; draw line from A towards or through the 110 mark. Along this line from A measure scale equivalent of 80 ft. This locates B. Now with B as starting point draw in BC and so on. When a routine has been established this procedure can be streamlined somewhat, but this forms the basic drill for staff in training.

(c) A device which can speed up this work without appreciable effect on accuracy can be made from certain types of large semicircular protractor. One is needed in which the straight side passes through the centre point; any part of the protractor lying outside the diameter represented by this straight side should be pared or sandpapered away. A scale in centimetres or inches (depending on the units being used) needs to be laid out along this straight edge, starting from the centre point and going in both directions. Thus a protractor of 8 inches in diameter might have scales of zero at the centre extending to 4 inches at the intersection with the curved edge. Depending on the scaling of the angles some adjustment of the numbering of the degrees may be needed; this can be done easily by covering with self-adhesive labels any numbers to be changed. The linear scale is best put on the underside of the protractor. It may be possible to incise a suitable scale but it is probably easier to mark a scale on a ½ inch wide strip of squared paper and glue or tape it on the underside. This modified protractor can be used to get the angles and the length of the sides of plotted fields in one step. If a single scale for converting field length to plotted length is used the linear scale could be put in these units direct.

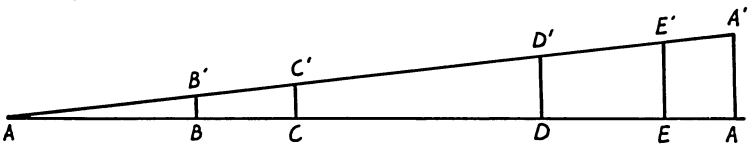
2. *Adjustments for simple cases of failure to close the traverses*

The figure A′, B, C, D, E, A is one in which A′ and A should have coincided but fail to do so because of errors in field reading and plotting.



The aim is to adjust A back to A′ and to "smooth" this adjustment back round E, D, C, B to A.

This can be easily done graphically by drawing a line A^1,–A, below.



equal, or proportionate, to the length of the traverse, marking the positions of B, C, D, E, along it. Put up a line A′–A equal in length to the required distance to be closed.

Perpendiculars put up at B, C, D and E will provide the distances by which B, C, D, and E should be adjusted on the plan. At B, C, D and E on the length of the traverse, lines are drawn parallel to A–A′.

Distances BB′, CC′ are then marked off to give the corrected positions of the corners of the figures. A′, B′, C′, D′, & E′, are then joined.

3. *Formulae useful in estimating areas*

Area of rectangle	length x breadth.
Trapezium	½ sum of two parallel sides times perpendicular distance between them
Parallelogram	perpendicular distance between two sides times length of one of them.
Triangle	½ base times vertical height or $s(s-a)(s-b)(s-c)$ where $s = \frac{a+b+c}{2}$ (semiperimeter formula)
Circle	3.1416 times radius squared 0.0795 times circumference squared

4. *Measurement of areas on maps and photographs*

(a) *Planimeter.* Full instructions are available with this instrument. By making its pointer follow the circumference of the part concerned the total area so traced is recorded on the register. Re-entrant sections must, however, be treated separately.

(b) *Squared paper.* It is sometimes convenient to prepare the diagram from which area measurements are to be taken on squared paper, (e.g. tracings of major features can sometimes be conveniently made on thin squared paper). Areas can then be computed by counting squares. Squares of which half or more are within the figures are counted, those with less than half ignored.

(c) *Transparent squared grid.* Areas can be computed on any paper, map or photograph by superimposing a transparent grid of suitable sized squares. These can be prepared from celluloid or acetate to a suitable scale. For odd jobs of this sort a spot of oil on a piece of ordinary graph paper will produce a transparent, though fragile, grid. One inch square on a one inch to the mile map represents about 640 acres or 1/8th inch square 10 acres. One inch squares falling wholly within the area should be counted first then the broken squares. When doing much counting it is easy to become muddled at some point between 40 and 80. A convenient device is to count in 20's, booking in groups of 4 strokes with the fifth across them, each such group representing 100.

(d) *Dot grid.* Regularly placed dots (e.g. formed by placing an ink dot at the point of intersection of lightly scored lines on a celluloid grid), can often be usefully substituted for the grid. (Photographic retouching fluid, or some varnishes, applied to celluloid or acetate will give a surface on which ink will "take".) Dots are then counted instead of squares. Those cut by the boundary are counted or not according to where the line falls but these decisions are few compared with those in square counting. When estimating areas (or taking any measurement) from photographs it is advisable before starting measurements to mark on the photograph the boundary or points concerned. A chinograph (wax) pencil is useful on glossy prints, though a pencil or pen will "take" on matt prints. Points are best marked by the crossing of two short lines.

It is worth bearing in mind that the *'proportion'* that part of an area is of the whole can often be judged quite accurately by eye. Occasionally, when very many readings are involved, eye judgement may be worth trying in preference to a detailed grid count. But where each map record represents, in recorders' time, etc., many times the cost of the time needed to do a count it would be false economy not to use the more accurate method.

5. *Assessment from air survey photographs*

Accurate and comprehensive interpretation is an expert job which lies outside the scope of this bulletin. However, certain points might usefully be made.

(a) Interpretation without reference to the ground is fraught with risk for all but the most obvious items.

(b) There are numerous cases where having air photos of the area concerned has been very valuable to field control staff in handling agricultural surveys. But there are also cases where the help was patchy either because the photos for certain parts of the area were obscure or because some of the staff found difficulty in using them. Generally speaking they are better looked on as aids than as basic tools, so far as agricultural statistics are concerned.

(c) Air photography is usually done at different scales for different purposes—perhaps from 1:6000 or larger in near urban areas to 1:30,000 for very open country. There seems a fair measure of agreement that for crop identification a scale of at least 1:10,000 is needed. As has been said elsewhere it is hardly likely that a general purpose air survey will have been made at a time ideal for the crops for which area estimated are required. Even where a good range of crops were in the ground at the time it is to be expected that only one or two can be positively identified in ordinary mixed cropping, though "bunded" lowland crops may be easy

(d) If assessment of areas is to be made from air photos, the portions of concern should be outlined and identified using chinograph pencil (yellow) before any measurements are attempted. How the measurement is to be done depends on the detail required and number of prints to be handled. Small numbers can be measured by planimeter, counting squares from a transparent grid or dots on a dot grid. Large numbers have been dealt with either by dot sampling or line intersect. For the former a randomly placed number of dots (minimum 2 per photo) should be disposed on the print and the vegetation under each identified. Randomly located points on half a dozen or so transparent overlays might be the easiest arrangement, drawing the overlay for use on any particular print at random. The relative frequency with which the points fall on the various land use types would be a proportion of their occupancy of the surface.

A line intersect method might have parallel lines across an overlay. This is laid on the print and the lengths of line lying over the various types of use gives the relative incidence in ground cover.

APPENDIX F

CHECKING AND TABULATION

1. *Checking*

Generally speaking, this falls into three sections.

(a) *Field Checking*

The only place to make sure that the details on the forms correspond to what actually exists in the field is at the site of the recording; Records should preferably be checked by a supervisor each night—or as soon as may be—and be subjected to general 'editorial' scrutiny for consistency and completeness before they leave the locality on their way to headquarters. Curious divergencies and suspicious similarities, many blanks and roundings, should be watched for. The making of fair copies of field returns should be discouraged; where they are made they should be clearly marked 'copy' *and the original attached.*

(b) *Checking receipt of returns*

There should be available at headquarters a record of the districts or other units from which forms are to be expected and how many should be returned by each. The supervisor responsible for the collection of a given block of records should be required to make a return stating how many records he is returning and accounting for any discrepancies compared with the number planned. At headquarters a check should be made that a consignment of records has been received from each supervisor who should be sending one and that the number in the consignment corresponds to the number he has noted that he is sending. In fact it would probably make little difference to the accuracy of the return if a few records were lost but unless a rigid check is maintained the loss of returns from whole districts is liable to pass unnoticed until they are needed for final working.

(c) *Checking of tabulation of records*

It is of the greatest importance that all computations should be checked stage by stage. It is wasteful to complete a series of calculations using figures derived from each other only to find on checking that there was an error in an early stage. The most useful checking methods in ordinary work are:

(a) checking of copied entries by a second person; this is critically important. A large proportion of errors arise when figures are transferred from one form to another;

(b) common knowledge and rough checks for misplaced decimals or wrong units of measurement;

(c) repetition, preferably by another worker;

(d) use of another method, e.g. to check a written calculation by repeating with a slide rule; or when multiplying by machine to use first one then the other of the pair of numbers on the key board;

(e) reversing the operation—e.g. checking a division by resetting the answer *as written down* from the first calculation, multiplying by the divisor and checking against the original figure;

(f) overall checks—these have the special advantage of not only checking individual computations but also detecting omissions and copying errors:

(i) summing series both directly to a grand total and by dividing the series into, say, 5 parts, summing each to a subtotal, then summing the subtotals;

(ii) arranging tabulations so as to give both line and column totals whose sums must be identical. These can usually be planned to give details for districts, tribes, etc., which are themselves interesting;

(iii) selecting any identities which may occur in the particular material.

The planning of what might be termed theoretical checks is not, however, sufficient to maintain efficiency in tabulation. Checking is usually tedious and offers little scope for

personal pride on the part of the workers. Some care may be needed to minimize such work as the preparation of straightforward summary sheets by copying from original forms. Theoretically it should make few demands on the staff but its sheer monotony invites scamping.

Again, it is not sufficient to arrange that totals are checked by totalling both individual items and subtotals–or cross totals. Staff are liable when they find a discrepancy to pass it over, excusing it to themselves by saying that "the man who did the first calculation was pretty good and usually right". The work must be arranged so that the totals of the items and of the subtotals are completed by different people and that these come together on the desk of their superior who compares the results. It helps in such work if the two figures can be brought, physically, very close together,–preferably one above the other, since it is by no means easy to be sure that two figures on two sheets are identical. (It is particularly easy to invert digits,–calling 1,739,897 identical with 1,738,987.)

2. *Tabulation*

Some enquiry organisers can depend on access to computing time more or less when they want it. It can be assumed that for them there will be no special problems in handling figures. At the other extreme are those who must manage with one or two desk machines. It is important not to despise simple aids, particularly not to despise them so far that work which could be done with them is not done simply because desired sophisticated facilities are not available.

The following suggestions do not pretend to put forward ideal arrangements. Rather, they are directed to the situation where computing services are not available for an enquiry either because total resources are small or because the enquiry is a lower priority one in a programme for which substantial resources are available.

(a) For the most part the operations needed for working-up agricultural statistics involve the grouping of records, the preparation of single-way and two-way frequency distributions, the calculation of totals and averages. With a small key-operated adding machine (non-electrical) a clerk can soon learn to handle addition and subtraction rapidly and accurately. Multiplication and division can be done for many purposes on a cylindrical slide rule.

(b) It may be possible to get help for simple calculations without involving a continuing commitment, by employing secondary school boys in the holidays, or agricultural college students. They may need careful supervision but, nonetheless, they make it possible to do a tabulation which would otherwise be left undone. Sometimes this may apply to the primary tabulation, sometimes to a secondary one omitted in the first round for lack of time.

(c) Carefully designed forms on which ruling and positioning of items make it easy to locate items can speed handwriting of forms.

(d) It may be useful to clip or punch holes at different positions round a form corresponding to certain items of information. When sorted for such an item the batch of such cards will show a continuous hole or groove at the appropriate place, providing a check that no wrong cards are included. A ball-point pen line down the edge of a batch of cards can give a similar check.

(e) Peg-board arrangements in which one or two edges of a form are punched with holes at regular intervals corresponding to pegs along the edge of a backing board greatly speed summation. The possibility of having all the right hand columns (or the tops and bottom lines) from a score of forms displayed side by side by such a method is fairly obvious. However, by suitable paper, folding, (and prior planning in positioning the items) practically any column or row can be brought to the edge and treated in this way.

(f) Cards which are designed for sorting by using a simple needle or rod offer scope for sorting and subsorting by any category for which information can be coded. They are useful when access to mechanised and electronic equipment cannot be relied on. The limiting factor in their use is usually getting the data on the cards since they may not

lend themselves to direct entry of data in the field. They are probably likely to be most useful in practice in small enquiries where there is prospect of continuing to use the records for some considerable period for repeated reference, since the relative importance of the cost of preparing the card in the first place will be less.

APPENDIX G

CHECK LISTS OF STAGES, PROCESSES AND EQUIPMENT

1. *Steps in an enquiry for which plans have to be made*

A survey of any appreciable size can be expected to have at least the following stages for which time and resources must be provided. (The items are not in time sequence).

(i) choosing the scope of the subject
(ii) defining and checking the frame
(iii) drawing the sample
(iv) preparing the timetable
(v) designing the forms or questionnaires
(vi) testing the forms or questionnaires
(vii) preparing instructions
(viii) reproducing forms, etc.
(ix) distribution of forms
(x) provision of equipment
(xi) distribution of equipment
(xii) training those who will train staff
(xiii) selection of staff
(xiv) training of staff
(xv) pilot surgery
(xvi) conducting enquiry
(xvii) collecting, checking and cleaning up completed forms
(xviii) tabulation, with or without preparatory coding
(xix) preparation of summary tables for publication
(xx) preparation of explanatory text for publication
(xxi) preparation of critique of the enquiry, its lessons and recommendations for the next one.

2. *Check list of equipment for area measurement and village surveys*

Lists of sample villages
Alternative list of villages
Instructions–for use of list
–for use of alternative
–for carrying out survey
Maps
Pads
Field notebook for each village
Pencils
Millboard
Tables, or pairs, of random numbers
Folder or envelope for each village to carry papers
Conversion table
Copy Paramount chief's, etc., instruction circular

1 Prismatic compass
3 Ranging rods
Chain
50 yard measuring wire graduated in 5 yards
Machete for bush cutting, peg making, etc.
List of vernacular names of crops
Farmer's calendar
Money
Requisite indent authority
First aid kit
Water bottle, etc. and such other kit as it is usual to carry on tour
Bicycle, tools and tyre repair kit

3. *Items to be covered in instructions to field teams for general survey*

To whom is the team responsible?
Whom should they contact if instructions do not cover the case? and how?
What types of case should specifically be referred to senior officer?
What are the arrangements for scrutinising completed schedules, forwarding them to headquarters?
What are the arrangements for travelling, subsistence, sickness relief, etc.
What arrangements have been made to notify villages?
How to contact village authorities and introduce team?
How to use alternative lists of sample villages?
Which record sheets are to be used for which purpose?
What degree of detail must be entered on sketch maps?
How are plots for area measurement to be chosen?
What cross-checks are to be applied to ensure completeness of records?

4. *Check list of equipment for crop cutting*

1 spring balance
1 chain, tape and/or calibrated wire; baskets–say 4
produce bags–say 50
ball of string
labels–say 100
prismatic compass
marker rods
materials for tripod
records of villages required

5. *Items to be covered in instructions for crop cutting survey*

Terms used
What has to be done (cutting, weighing, bagging, storing or transporting)
What has to be arranged with chief, farmers
Financial arrangements (if any)
Procedure for selecting crop patches on which sample cuts located
How to decide when cut shall be made
Procedure for selecting site of cut
Procedure for demarcating cut
How to adjust position of a cut which initially is sited partly outside the patch
Cutting and weighing

Threshing or rubbing out
Labelling - one inside and one outside each bag, each giving name of crop, weight cut, patch number, landholder's name
Which books and pads are to be used for which records
If printed forms are not provided how is the notebook to be ruled
What data must be recorded, etc.

For all cuts	For each cut
Village	Crop
Headman	Date of Harvest
Date	Name of farmer
Recorder	Size of cut
	Field weight of produce
	Leave space for later recording of dry and/or shelled weight

What detail must weights be recorded to?
Ticking off patch in reference list as the cuts are taken
What processing required and what does it consist of
- harvesting (e.g. what amount of straw)
- drying
- threshing
- winnowing
- shelling

(Any specially awkward process should be noted).

(Processing of crops in the ordinary way is often carried out by women and recorders may therefore need to arrange their help through the village head).

RELEVANT PUBLICATIONS

In preparing the following list considerable weight has been given to the question whether a publication is likely to be available or readily obtainable by the users of this manual. Abstracts of articles and other publications on agricultural economics and agricultural economic statistics, other than reference books of statistics, are published in the *World Agricultural Economics and Rural Sociology Abstracts* published by the Commonwealth Bureau of Agricultural Economics, and obtainable from The Commonwealth Agricultural Bureaux, Farnham House, Farnham Royal, Slough, Bucks, England. The scope of the abstracting service is worldwide—it is not confined to publications from Commonwealth countries—and began in 1964/5.

A. *Textbooks on sampling methods*

Cochran, William G. *Sampling techniques* New York: Wiley 1965 pp 413.

Stampford, M.R. *An introduction to sampling theory with applications to agriculture* Edinburgh: Oliver and Boyd 1962 pp 292.

Sukhatme, P.V. *Sampling theory of surveys with applications* New Delhi: Indian Society of Agricultural Statistics 1954 pp 492.

Yates, Frank *Sampling for censuses and surveys* 3rd edition. London: Griffin 1961 pp 440.

B. *Background* – indicating requirements of statistics

F.A.O. The use of agricultural census results as basic data for planning, formulating, and appraising agricultural development projects. F.A.O. *Monthly Bulletin of Economics and Statistics* 13 No 7/8 July/Aug. 1964.

F.A.O. 1970 *World Census of Agriculture*
Regional programme for Africa, 1967 pp 105.
Regional programme for Near East, 1967 pp 98.
Regional programme for Asia and Far East, 1967 pp 100.
Rome: F.A.O.

O.E.E.C. *Regional economic planning, Techniques of analysis for less developed areas.* O.E.E.C. Paris 1961.

U.N./E.C.A.F.E. *Basic statistics for formulating and implementing plans of economic and social development in countries of Asia and the Far East.* U.N.O. 1965.

C. *Comprehensive accounts of agricultural statistical services*

Theodore, G. *Statistiques Agricoles dans les Pays en voie de Developpement.* Centre European de Formation des Statisticiens Economistes des Pays en voie de Developpement 1966.

United States Department of Agriculture
Statistical Reporting Service of the U.S. Dept. of Agriculture Scope – Methods Washington 1964
Miscellaneous Publication No 967.

D. *Experience in agricultural statistical work in developing countries*

Centro *Symposium on Agricultural Statistics*
Ankara, Turkey Oct. 17–28, 1967.
(Basic concepts and needs for agricultural statistics. Description of current practices and problems, experiences and future plans for agricultural census, new techniques and methods including photogrametry).

F.A.O. *Report on the improvement of coconut statistics in countries of Asia and the Far East* (1960–65)
Rome: F.A.O. 1967 pp 150 TA 2313.

F.A.O. *FAO Regional Project on cocoa statistics for evolving a suitable technique to forecast the cocoa crop production in West Africa.*
Rome: F.A.O. 1965 pp 30 E.P.T.A. No 2093.

Panse, V.G. *Some problems of agricultural census taking with special reference to developing countries.*
F.A.O. Rome 1965 (mimeo) pp 76.

Sukhatme, P.V. Recent experiences in sample censuses in agriculture.
F.A.O. *Monthly Bulletin of Economics and Statistics*
Vol. 17 No 7/8 July–August 1968.

Zarkovich, S.S. *Estimation of areas in agricultural statistics*
Rome: F.A.O. 1956 pp 203.

– *Sampling methods and censuses*
Rome: F.A.O. 1965 pp 214.

– *The quality of statistical data*
Rome: F.A.O. 1966 pp 396.

E. *Crop yield estimating and forecasting* (see also D)

Houseman, E.E. and Huddleston, H.F.
Forecasting and estimating crop yields from plant measurements
F.A.O. *Monthly Bulletin of Economics & Statistics 15*
No 10 October 1968.

Sanderson, Fred H. *Methods of crop forecasting*
Harvard University Press 1954 pp 260.

F. *Food consumption studies*

Reh, Emma *Manual on household food consumption surveys*
Rome F.A.O. 1962.
F.A.O. Nutritional Studies No 18.

F.A.O. *Handbook for the preparation of Food Balance Sheets*
Washington, U.S.A. F.A.O. 1949 pp 36.

G. *Data handling*

F.A.O. *Handbook on Data Processing Methods,* Provisional edition
Part I. 1959 pp 112. Part II 1962 pp 140. F.A.O. Rome.
(concerned with practical aspects of obtaining, housing and operating data-processing equipment; forms; cards; organisation of work; etc. etc.)

F.A.O. *Report on the 1960 World Census of Agriculture.*
Processing and tabulation. Rome, F.A.O. 1918.
(More an account of what was done than of how to do).